

INTELLISOURCE

AI - DRIVEN SOURCING & CONTRACTING PLATFORM

PLANNING PIPELINE REVIEW — PROCUREMENT BUSINESS UNIT

Full planning artifact set: Product Brief · PRD · Architecture · UX Specification · Epics & Stories · Research · Readiness (GO)

Version 01.00 · 24 July 2026 · Confidential

Gijima

INTELLISOURCE AI - DRIVEN SOURCING & CONTRACTING PLATFORM

Issued by	Gijima Group, Midrand Gate: Gijima, 1 Pioneer Avenue, South Wing, Midrand, Johannesburg
Area of application	Gijima Procurement Business Unit — IntelliSource product initiative
Document purpose	Consolidated review pack of the complete BMad planning pipeline for the IntelliSource platform, for Procurement team review and sign - off ahead of implementation.
Version / Status	01.00 / Issued for review
Author	Andre Vorster (andre.vorster@gijima.com) — BMad Enterprise SDLC pipeline
Reviewer	Procurement Business Unit (this review)
Involved parties	Gijima Procurement BU (Customer Zero) · Product · Architecture · UX · Delivery
Contact	andre.vorster@gijima.com
Brief details	Seven artifacts produced 24 July 2026; readiness decision GO at aggregate quality 95.7/100; 124 functional requirements traced 100% to 79 implementation stories.

Copyright (C) 2026 by Gijima Group.

This document and any information contained in this document is proprietary to Gijima and/or its affiliates. Any information gained from the use, viewing, reading or in any other manner becoming aware of the contents of this document must be retained as confidential and no person may disclose, publish, or use the confidential information for their own or any other persons' gain, direct or indirect benefit or interest, or incorporate, modify and create derivative works thereof, or otherwise employ, exploit or use this document in any manner except for the purpose as may be agreed in writing by Gijima and/or its affiliates.

VERSION HISTORY AND DISTRIBUTION

Version	Date revised	Change No.	Edited by	Changes / Comments
01.00	24.07.2026	—	A. Vorster	Initial issue — full planning pipeline consolidated for Procurement review

Distribution list

Name / Role	Organisation	Purpose
Procurement Business Unit leadership	Gijima	Review & sign - off (business requirements, process fit)
Category managers & sourcing leads	Gijima Procurement BU	Review journeys, stories, mockups
Head of Procurement	Gijima	Governance, audit & compliance review
Product & delivery team	Gijima	Reference baseline for implementation

TABLE OF CONTENTS

INTELLISOURCE AI - DRIVEN SOURCING & CONTRACTING PLATFORM	2
VERSION HISTORY AND DISTRIBUTION	3
TABLE OF CONTENTS	4
EXECUTIVE SUMMARY	7
1. Product Brief	8
1.1 Vision	8
1.2 Problem Statement	8
1.3 Target Users (Personas)	9
1.4 Value Proposition	10
1.5 Differentiators (defensible order)	10
1.6 MVP Scope (Stage 1, 0–12 months)	10
1.7 Cross - Cutting MVP Decisions	12
1.8 Success Metrics (KPIs)	12
1.9 Constraints	13
1.10 Risks (top 5, from research)	14
2. Product Requirements Document (PRD)	15
2.1 Document History	15
2.2 Executive Summary	15
2.3 Glossary	15
2.4 Personas (from Product Brief)	16
2.5 User Journeys	16
2.6 Functional Requirements (MVP — Stage 1)	22
2.7 Stage 2–3 Requirements (Outline — for roadmap epics)	29
2.8 Non - Functional Requirements	30
2.9 Data Model (core entities)	33
2.10 API Overview (representative /api/v1 endpoints)	35
2.11 Assumptions, Dependencies, Out of Scope, Open Questions	36
3. Technical Architecture	37
3.1 Architecture Overview	37
3.2 Gijima Architecture Principles → Design Response	38
3.3 Service Decomposition (bounded contexts)	39
3.4 Clean Architecture Layout (per service)	40
3.5 Architecture Decision Records	40
3.6 Data Architecture	43
3.7 Testing Architecture	43

3.8 Error Handling Architecture	44
3.9 Security Architecture (Zero Trust + defence - in - depth)	44
3.10 Infrastructure & CI/CD	44
3.11 Performance Architecture	45
3.12 Accessibility Architecture (WCAG 2.1 AA, target 2.2 AA)	45
3.13 Coding Standards	46
4. UX Design Specification.....	47
4.1 Design Principles	47
4.2 Design Tokens	47
4.3 Component Library	50
4.4 Page Layouts (12 core).....	53
4.5 Interaction & State Patterns	54
4.6 Accessibility Specification (WCAG 2.1 AA — gate)	55
4.7 Frontend - Design Readiness Checklist	55
4.9 Reference UX mockups (generated from the design system)	56
5. Epics & User Stories	59
5.1 Epic Overview.....	59
5.2 EPIC - 01 — Foundation & Platform Infrastructure (P0)	60
5.3 EPIC - 02 — Intake & AI Specification (P0)	64
5.4 EPIC - 03 — RFX Authoring & Lifecycle (P0)	67
5.5 EPIC - 04 — Review, Approval & Publication (P0).....	70
5.6 EPIC - 05 — Supplier Portal & Sealed Responses (P0)	72
5.7 EPIC - 06 — Change Request Management (P1).....	75
5.8 EPIC - 07 — Evaluation & Award (P0)	76
5.9 EPIC - 08 — Teams Orchestration (P1).....	79
5.10 EPIC - 09 — Audit Query & Evidence Packs (P0)	80
5.11 EPIC - 10 — Dashboards, Analytics & Notifications (P1).....	81
5.12 EPIC - 11 — Identity, Admin & POPIA (P0).....	83
5.13 EPIC - 12 — Hardening, Performance & GA Readiness (P1)	85
5.14 FUTURE - 13 — Stage 2 Epics (12–24 months, outline)	86
5.15 FUTURE - 14 — Stage 3 Epics (24–36 months, outline)	87
5.16 Dependency Graph	87
5.17 Sprint Plan (2 - week sprints, indicative — 3 - dev + 1 - QA team)	87
5.18 Requirement Traceability Matrix (124/124 FRs mapped — 100%)	88
6. Supporting Research: Market, Competitors & Compliance (Annex)	92
6.1 Executive Summary.....	92
6.2 Research Questions & Answers	92

6.3 Competitor Comparison Table — Legacy Suites (July 2026)	93
6.4 Challenger Comparison Table — AI - Natives (feature blueprint)	94
6.5 Feature Capability Matrix (market scan → IntelliSource backlog candidates)	95
6.6 RFX Reuse Inventory (seed assets)	97
6.7 Microsoft - Native Strategy Assessment	98
6.8 South Africa / Africa Compliance Requirements (moat features)	98
6.9 Technology Stack Options	98
6.10 Best Practice Patterns	99
6.11 Benchmarks (targets to match or beat)	100
6.12 Key Findings & Recommendations	100
6.13 Risks & Constraints	101
6.14 Sources	101
7. Implementation Readiness Report — GO/NO - GO	103
7.1 Executive Summary — Decision: ✓ GO	103
7.2 Artifact Inventory	103
7.3 Cross - Reference Validation (programmatically verified where possible)	104
7.4 Findings	105
7.5 Quality Score Detail	105
7.6 Recommendations	106

EXECUTIVE SUMMARY

IntelliSource is Gijima's AI - driven, Microsoft - native, end - to - end sourcing and contracting platform: plain - language needs become governed RfX events, suppliers are engaged through a free portal, bids are evaluated with explainable AI, stakeholders act inside Microsoft Teams, and every step lands in a court - grade audit trail with built - in South African compliance (POPIA, King V, B - BBEE readiness).

This pack consolidates the **complete planning pipeline** produced on 24 July 2026 for Procurement review. Each chapter reproduces one planning artifact in full, in pipeline sequence; supporting market research is annexed before the closing readiness report. The pipeline concluded with a **GO decision at an aggregate quality score of 95.7/100**, with 124 functional requirements traced 100% to 79 implementation - ready stories.

BMad Planning Pipeline

Seven quality-gated agents take IntelliSource from business case to implementation-ready backlog

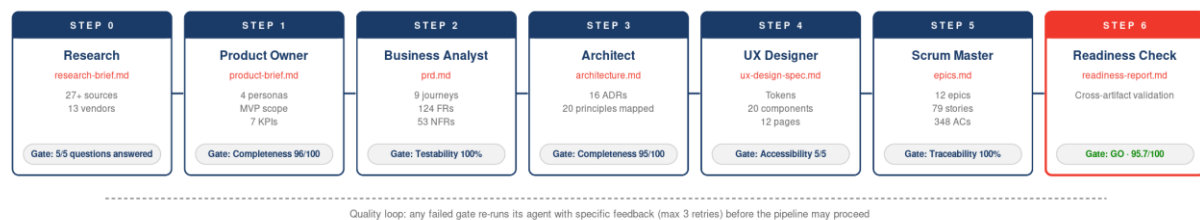


Figure 1 — The seven - step BMad planning pipeline and its quality gates

How to review this document. Procurement stakeholders should focus on: Chapter 1 (what we are building and why), Chapter 2 sections on user journeys and functional requirements (the business requirements), Chapter 4's reference mockups (what users will see), and Chapter 5's epics (scope and sequencing). Architecture (Chapter 3) is included in full for technical reviewers. The readiness report (Chapter 7) is the formal gate evidence.

Chapter	Artifact	What it answers	Key numbers
1	Product Brief	Why build it; for whom; what is in the MVP	4 personas · 10 must - have capabilities · 7 KPIs
2	PRD	What the platform must do, testably	9 journeys · 124 FRs · 53 NFRs · 23 entities
3	Architecture	How it is built and secured	16 ADRs · all 20 Gijima principles mapped
4	UX Specification	How it looks and behaves (Gijima brand)	Full token set · 20 components · WCAG 2.1 AA
5	Epics & Stories	How it will be delivered	12 epics · 79 stories · 348 ACs · 182 dev - days
6	Research Brief (annex)	Market, competitors, compliance evidence	13 vendors · SA compliance moat analysis
7	Readiness Report	Formal GO/NO - GO with gate scores	GO · 95.7/100 · 0 blockers

Decision requested from Procurement: confirm the business requirements (Chapter 2), personas and process fit (Chapters 1–2), and MVP scope boundaries (Chapters 1 and 5), or return consolidated comments for a pipeline change cycle before implementation starts.

1. PRODUCT BRIEF

Source artifact: product - brief.md

The product vision, target users, value proposition, MVP scope and success metrics — the foundation every later artifact builds on.

Project: IntelliSource — AI - Driven Sourcing & Contracting Platform

Owner: Gijima Procurement Business Unit

Author: BMad Product Owner Agent

Date: 2026 - 07 - 24

Status: Approved for PRD

Inputs: research - brief.md, IntelliSource Expanded Business Case v1, Gijima Architecture Design Principles

1.1 Vision

IntelliSource is the Autonomous Procurement Orchestrator: an AI - driven, Microsoft - native, end - to - end sourcing and contracting platform that turns plain - language business needs into governed RFx events, discovers and invites suppliers, evaluates bids with explainable AI, orchestrates stakeholders inside Microsoft Teams, and drives to contract award — with a court - grade audit trail and built - in South African compliance.

One sentence: From "I need 200 laptops" typed in Teams to a signed, auditable contract award — in days, not months.

IntelliSource End-to-End Business Process

From plain-language need to audited contract award — every action recorded in the immutable audit trail

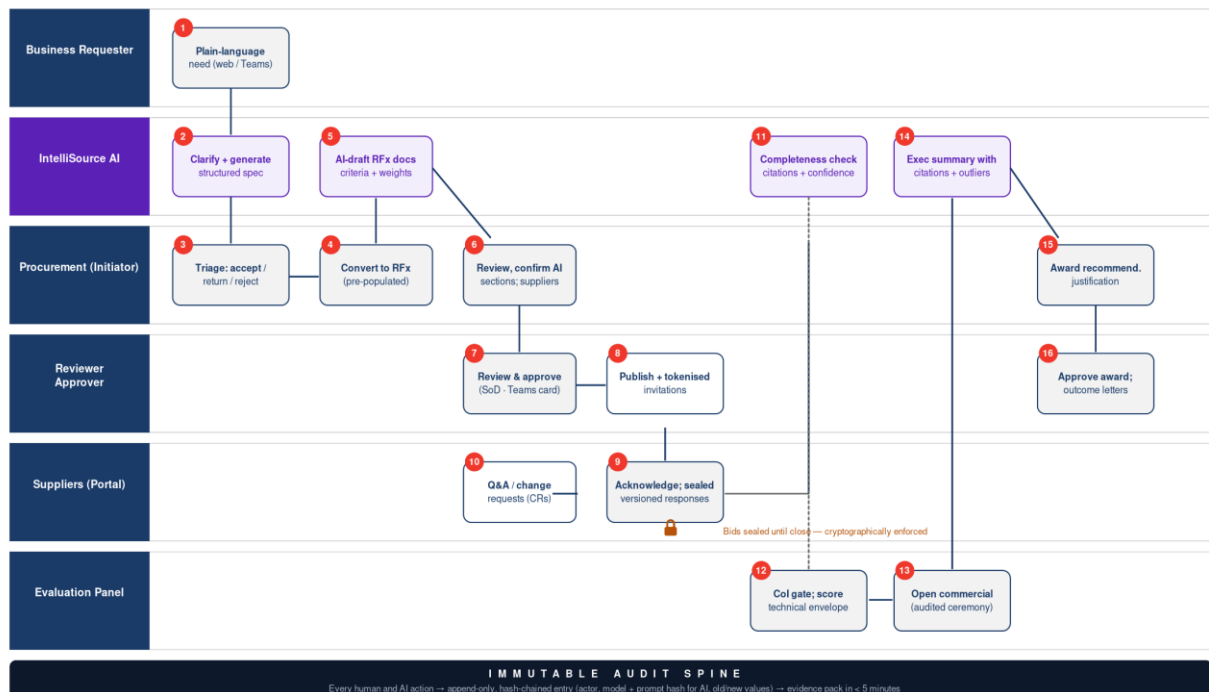


Figure 2 — IntelliSource end - to - end business process (intake to audited award)

1.2 Problem Statement

Organizations waste 60–70% of sourcing cycle time on manual intake, RFx preparation, supplier lookup, and committee approvals. Requirements arrive as vague emails; buyers re-type them into Word templates; supplier lists are stale ERP extracts; bids arrive as email attachments viewed prematurely (compliance risk); evaluation happens in spreadsheets with no scoring audit trail; committee approvals crawl through calendars; and when auditors ask "prove this award was fair," the evidence is scattered across mailboxes.

Existing platforms don't fix this for our market:

- **Legacy suites (SAP Ariba, Coupa, GEP, Ivalua, Jaggaer)** cost \$250k+/year, take 6–18 months to deploy, require consultants, and charge suppliers network fees. All now have AI copilots — but none are Microsoft Teams - native, none automate South African B - BBEE/PPFA preference scoring, and none fit mid - market budgets.
- **AI - native challengers (Zip, Levelpath, Fairmarkit)** prove the demand for conversational intake and autonomous sourcing but are US - centric, lack RFx - to - contract depth, and have zero African compliance coverage.
- **Manual/email processes** — the actual incumbent for most African mid - market and public - sector organizations — offer no auditability, no cycle - time control, and growing regulatory exposure (POPIA, King V, Public Procurement Act 2024).

1.3 Target Users (Personas)

1.3.1 Persona 1 — Nomsa, Business Requester (Operations Manager)

Needs goods/services but is not a procurement professional. **Pain points:** (1) Doesn't know how to write a specification — her email requests bounce back with questions for weeks; (2) has no visibility into where her request sits once submitted; (3) resorting to maverick spend because "procurement takes too long." **Goal:** describe what she needs in plain language in Teams and watch it progress. **Current workflow:** email to procurement → wait → clarification ping - pong → escalate to her director.

1.3.2 Persona 2 — Thandi, Senior Category Manager (RFx Creator & Sourcing Lead)

8+ years in strategic sourcing, owns high - value events. **Pain points:** (1) Spends 3–4 hours/week converting vague requests into RFx documents and chasing email submissions; (2) cannot prove invitation delivery or keep bids sealed until close — audit exposure; (3) evaluation consolidation in Excel takes days and invites scoring - bias accusations. **Goal:** launch governed events fast, with AI drafting the RFx and scoring support she can defend. **Current workflow:** Word templates + Outlook + shared drives + Excel scoring sheets.

1.3.3 Persona 3 — Pieter, Head of Procurement / CPO (Governance Owner)

Reports to CFO; faces external auditors and, for SOE clients, Parliament - grade scrutiny. **Pain points:** (1) No real - time pipeline visibility without chasing buyers; (2) cannot rapidly produce audit - ready evidence of fair process (King V, PFMA); (3) B - BBEE preference calculations done by hand and challenged by losing bidders. **Goal:** dashboard oversight, provable governance, automated preference scoring. **Current workflow:** monthly Excel reports, manual audit file assembly.

1.3.4 Persona 4 — Ahmed, Supplier Respondent (Supply - Chain Manager at mid - sized vendor)

Responds to RFQs/RFPs for multiple customers. **Pain points:** (1) Juggles 8+ different customer portals, each with fees or clunky logins (industry average 8.4); (2) email submissions vanish — no receipt, no deadline clarity, reminders after he's already submitted; (3) version confusion when he updates pricing. **Goal:** one free, mobile - friendly portal with acknowledgement, receipts, versioning, and fair Q&A. **Current workflow:** email attachments and hope.

1.4 Value Proposition

For mid - to - large enterprises and public entities standardized on Microsoft 365

Who lose months and audit defensibility to manual, fragmented sourcing

Our product IntelliSource

Is an AI - driven autonomous procurement orchestration platform (Source - to - Contract)

That cuts sourcing cycle time $\geq 30\%$ by turning plain - language needs into governed RFx events, AI - evaluated bids, and Teams - orchestrated decisions with immutable auditability

Unlike SAP Ariba/Coupa/GEP (expensive, slow to deploy, not Teams - native, no African compliance) and email/Excel (no governance at all)

Our product is Microsoft - native (Teams, Outlook, Entra ID, Agent Store), deploys in weeks at mid - market pricing with zero supplier fees, and ships POPIA/King V/B - BBEE compliance as first - class features.

1.5 Differentiators (defensible order)

- 1. South African/African compliance moat** — automated B - BBEE 80/20 & 90/10 preference - point scoring, CSD/SARS/CIPC verification hooks, POPIA - by - design (SA data residency, s72 - aware AI routing), King V - grade audit evidence, Public Procurement Act - ready rule packs. No global suite does this.
- 2. Microsoft - native orchestration depth** — approvals, committee meetings, e - voting, and notifications inside Teams; Entra ID + MFA; distribution via M365 Agent Store; agents built on Microsoft Agent Framework with MCP/A2A interop so IntelliSource extends (not fights) customer Copilot estates.
- 3. End - to - end AI workflow, human - governed** — intake→spec→RFx draft→completeness check→weighted scoring→executive summaries, every AI output with rationale + citations, human confirmation audited. Explainability designed for tender - dispute defensibility.
- 4. Speed - to - value economics** — weeks to deploy, \$8k/mo mid - market / \$18k/mo enterprise, zero supplier fees (explicitly anti - Ariba), free mobile - first supplier portal.
- 5. Proven foundation** — RFx lifecycle, supplier portal, and audit framework battle - tested in the RFx Supplier Sourcing System (GO - rated 95.9/100), de - risking the MVP build.

1.6 MVP Scope (Stage 1, 0–12 months)

1.6.1 Must - Have (product doesn't work without these)

M1. Conversational requirement intake — plain - language need capture (web + Teams), AI - guided clarification, structured specification generation, category auto - tagging, approval routing to sourcing.

M2. AI RFx authoring — generate RFQ/RFP/RFI drafts from the approved spec using template libraries; buyer edits and owns the final document; versioned templates (RFx pattern).

M3. RFx lifecycle & governance engine — statuses Draft → PendingReview → PendingApproval → Published → Closed (+ Rejected/Returned/Clarification), segregation of duties, mandatory comments on reject/clarify, auto - approval when reviewer=approver, RFQ short path; RFx - YYYY - NNN numbering; clone/edit/delete rules by status. (*RFx reuse*)

M4. Supplier invitation & portal — invite from supplier master or manual entry; tokenized email invitations (Outlook/Graph); free supplier portal with first - time password setup, acknowledgement - gated document access, versioned draft/final submissions, deadline sealing (bids hidden until close), change requests. (*RFx reuse*)

M5. Sealed - bid evaluation workspace — two - envelope - capable data model (technical/commercial separation); AI completeness check (missing docs, certs, forms) with gap flags; configurable weighted scoring matrix; AI evaluation summaries with rationale + citations to bid text; side - by - side comparison; human confirm/override with audited reasons.

M6. Teams orchestration (wedge) — approval requests as actionable Teams cards; auto - scheduled review/evaluation meetings with agenda + links; event notifications in channels; adaptive - card status queries.

M7. Immutable audit & evidence — append - only, hash - chained audit log of every human and AI action (model version + prompt hash for AI); status history; evidence - pack export (PDF/CSV) per event. (*RFx framework, hardened*)

M8. Dashboard & reporting — real - time pipeline view by status/owner; KPI cards (cycle time, open events, responses); charts with CSV/PDF export; SignalR live updates. (*RFx reuse + charts*)

M9. Identity & admin — Entra ID SSO + MFA for internal users; roles Initiator/Reviewer/Approver/Administrator/ViewOnly (extensible); user & supplier administration; POPIA data - subject tooling (access/correction/deletion).

M10. Platform & API foundation — .NET 10 Clean Architecture, EF Core 10 + PostgreSQL, React + Tailwind (Gijima brand tokens), OpenAPI 3.2 published spec, Docker→AKS deployment, multi - tenant with in - tenant deployment option, virus - scanned uploads, RFC 7807 errors.

1.6.2 Should - Have (MVP if capacity allows; else first fast - follow)

S1. Multi - level/configurable approval chains (RFx gap).

S2. B - BBEE data capture on supplier profiles (certificate/affidavit upload + level) — scoring engine lands Stage 2, but capture starts now.

S3. Anonymous supplier Q&A with addenda broadcast.

S4. Notification engine with scheduled reminders (7 - day/1 - day) and retry management. (*RFx reuse — likely in MVP*)

S5. Basic savings & cycle - time instrumentation (baseline vs actual per event) — feeds the ROI story.

1.6.3 Nice - to - Have (explicitly deferred)

- Stage 2 (12–24 mo): AI supplier discovery (web/ESG/EcoVadis), committee blind scoring & e - voting in Teams, multi - round/BAFO, B - BBEE 80/20–90/10 scoring engine + PPPFA staged evaluation, ERP connectors (SAP/Oracle/D365/Sage), demand bundling, savings ledger, ISO 27001/SOC 2 certification work.

- Stage 3 (24–36 mo): contract drafting from award + clause libraries + AI redlining, eSignature, obligation tracking, auction engine, predictive analytics, autonomous negotiation, contract - expiry - triggered re - sourcing, Copilot Agent Store listing.

1.6.4 Out of Scope (MVP)

Purchase orders/P2P execution, invoicing, inventory, supplier performance management, payment processing, non - Microsoft collaboration surfaces (Slack/GChat).

Three-Stage Product Roadmap.

MVP detailed in this plan (Stage 1) — Stages 2–3 outlined as future epics FUTURE-13 / FUTURE-14 for traceability

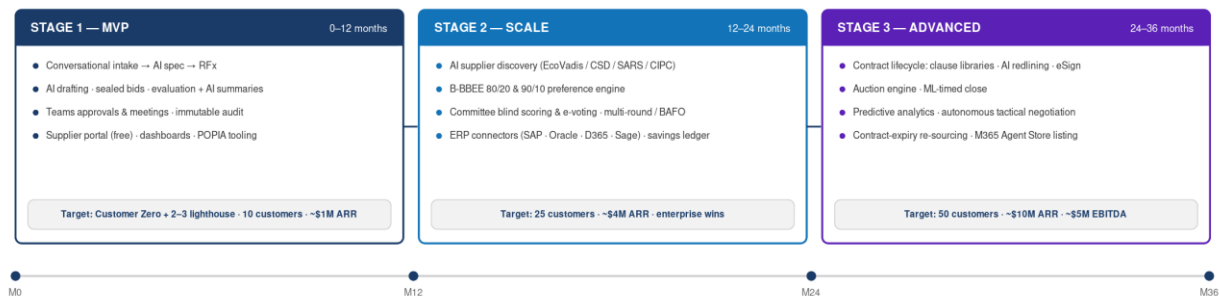


Figure 3 — Three - stage roadmap: MVP detailed in this plan; Stages 2–3 outlined as future epics

1.7 Cross - Cutting MVP Decisions

- **Error recovery:** automatic retry with exponential backoff (100/300/900ms, max 3) on transient failures; graceful AI degradation — if AI services are down, every AI - assisted step falls back to manual authoring/scoring, never blocking the sourcing process; queued outbound email/Teams messages with retry + dead - letter; autosave drafts.
- **Accessibility:** WCAG 2.1 AA compliance target (design toward 2.2 AA), enforced in CI (axe scans) + manual screen - reader passes; supplier portal is the highest - exposure surface; publish an ACR — sales asset for public sector.
- **API - first:** the API is a first - class product (Gijima principle #7). OpenAPI 3.2 spec is an MVP deliverable, published at /api/docs; versioned v1; webhooks for event lifecycle.
- **Performance targets:** API p95 <500ms; page LCP <2.5s / INP <200ms; initial bundle <1MB gzipped; architecture sized for 10k+ concurrent users <2s (principle #13); 99.5% app uptime.
- **AI governance:** agents act strictly under the invoking user's permissions; every AI action logged with model version + prompt hash; no customer data used for model training; human - in - the - loop on all award - affecting outputs; SA - resident inference preferred, s72 - papered otherwise.

1.8 Success Metrics (KPIs)

#	Metric	Target	Timeframe	Type
K1	Sourcing cycle time (intake→award decision), instrumented in - product	≥30% reduction vs customer baseline	Within 6 months of Customer Zero go - live	Business value
K2	Pilot adoption	Gijima Customer Zero live + 2–3 lighthouse	Month 12	Acquisition

#	Metric	Target	Timeframe	Type
		clients; 10 paying customers / ~\$1M ARR		
K3	Intake conversion	≥70% of plain - language intakes reach published RFX without manual re - work escalation	Month 9	Engagement
K4	Supplier experience	≥90% invited suppliers acknowledge via portal; supplier SUS ≥75; 100% submissions receipted	Month 9	Engagement
K5	AI evaluation utility	≥80% of AI completeness flags accepted by evaluators; 100% of AI summaries carry citations; 0 unexplained AI decisions in audit	Continuous	Trust/quality
K6	Technical SLOs	API p95 <500ms; LCP <2.5s; 99.5% uptime; error - recovery success ≥95%; axe a11y score ≥95, 0 critical	Continuous from sprint 1	Technical
K7	Audit readiness	Evidence pack for any event generated <5 min; 100% actions (human+AI) in immutable log	Continuous	Governance

1.9 Constraints

- **Architecture:** Gijima's 20 Architecture Design Principles are contractual: .NET 10 Clean Architecture; MVC + React + Tailwind; EF Core→PostgreSQL; Docker local / AKS production; Zero Trust + defence - in - depth; OpenAPI 3.2.0; POPIA; Entra ID SSO + MFA; full audit; microservices with strict boundaries; 10k+ users <2s; 99.5%/99.99% uptime.
- **Branding:** UI implements gijima - styles.css tokens (web palette red #F20023 / navy #0E355A, Proxima Nova, pill buttons/forms, eyebrow labels, red - dot heading accent).
- **Budget/timeline:** Stage 1 MVP in 9–12 months, lean team (business case: R36.9m / ~\$2.05M Year 1).
- **Deployment:** multi - tenant SaaS + in - tenant option for regulated clients; Azure South Africa North primary region.
- **Reuse mandate:** RFX planning artifacts and patterns seed the RFX core (~70% requirement reuse) — do not reinvent.

1.10 Risks (top 5, from research)

1. Microsoft first - party agents commoditize light procurement AI → counter with domain depth + MCP interop.
2. Azure OpenAI model gaps in SA North vs POPIA s72 → model - routing abstraction + contractual cover.
3. AI evaluation challenged in tender disputes → explainability + human - in - the - loop + immutable overrides.
4. Enterprise sales cycles (9–18 mo) → Customer Zero internal pilot proves ROI while lighthouse deals mature.
5. Suite consolidation absorbs challenger partners → multi - source supplier - data strategy.

Quality Gate — Product Owner (artifact_completeness):

- ✓ Problem statement clear and specific (quantified pain, named alternatives)
- ✓ 4 personas with pain points, goals, current workflows
- ✓ MVP scope: 10 must - have / 5 should - have / staged nice - to - have + explicit out - of - scope
- ✓ 7 success metrics, each with number + timeframe
- ✓ Error recovery, a11y (WCAG 2.1 AA), API - first, performance targets defined
- ✓ No placeholders
- **Score: 96/100** ✓ (≥90 gate passed)

Next: Business Analyst agent → prd.md

2. PRODUCT REQUIREMENTS DOCUMENT (PRD)

Source artifact: prd.md

The complete business requirements: user journeys with failure and edge cases, 124 testable functional requirements, 53 measurable NFRs, the data model and the API surface.

Project: IntelliSource — AI - Driven Sourcing & Contracting Platform

Author: BMad Business Analyst Agent

Date: 2026 - 07 - 24

Version: 1.0

Status: Approved for Architecture

Inputs: product - brief.md, research - brief.md, RFX PRD (reuse baseline), Gijima Architecture Design Principles

2.1 Document History

Version	Date	Author	Changes
0.1	2026 - 07 - 24	Business Analyst Agent	Initial draft from product brief
1.0	2026 - 07 - 24	Business Analyst Agent	Full FR/NFR set, journeys, data model, API overview

2.2 Executive Summary

IntelliSource turns plain - language business needs into governed RFX events, automates supplier engagement through a free portal, evaluates bids with explainable AI, orchestrates approvals and committees in Microsoft Teams, and produces court - grade audit evidence — deployed on Gijima's mandated architecture (.NET 10 Clean Architecture, React + Tailwind, EF Core + PostgreSQL, AKS, Zero Trust, OpenAPI 3.2).

This PRD specifies the **Stage 1 MVP** in full detail (10 must - have capability areas M1–M10 from the product brief) and outlines Stage 2–3 requirements for roadmap traceability. Approximately 70% of the RFX execution core reuses proven requirements from the RFX Supplier Sourcing System (GO - rated); net - new scope concentrates on AI intake, AI RFX authoring, AI evaluation, Teams orchestration, and evidence packaging.

Scope pillars: (1) Intake - to - Spec AI, (2) RFX Lifecycle & Governance, (3) Supplier Portal, (4) Sealed Evaluation with AI assistance, (5) Teams Orchestration, (6) Immutable Audit & Evidence, (7) Dashboards & Reporting, (8) Identity, Admin & POPIA, (9) API - First Platform.

2.3 Glossary

Term	Definition
RFX	Collective term for RFQ (Request for Quotation), RFP (Request for Proposal), RFI (Request for Information)
Intake	A plain - language business need captured before it becomes an RFX
Specification (Spec)	Structured requirement document generated from an intake (scope, quantities, criteria, timelines)
Sealed bid	Supplier response hidden from all internal users until the event closing time

Term	Definition
Two - envelope	Separation of technical and commercial response content; commercial opened only after technical gate
BAFO	Best and Final Offer (multi - round negotiation) — Stage 2
CR	Change Request against a published RFx (initiator - or supplier - raised)
SoD	Segregation of Duties (creator ≠ approver)
Evidence pack	Exported, tamper - evident bundle of all event records for audit
B - BBEE	Broad - Based Black Economic Empowerment (SA preferential procurement) — scoring engine Stage 2, data capture MVP
Customer Zero	Gijima's internal Procurement BU as first production user
Operator (POPIA)	Party processing personal information on behalf of a Responsible Party — IntelliSource's legal role
Agent	An AI process acting under a human user's identity and permissions

2.4 Personas (from Product Brief)

Nomsa — Business Requester · Thandi — Senior Category Manager (Initiator/Sourcing Lead) · Pieter — Head of Procurement (Governance/ViewOnly+Approver) · Ahmed — Supplier Respondent. Internal system roles: **Requester, Initiator, Reviewer, Approver, Administrator, ViewOnly**; external: **SupplierUser**.

2.5 User Journeys

2.5.1 J1 — Plain - Language Intake to Approved Specification (Nomsa, Thandi)

Trigger: Nomsa needs goods/services (e.g., "We need 200 mid - range laptops for the Durban office by October").

#	Actor	Action	System response
1	Nomsa	Types need in Teams bot or web intake form	AI asks 3–7 clarifying questions (quantity, budget range, delivery, constraints, category confirmation) one group at a time
2	Nomsa	Answers; uploads any reference docs	Draft structured spec generated (scope, line items, evaluation criteria suggestions, timeline); category auto - tagged
3	Nomsa	Reviews spec summary; submits	Intake # INT - YYYY - NNN created, status PendingTriage; confirmation with tracking link
4	Thandi	Opens triage queue; reviews spec	Can accept (convert to RFx), return to requester with comments, or reject with reason

#	Actor	Action	System response
5	Thandi	Accepts → "Convert to RFx"	New RFx pre - populated from spec; intake linked; Nomsa notified with status link

Success: Intake converted to draft RFx ≤2 days; Nomsa tracks status without emailing anyone.

Failure handling: AI service unavailable → form falls back to structured manual template, banner "AI assist unavailable — standard form shown", intake still submittable. AI misclassifies category → Thandi corrects at triage; correction logged and used for prompt tuning. Requester abandons mid - dialog → autosaved draft retrievable 30 days.

Edge cases: Duplicate intake detection (similar open intake within same category/requester → warn with link); intake in another language (respond in kind if supported, else English + notice); budget exceeds requester's delegation → auto - adds financial approver at triage; attachment fails virus scan → rejected with reason, intake continues without file.

2.5.2 J2 — AI - Assisted RFx Authoring (Thandi)

Trigger: Converted intake or blank "New RFx".

#	Actor	Action	System response
1	Thandi	Selects sourcing method (RFQ/RFP/RFI)	Method - specific required fields + template list shown
2	Thandi	Clicks "Draft with AI"	AI drafts RFx document from spec + selected template: background, scope, line items/pricing schedule, submission instructions, proposed evaluation criteria & weights; each section marked "AI - drafted — review required"
3	Thandi	Edits sections; sets dates (BiddingStart, ResponseDue); uploads attachments	Validation: ResponseDue > BiddingStart > now; file type/size/virus checks
4	Thandi	Defines evaluation criteria & weights; envelope assignment (technical/commercial)	Weights must total 100%; criteria locked at publish
5	Thandi	Adds suppliers (master lookup or manual) & contacts	Duplicate detection on entity + email
6	Thandi	Saves	RFx - YYYY - NNN assigned, status Draft

Success: Publishable draft in <30 min for a standard RFQ.

Failure: AI draft fails/timeout (>60s) → retry once automatically, then offer blank template; nothing auto - publishes. Template missing → generic type template offered.

Edge cases: Clone existing RFx (copies structure, docs, suppliers; resets dates/status per RFx rules); concurrent edit by two users → optimistic concurrency, second save gets conflict prompt with diff; AI proposes criteria weights summing ≠100% → blocked client+server side.

2.5.3 J3 — Review, Approval & Publication (Thandi, Reviewer, Approver)

Trigger: Thandi submits Draft for review.

#	Actor	Action	System response
1	Thandi	"Submit for Review"	Mandatory - field validation; status → PendingReview; RFx locks for editing; reviewer notified (Teams card + email)
2	Reviewer	Opens read - only review view	Full RFx + AI pre - review checklist (completeness of dates, docs, criteria, supplier list)
3a	Reviewer	RFQ: "Approve"	Status → Published (short path); invitations dispatched
3b	Reviewer	RFI/RFP: "Submit for Approval"	Status → PendingApproval; approver notified
3c	Reviewer	"Reject" or "Send Back for Clarification"	Comments mandatory; status → Returned (editable) / PendingClarification; initiator notified
4	Approver	"Approve" (RFI/RFP)	Status → Published; invitations auto - dispatched; publish date recorded
5	System	Reviewer = Approver case	Auto - approval short - circuit with two distinct audit entries

Success: Governed publication with SoD enforced; suppliers invited within 5 minutes of approval.

Failure: Invitation email bounce → per - recipient retry (3x exponential backoff), failed recipients flagged on dashboard with manual resend. Approver vacancy → Administrator reassigns; reassignment audited.

Edge cases: Approve attempt by RFx creator blocked (SoD, configurable); skip - stage transitions rejected server - side ("RFx must be reviewed before publication"); withdrawal before close → status Withdrawn, suppliers notified.

2.5.4 J4 — Supplier Response Submission (Ahmed)

Trigger: Invitation email with tokenized portal link.

#	Actor	Action	System response
1	Ahmed	Clicks link; first - time → sets password (+ optional MFA)	Account created against invited contact email; lockout 5 fails/30 min

#	Actor	Action	System response
2	Ahmed	Lands on the specific RFx	Sees only RFx he's invited to
3	Ahmed	Acknowledges participation (or declines with reason)	Acknowledgement timestamped + IP; unlocks documents; buyer notified
4	Ahmed	Downloads docs; raises questions via Q&A (if enabled) or CR	Q&A anonymized to other suppliers; answers broadcast as addenda
5	Ahmed	Uploads response files per envelope (technical/commercial), saves drafts	Type/size (≤50MB/file)/virus validation; draft versions retained
6	Ahmed	Submits final before deadline	Version N recorded, receipt with timestamp; can revise until close (new version)
7	System	Deadline passes	Submissions locked; late attempts blocked with clear message; closing notice sent; buyer sees sealed count only until close

Success: 100% receipted submissions; zero premature bid visibility.

Failure: Upload interrupted → resumable/chunked upload, retry prompt; portal down at deadline (P1) → incident process may extend via CR, all extensions audited.

Edge cases: Contact left supplier company → supplier admin (or buyer via CR) transfers contact; multiple contacts per supplier collaborate on one response workspace; declined supplier cannot submit; deadline extension CR mid - drafting → live banner update + email.

2.5.5 J5 — Change Requests (Thandi or Ahmed)

Trigger: Post - publication change needed (extend due date, clarification, document update/issue, add supplier, update requirement, other).

Flow (RFx pattern): raise CR (CR - YYYY - NNN, dynamic form per type, attachments allowed) → status PendingReview → reviewer Approve/Reject/Return - for - Clarification (comments mandatory on reject/clarify) → on approval, system auto - applies effects (extend ResponseDueDate + notify all; new document version active, old archived; invite added supplier only; update requirement text) → affected parties notified; all steps audited.

Failure/edge: CR against Closed/locked RFx blocked; one active CR per RFx per type; new due date must be future & ≥ current; supplier CRs only on RFx they're invited to and after acknowledgement; auto - reject open CRs when RFx closes.

2.5.6 J6 — Sealed Evaluation & Award Decision (Thandi, Evaluators, Pieter)

Trigger: RFx closes with ≥1 submission.

#	Actor	Action	System response
1	System	At close	Technical envelope unsealed to evaluation team; commercial stays sealed (two - envelope mode)

#	Actor	Action	System response
2	System	AI completeness check	Per supplier: required docs/certs/forms present? Gaps flagged with citation to requirement; confidence level shown
3	Thandi	Confirms/edits evaluation panel	Evaluators get scoring assignments; Col declaration required before access (Stage 2: blind scoring)
4	Evaluators	Score criteria on defined scale with comments	AI draft scores/summaries per criterion offered as assist, marked "AI - suggested"; human score is the score of record
5	Thandi	Opens commercial envelope (after technical gate met)	Unsealing is a logged, permissioned event; price comparison table generated
6	System	Consolidation	Weighted totals, ranking, side - by - side matrix, AI executive summary with rationale + citations + outlier flags
7	Thandi	Records award recommendation	Recommendation + justification captured; sent to Approver
8	Approver	Approves/rejects recommendation (Teams card or web)	Status → AwardApproved/Returned; outcome letters queued to suppliers; event → Awarded

Success: Defensible, explainable award decision; evaluation cycle ≤5 days for standard RFQ.

Failure: AI summary service down → scoring proceeds fully manually; zero submissions at close → event marked Unsuccessful with re - issue shortcut (clone); evaluator misses SLA → reminder + escalation to Thandi.

Edge cases: Tie scores → configurable tie - break policy documented in event (price weight, B - BBEE level Stage 2); evaluator overrides AI - flagged gap → mandatory reason, audited; single - bid events → flagged for enhanced justification; abandoned evaluation (all bids non - compliant) → Unsuccessful with evidence retained.

2.5.7 J7 — Teams Orchestration (all internal actors)

Trigger: Any workflow event needing attention.

Approval requests arrive as actionable Teams adaptive cards (Approve/Reject/Clarify with comment box) that execute against the API under the actor's Entra identity; evaluation kick - off auto - schedules a Teams meeting with agenda + deep links; status queries answered by the IntelliSource bot ("show my pending approvals"); channel notifications for publish/close/award milestones.

Failure: Teams/Graph unavailable → email fallback within 60s, action links to web app; card action on stale state (already approved) → card refreshes with current state message.

Edge: Actor without Teams license → email - only path; guest/external users never receive internal cards; card actions idempotent.

2.5.8 J8 — Audit & Evidence (Pieter, Auditor)

Trigger: Audit query or dispute ("prove this award was fair").

Pieter filters the audit trail by event/user/date/action; views immutable entries (who/what/when/IP/old→new, AI entries with model + prompt hash); generates an **evidence pack** (PDF summary + CSV/JSON detail + document manifest with SHA - 256 checksums) in <5 min; export itself is audited. Status history reconstructed from the log. 7 - year retention.

Edge: Evidence for deleted (soft) drafts retained; POPIA data - subject deletion produces redacted - but - verifiable entries (hash chain preserved).

2.5.9 J9 — Administration (Administrator)

User/role management synced with Entra ID groups; supplier master administration (dedupe, merge, deactivate); template library CRUD with versioning; workflow configuration (SoD toggle, approval chains S1, Col requirement); notification template editing; POPIA tooling (subject access report, correction, deletion with legal - hold override); tenant configuration (multi - tenant SaaS or in - tenant).

Edge: Role change mid - workflow doesn't orphan in - flight approvals (reassignment prompt); deactivating a supplier with open invitations warns and requires CR.

RFx Lifecycle — Status State Machine

Server-enforced transition table (FR-RFX-04) — invalid transitions rejected with HTTP 409, every transition audited + timestamped

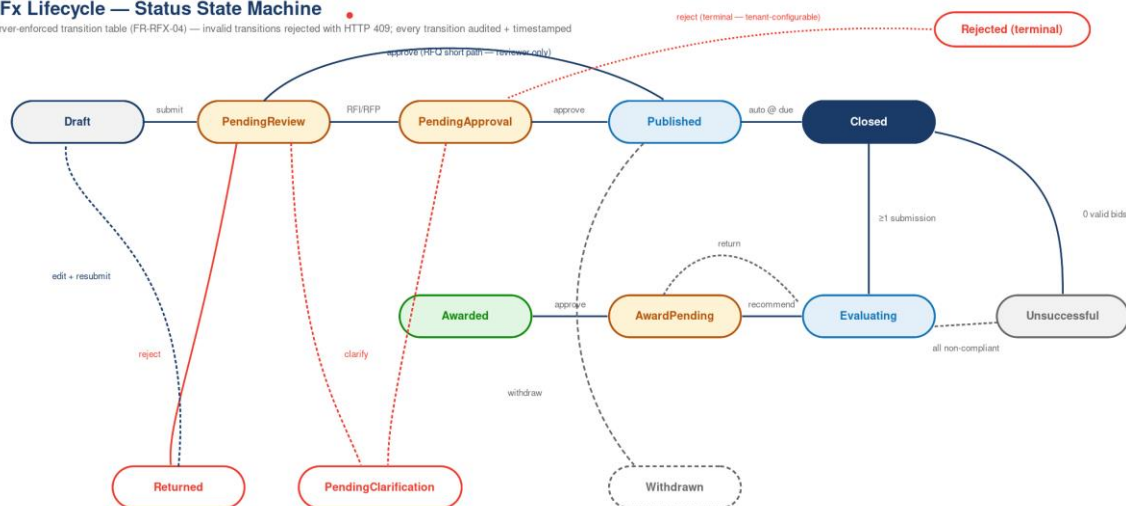
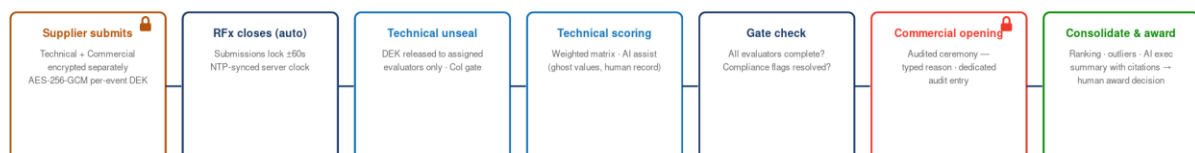


Figure 4 — RFx lifecycle state machine (server - enforced, fully audited)

Sealed-Bid & Two-Envelope Evaluation Flow

Premature access is structurally impossible: per-event data keys (DEK) wrapped in Key Vault; unwrap is policy-gated + audited (ADR-08)



Buyer view before close: submission count + submitted/not-submitted only — content endpoints refuse and attempts are logged (FR-RESP-04)
DB and storage administrators cannot read sealed content: keys, not permissions, enforce the embargo (NFR-SEC-05)

Figure 5 — Sealed - bid and two - envelope evaluation flow

2.6 Functional Requirements (MVP — Stage 1)

Requirement IDs: FR - {MODULE} - {NN}. Every FR is testable — a Given/When/Then can be written without ambiguity. Journey cross - references in brackets.

2.6.1 1 Intake & Specification (FR - INT) — net - new [J1]

ID	Requirement
FR - INT - 01	The system shall accept plain - language intake requests via web form and Teams bot, min 20 / max 5,000 characters, with up to 10 attachments (≤50MB each).
FR - INT - 02	On intake submission, the AI shall generate clarifying questions grouped in batches of ≤5, covering at minimum: quantity/scope, required - by date, budget range (optional), delivery location, and category confirmation.
FR - INT - 03	The AI shall generate a structured specification containing: title, background, scope statement, line items (description, quantity, unit), proposed evaluation criteria, and proposed timeline — each section individually editable by the requester.
FR - INT - 04	The system shall auto - assign a category from the configured taxonomy with a confidence score; confidence <70% shall display "category unconfirmed" requiring triage confirmation.
FR - INT - 05	Each submitted intake shall receive a unique sequential ID in format INT - YYYY - NNN.
FR - INT - 06	Intake status shall follow: Draft → PendingTriage → Accepted (converted) / Returned / Rejected. Returned intakes shall be editable and resubmittable by the requester.
FR - INT - 07	Triage users (Initiator role) shall be able to Accept - and - Convert, Return (comments mandatory), or Reject (comments mandatory) an intake; the requester shall be notified of each decision within 60 seconds via their notification channels.
FR - INT - 08	"Convert to RFx" shall create a Draft RFx pre - populated with the spec fields and maintain a permanent bidirectional link between intake and RFx.
FR - INT - 09	The requester shall see a status - tracking view of their intakes showing current status, assigned owner, and timestamped history.
FR - INT - 10	If AI services are unavailable or respond >30s, the intake form shall fall back to a structured manual template with the banner "AI assist unavailable"; submission must remain possible.
FR - INT - 11	The system shall detect potential duplicate intakes (same requester + category with open intake, or title cosine similarity ≥0.85) and display a warning with links before submission.
FR - INT - 12	Intake drafts shall autosave every 30 seconds and be retrievable for 30 days.
FR - INT - 13	When an intake's stated budget exceeds the requester's configured delegation - of - authority limit, the system shall automatically add the configured financial approver to the triage step.

2.6.2 2 RFx Creation & Lifecycle (FR - RFx) — RFx reuse+extend [J2, J3]

ID	Requirement
FR - RFx - 01	Users with Initiator role shall create RFx events of type RFQ, RFP, or RFI, capturing: title (≤200 chars), requirement/objective, sourcing method, bidding start date, response due date.
FR - RFx - 02	Each RFx shall receive a unique sequential number RFx - YYYY - NNN at creation; the dashboard shall additionally render a type - prefixed display form (e.g., RFQ - 2026 - 014).
FR - RFx - 03	ResponseDueDate must be > BiddingStartDate and > current time; violations shall block save with field - level error messages.
FR - RFx - 04	RFx status shall follow the state machine: Draft → PendingReview → (RFQ: Published \

ID	Requirement
FR - RFX - 05	RFX editing shall be permitted only in Draft and Returned statuses; all other statuses render read - only views (change via CR only).
FR - RFX - 06	Cloning an RFX shall copy structure, documents, evaluation criteria, and supplier list; reset status to Draft, clear dates and responses, and assign a new number.
FR - RFX - 07	Deleting an RFX shall be soft - delete, permitted only in Draft status, and require confirmation showing the RFX number and title.
FR - RFX - 08	Submission for review shall validate all mandatory fields (title, objective, method, dates, ≥1 document for RFP/RFI, ≥1 invited supplier) and list all failures in one response.
FR - RFX - 09	Evaluation criteria shall be definable per RFX: name, description, weight %, scale (1–5 or 1–10), envelope assignment (Technical/Commercial); weights must total exactly 100% (client + server validation).
FR - RFX - 10	Evaluation criteria and envelope assignments shall be immutable after publication.
FR - RFX - 11	An Initiator shall be able to withdraw a Published RFX before close with a mandatory reason; suppliers shall be notified within 5 minutes and the status set to Withdrawn.
FR - RFX - 12	The RFX shall auto - transition Published → Closed at ResponseDueDate (±60s); closing shall lock submissions and trigger closing notifications.
FR - RFX - 13	Concurrent edits shall use optimistic concurrency: the second writer receives a 409 with a field - level diff of conflicting changes.

2.6.3 3 AI Authoring & Governance (FR - AI) — net - new [J2, J6]

ID	Requirement
FR - AI - 01	"Draft with AI" shall generate a complete RFX document set from the linked spec + selected template within 60s: background, scope, line - item/pricing schedule, submission instructions, and proposed evaluation criteria with weights.
FR - AI - 02	Every AI - generated section shall carry a visible "AI - drafted — review required" badge until a human edits or explicitly confirms it; publication shall be blocked while any section remains unconfirmed.
FR - AI - 03	AI drafting failure or timeout (>60s) shall trigger exactly one automatic retry, then offer the blank template path; the failure shall be logged with correlation ID.
FR - AI - 04	All AI invocations shall execute under the requesting user's identity and permissions; an AI call shall never access data the invoking user cannot access (enforced by passing the user's authorization context to all downstream queries).
FR - AI - 05	Every AI action shall write an audit entry containing: user, timestamp, model identifier + version, prompt template ID + hash, input document references, and output hash.
FR - AI - 06	AI outputs affecting evaluation or award (completeness flags, scores, summaries) shall include: rationale text, citations to source bid/RFX content (document + location), and a confidence indicator (High/Medium/Low).
FR - AI - 07	Customer data shall not be used to train models; AI configuration shall document per - tenant model routing, and inference for SA - resident tenants shall default to Azure South Africa regions where the model is available, else route per the tenant's signed s72 configuration.
FR - AI - 08	Administrators shall be able to disable each AI feature per tenant (intake assist, RFX drafting, completeness check, scoring assist, summaries) independently; disabled features degrade to the manual path without error.

2.6.4 4 Documents & Templates (FR - DOC) [J2, J4]

ID	Requirement
FR - DOC - 01	Users shall upload multiple documents per RFx with type classification (RFxDocument, PricingSchedule, Specification, Addendum); allowed formats: pdf, docx, xlsx, pptx, csv, txt, png, jpg, zip; max 50MB per file.
FR - DOC - 02	100% of uploaded files (internal and supplier) shall be virus - scanned before storage commit; infected files shall be rejected with reason and the attempt audited.
FR - DOC - 03	Every stored document shall record a SHA - 256 checksum, uploader, and timestamp; downloads shall verify checksum integrity.
FR - DOC - 04	Documents shall be deletable only while the RFx is in Draft; after publication, replacement occurs only via approved Document - Update CR, which archives the prior version (version history retained).
FR - DOC - 05	Administrators shall maintain a versioned template library per RFx type and category; only one Active version per template; prior versions remain viewable.
FR - DOC - 06	Document storage shall be encrypted at rest (AES - 256) in Azure Blob with tenant - scoped containers/paths.

2.6.5 5 Supplier Management (FR - SUP) [J2]

ID	Requirement
FR - SUP - 01	Initiators shall add suppliers to an RFx from the tenant supplier master (search by name/registration/email) or by manual entry (name + central mailbox minimum).
FR - SUP - 02	Each supplier may appear once per RFx (unique RfxId + central mailbox); duplicates blocked with an explanatory message.
FR - SUP - 03	Suppliers shall have ≥1 contact (name + email required, phone optional); contact email uniqueness enforced per supplier.
FR - SUP - 04	After publication, suppliers may be added only via approved Add - Supplier CR; newly added suppliers receive invitations without re - notifying existing ones.
FR - SUP - 05	Removing a supplier (Draft only) shall cascade - remove its contacts from that RFx after confirmation.
FR - SUP - 06	The supplier master shall support profile fields: legal name, registration number, tax reference, B - BBEE level + certificate/affidavit upload with expiry date (capture only in MVP), contact directory, and status (Active/Inactive).
FR - SUP - 07	Supplier master administration shall support dedupe merge (surviving record selected, references re - pointed, merge audited).

2.6.6 6 Review & Approval Workflow (FR - WF) — RFx reuse+extend [J3]

ID	Requirement
FR - WF - 01	"Submit for Review" shall be available only in Draft/Returned and shall lock the RFx from editing.
FR - WF - 02	Reviewers shall access a read - only review view including an automated pre - review checklist (dates valid, documents present, criteria total 100%, ≥1 supplier with valid contact).
FR - WF - 03	Reviewer actions: Approve (RFQ → publish; RFI/RFP → submit for approval), Reject, Send Back for Clarification. Reject and Clarification require comments (≥10 chars).
FR - WF - 04	Approver actions on PendingApproval: Approve (→ publish), Reject (→ Returned), Send Back for Clarification; comments mandatory except Approve.

ID	Requirement
FR - WF - 05	When assigned Reviewer = Approver, approval of review shall auto - progress to Published, writing two distinct audit entries (review - approve, approval - approve).
FR - WF - 06	Segregation of Duties: the RFx creator shall not be assignable as its Reviewer or Approver (tenant - configurable toggle, default ON; toggle changes audited).
FR - WF - 07	Approval chains shall be configurable per tenant: 1 - step (Reviewer only, RFQ default) or 2 - step (Reviewer + Approver); chain resolved and frozen at submit time.
FR - WF - 08	Administrators shall reassign pending review/approval tasks (e.g., vacancy); reassignment requires reason and is audited; the new assignee is notified.
FR - WF - 09	All workflow decisions shall be actionable from both web UI and Teams adaptive cards with identical validation and audit results.

2.6.7 7 Publication & Invitations (FR - PUB) [J3]

ID	Requirement
FR - PUB - 01	On final approval, the system shall auto - publish: set PublishDate, generate per - contact tokenized portal links (single - use registration token, 14 - day expiry, re - issuable), and dispatch invitation emails within 5 minutes.
FR - PUB - 02	Invitation dispatch shall be tracked per recipient (Sent/Failed/Pending) with automatic retry (3 attempts, exponential backoff); failures surfaced on the RFx view with manual resend.
FR - PUB - 03	Invitation emails shall include: RFx number/title, closing date/time with timezone, portal link, and buyer contact mailbox; content from an admin - editable template.
FR - PUB - 04	Tokenized links shall deep - link authenticated suppliers directly to the specific RFx.

2.6.8 8 Supplier Portal (FR - SP) — RFx reuse [J4]

ID	Requirement
FR - SP - 01	Suppliers shall authenticate with email + password; first - time access via invitation token forces password setup meeting policy (≥10 chars, upper+lower+digit+special).
FR - SP - 02	Account lockout after 5 failed attempts for 30 minutes; lockout and unlock events audited; no user - enumeration in error messages ("invalid credentials" only).
FR - SP - 03	Supplier sessions shall time out after 60 minutes of inactivity with a 5 - minute warning; internal sessions after 30 minutes.
FR - SP - 04	A supplier user shall see only RFx events they are invited to; direct URL access to others returns 404 (not 403).
FR - SP - 05	Suppliers must acknowledge participation (or decline with reason) before accessing RFx documents; acknowledgement records timestamp, user, and IP, and notifies the buyer.
FR - SP - 06	The portal shall be free for suppliers — no registration, subscription, or transaction fees anywhere in the flow.
FR - SP - 07	The portal shall be fully responsive (mobile - first) and WCAG 2.1 AA conformant.

2.6.9 9 Responses & Sealing (FR - RESP) [J4, J6]

ID	Requirement
FR - RESP - 01	Suppliers shall upload response files per envelope (Technical/Commercial when two - envelope mode is ON; single envelope otherwise), save drafts, and submit final.
FR - RESP - 02	Final submission before deadline shall create an immutable version (ResponseVersion N), timestamped to the second (NTP - synced), with an on - screen + email receipt including version and checksum.
FR - RESP - 03	Suppliers may revise until deadline; each revision creates version N+1; all versions retained; only the latest final version enters evaluation.
FR - RESP - 04	Submission content (files and metadata beyond count) shall be inaccessible to ALL internal users until RFx close ("sealed"); attempts logged. Buyer view shows submission count and per - supplier submitted/not - submitted only.
FR - RESP - 05	In two - envelope mode, Commercial envelope content shall remain sealed after close until a permissioned "Open Commercial Envelope" action, which requires the technical gate flag and writes a dedicated audit entry.
FR - RESP - 06	Submissions after deadline shall be blocked server - side with message "This RFx closed on {date time}"; the attempt is logged.
FR - RESP - 07	Uploads shall be chunked/resumable; an interrupted upload shall resume without restarting the file.

2.6.10 10 Change Requests (FR - CR) — RFx reuse [J5]

ID	Requirement
FR - CR - 01	Initiators (own RFx) and acknowledged suppliers (invited RFx) shall raise CRs with types: ExtendDueDate, Clarification, DocumentIssue, UpdateDocuments (initiator), AddSupplier (initiator), UpdateRequirement (initiator), Other.
FR - CR - 02	Each CR shall receive number CR - YYYY - NNN, dynamic form fields per type, optional attachments (validated per FR - DOC - 01/02), and start in PendingReview.
FR - CR - 03	CR reviewers shall Approve, Reject, or Return - for - Clarification; comments mandatory for Reject/Return; requesters may edit and resubmit returned CRs.
FR - CR - 04	Approved CRs shall auto - apply effects atomically: ExtendDueDate updates ResponseDueDate (must be future and $\geq$ current) and notifies all invited suppliers + stakeholders; UpdateDocuments activates new version and archives old; AddSupplier invites only the new supplier; UpdateRequirement updates the objective text.
FR - CR - 05	Only one active (non - terminal) CR per type per RFx; CRs against Closed/Withdrawn RFx blocked; open CRs auto - reject with system comment when the RFx closes.
FR - CR - 06	All CR actions shall be audited with old/new values; affected suppliers notified within 5 minutes of an applied change; live views refresh via SignalR.

2.6.11 11 Evaluation & Award (FR - EVAL) — net - new [J6]

ID	Requirement
FR - EVAL - 01	At close, the system shall create an evaluation workspace listing all final submissions with metadata, unsealing Technical content to assigned evaluators only.
FR - EVAL - 02	AI completeness check shall verify each submission against the RFx's required - document checklist and flag: missing documents, missing/expired certificates, unsigned forms — each flag with a citation and confidence level; evaluators may accept or override each flag with a mandatory reason (audited).

ID	Requirement
FR - EVAL - 03	The Initiator shall assign an evaluation panel (≥ 1 evaluator); each evaluator must complete a conflict - of - interest declaration (none / declared+description) before accessing submissions; declared conflicts block access pending Initiator decision (audited).
FR - EVAL - 04	Evaluators shall score each criterion on the defined scale with optional comment; a submission is fully scored when all criteria have scores from all assigned evaluators.
FR - EVAL - 05	AI scoring assist (per criterion: suggested score + supporting excerpt citations) shall be visually distinct ("AI - suggested") and never auto - populate the score of record; the human - entered score is authoritative.
FR - EVAL - 06	The system shall compute weighted totals per supplier (avg evaluator score $\times$ weight, normalized to 100), produce a ranked side - by - side comparison matrix, and flag statistical outliers (any evaluator's score $> 2\sigma$ from the mean for that criterion).
FR - EVAL - 07	The AI executive summary shall cover: ranking rationale, per - supplier strengths/gaps with citations, price comparison (after commercial opening), completeness posture, and outliers — regenerable, versioned, and marked AI - generated.
FR - EVAL - 08	The Initiator shall record an award recommendation (supplier(s) + justification ≥ 50 chars); submission routes to the Approver as AwardPendingApproval.
FR - EVAL - 09	The Approver shall Approve ($\rightarrow$ Awarded) or Return (comments mandatory, $\rightarrow$ Evaluating); on award, outcome notifications (award + regret letters from templates) are queued to all responding suppliers.
FR - EVAL - 10	Zero - submission or all - non - compliant events shall be closable as Unsuccessful with reason; a "re - issue" action clones the event.
FR - EVAL - 11	Single - response awards shall require an additional justification field ("single - bid justification") before approval.
FR - EVAL - 12	Scoring progress (per evaluator, per supplier) shall be visible to the Initiator without revealing individual scores until consolidation (configurable: open vs hidden until all complete).

2.6.12 12 Teams Orchestration (FR - TEAMS) — net - new [J7]

ID	Requirement
FR - TEAMS - 01	Review, approval, CR, and award - approval requests shall be delivered as Teams adaptive cards with action buttons (Approve/Reject/Clarify + comment box) executing against the API under the actor's Entra identity.
FR - TEAMS - 02	Card actions shall be idempotent and stale - safe: acting on an already - decided item refreshes the card to current state with an explanatory note, without error.
FR - TEAMS - 03	On evaluation kick - off, the system shall auto - schedule a Teams meeting for the panel (title, agenda, deep link to workspace) via Graph API, at the first common free slot within 3 business days (Initiator can override).
FR - TEAMS - 04	Milestone notifications (Published, Closing in 24h, Closed, Awarded) shall post to a configured Teams channel per category/BU.
FR - TEAMS - 05	The IntelliSource bot shall answer authenticated status queries: "my pending approvals", "status of {RFx number}", "events closing this week" — scoped to the user's permissions.
FR - TEAMS - 06	If Teams delivery fails (Graph error/timeout 10s), the equivalent email shall be sent within 60 seconds and the fallback logged.
FR - TEAMS - 07	The plain - language intake dialog (FR - INT - 01/02) shall be available through the Teams bot with identical validation and outcomes.

2.6.13 13 Audit & Evidence (FR - AUD) — RFX reuse+hardened [J8]

ID	Requirement
FR - AUD - 01	Every material action (create/update/delete/submit/review/approve/reject/publish/acknowledge/submit - response/unseal/score/override/award/export/login - fail/lockout/config - change and every AI invocation) shall write an audit entry: actor (user or supplier user or system), action, entity type + ID, timestamp (UTC, NTP - synced), IP, old/new values (JSON), and details.
FR - AUD - 02	The audit store shall be append - only: no UPDATE or DELETE grants at the database level; schema enforced by migration tests.
FR - AUD - 03	Audit entries shall be hash - chained (each entry stores SHA - 256 of previous entry + own content); a verification job shall validate chain integrity daily and on evidence export, alerting on breaks.
FR - AUD - 04	Authorized users (Administrator, ViewOnly - Audit) shall filter/search the trail by entity, actor, action, and date range, with results <3s for any single - event query.
FR - AUD - 05	Evidence - pack export per RFX shall produce within 5 minutes: PDF chronology, CSV/JSON full detail, document manifest with checksums, and chain - verification statement; the export itself is audited.
FR - AUD - 06	RFX status history shall be derivable from the audit trail and rendered as a timeline on the RFX view.
FR - AUD - 07	Audit retention: 7 years minimum (tenant - configurable upward); POPIA data - subject deletions produce redaction entries that preserve chain verifiability.

2.6.14 14 Dashboard & Reporting (FR - DASH) — RFX reuse+extend [J8, J9]

ID	Requirement
FR - DASH - 01	The internal dashboard shall list RFX events with columns (number, title, type, status, owner, due date, responses) filterable by status, type, owner, category, and date range, sortable, paginated (25/page default).
FR - DASH - 02	KPI cards shall show: open events by status, events closing ≤7 days, average cycle time (intake→award) rolling 90 days, response rate (responses/invitations), and pending approvals for the current user.
FR - DASH - 03	Charts (bar/line/pie/slice) shall visualize pipeline by status, cycle - time trend, response rates by category, and supplier participation; every chart exportable to CSV and PDF.
FR - DASH - 04	Dashboard data shall update in real time via SignalR (≤5s from event) without page refresh.
FR - DASH - 05	Role - scoped views: Requesters see own intakes; Initiators own+team events; Pieter (ViewOnly/Head) sees all tenant events; Administrators see config + health.
FR - DASH - 06	Cycle - time instrumentation shall record timestamps for each stage transition and compute per - event and aggregate durations (intake→RFX, draft→publish, publish→close, close→award).
FR - DASH - 07	Action buttons per row shall reflect status + role permissions exactly (e.g., no Edit on Published; Approve only for assigned approver).

2.6.15 15 Notifications (FR - NOTIF) — RFX reuse [J1–J7]

ID	Requirement
FR - NOTIF - 01	Event - driven notifications (email + Teams where applicable) for: intake decisions, review/approval requests and outcomes, publication, acknowledgement, submission receipt, CR decisions, closing reminders, close, award/regret.
FR - NOTIF - 02	Closing reminders to non - responding acknowledged suppliers at T - 7 days and T - 1 day (suppressed for already - submitted suppliers).
FR - NOTIF - 03	All notification templates shall be admin - editable with variable placeholders and preview; template changes versioned and audited.
FR - NOTIF - 04	Delivery status per notification (Sent/Failed/Pending) with automatic retry (3x exponential backoff) and dead - letter queue surfaced to Administrators.
FR - NOTIF - 05	Bulk invitation dispatch shall handle $\geq 1,000$ emails/batch without loss (queue - based, at - least - once with idempotent send guard).

2.6.16 16 Identity, Admin & POPIA (FR - ADM) [J9]

ID	Requirement
FR - ADM - 01	Internal users shall authenticate via Entra ID SSO with MFA enforced by conditional access; local passwords shall not exist for internal accounts.
FR - ADM - 02	Roles (Requester, Initiator, Reviewer, Approver, Administrator, ViewOnly) shall be assignable directly or mapped from Entra groups; role checks enforced at API and UI; changes audited.
FR - ADM - 03	Administrators shall manage users (activate/deactivate, roles, delegation limits), supplier master (FR - SUP - 06/07), templates (FR - DOC - 05), notification templates, and workflow configuration (SoD toggle, chains, two - envelope default, Col requirement).
FR - ADM - 04	POPIA tooling: generate a data - subject access report (all personal data + processing log for a person) within 24h SLA; support correction; support deletion producing redaction with legal - hold override — all audited.
FR - ADM - 05	Tenant provisioning shall support multi - tenant SaaS (shared cluster, isolated schema/data) and in - tenant deployment (dedicated stamp) from the same codebase via configuration.
FR - ADM - 06	All admin configuration changes shall take effect without redeployment and be captured in the audit trail with old/new values.

2.7 Stage 2–3 Requirements (Outline — for roadmap epics)

Stage 2 (12–24 mo): **FR - S2 - DISC** AI supplier discovery (web/ESG/EcoVadis/D&B enrichment, sanctions screening, CSD/SARS/CIPC verification); **FR - S2 - BEE B** - BBEE preference engine (80/20 & 90/10 formulas, staged compliance→functionality→price+preference evaluation, certificate validation, level - based reporting); **FR - S2 - COMM** committee blind scoring, consensus meetings, e - voting in Teams, Col registers; **FR - S2 - ROUND** multi - round/BAFO workflows and scenario analysis; **FR - S2 - ERP** connectors (SAP, Oracle, Dynamics 365, Sage) for supplier master, GL/budget check, and award handoff; **FR - S2 - SAVE** savings ledger (baseline vs awarded vs invoiced) and demand bundling; **FR - S2 - QA** anonymous supplier Q&A with addenda (may pull into MVP as S3).

Stage 3 (24–36 mo): **FR - S3 - CLM** contract auto - draft from award, clause libraries, AI redline vs playbooks, eSignature (DocuSign/Adobe), obligation tracking; **FR - S3 - AUC** auction engine (English/Dutch/Japanese, multi - attribute) with ML - timed close; **FR - S3 - PRED** predictive analytics (supplier risk, price trends, contract leakage) and autonomous negotiation for tactical spend; **FR - S3 -**

RENEW contract - expiry - triggered re - sourcing; **FR - S3 - STORE** M365 Copilot Agent Store listing with MCP/A2A interoper.

2.8 Non - Functional Requirements

2.8.1 1 Performance (NFR - PERF)

ID	Requirement
NFR - PERF - 01	API responses: median <100ms, p95 <500ms, p99 <2s under normal load; GET list endpoints p95 <200ms.
NFR - PERF - 02	Page performance: LCP <2.5s, INP <200ms, CLS <0.1 on 4G - class connection; dashboard time - to - interactive <2s.
NFR - PERF - 03	Platform shall sustain 10,000+ concurrent users with p95 end - to - end response <2s (Gijima principle #13), demonstrated by load test before GA.
NFR - PERF - 04	50MB file upload completes <10s on 100Mbps; chunked resumable protocol.
NFR - PERF - 05	Database queries p95 <200ms; no endpoint issues >10 queries per request (N+1 guard in CI).
NFR - PERF - 06	AI operations: intake clarifications <10s; RFx draft <60s; completeness check <120s per 10 submissions; all long - running AI jobs stream progress or run async with notification.
NFR - PERF - 07	SignalR fan - out latency ≤5s from committed event to dashboard update at 1,000 concurrent dashboard sessions.

2.8.2 2 Performance Budgets (NFR - BUDGET)

ID	Requirement
NFR - BUDGET - 01	Initial JS bundle <300KB gzipped; total initial payload <1MB gzipped; CSS <50KB gzipped; budgets enforced in CI (build fails on breach).
NFR - BUDGET - 02	Route - level code splitting; no route chunk >200KB gzipped.
NFR - BUDGET - 03	Images served in modern formats (WebP/AVIF) with responsive sizes; no single image >300KB.

2.8.3 3 Security (NFR - SEC)

ID	Requirement
NFR - SEC - 01	Zero Trust: every request authenticated + authorized; no network - location trust; service - to - service auth via managed identities/mTLS.
NFR - SEC - 02	Internal auth: Entra ID OIDC + MFA. Supplier auth: email/password per FR - SP - 01/02 (optional Entra External ID later).
NFR - SEC - 03	TLS 1.2+ everywhere; HSTS; AES - 256 at rest (DB + blobs + backups).
NFR - SEC - 04	RBAC enforced at API and UI; authorization tests cover every endpoint × role matrix (deny - by - default).
NFR - SEC - 05	Sealed - bid enforcement at data layer: response content encrypted with per - event keys; decryption keys released to application only on close (technical) / envelope - open action (commercial); DB administrators cannot read sealed content.
NFR - SEC - 06	Input validation server - side on all endpoints; parameterized queries only; output encoding; CSP without unsafe - inline; OWASP ASVS L2 verification before GA.

ID	Requirement
NFR - SEC - 07	100% uploads virus - scanned (Defender/ClamAV) pre - commit; infected files quarantined + audited.
NFR - SEC - 08	Secrets in Azure Key Vault; no secrets in code/config/logs; quarterly rotation policy.
NFR - SEC - 09	Rate limiting tiered: authenticated API 1,000 req/min/user; supplier portal 100 req/min/IP; file - upload endpoints 50 req/min/supplier (RFx lesson); 429 with Retry - After.
NFR - SEC - 10	Defence - in - depth logging: authn/authz failures, lockouts, sealing violations, admin changes → SIEM - forwardable structured logs; no PII in logs.
NFR - SEC - 11	Tenant isolation: every query tenant - scoped via global EF filters + row - level security; cross - tenant access attempts return 404 and alert.
NFR - SEC - 12	No user enumeration on any auth/reset endpoint (uniform responses/timing).

2.8.4 4 Reliability & Error Handling (NFR - REL / NFR - ERR)

ID	Requirement
NFR - REL - 01	Application uptime ≥99.5% monthly (≤3.6h downtime); infrastructure ≥99.99% (multi - AZ AKS + zone - redundant Postgres).
NFR - REL - 02	DR: RTO ≤1h, RPO ≤15 min (PITR + geo - paired backups, SA West secondary); restore rehearsal quarterly.
NFR - REL - 03	Maintenance windows avoid the 48h before any tenant's event close (scheduler - aware deploys).
NFR - ERR - 01	Transient - failure retries: exponential backoff 100/300/900ms max 3 attempts; ≥95% transient errors auto - recover.
NFR - ERR - 02	All API errors follow RFC 7807 problem+json with correlation ID; user - facing errors show inline message + recovery action (retry button, fallback path) and reference ID ERR - YYYYMMDD - NNNNN.
NFR - ERR - 03	Timeouts: external HTTP 10s, AI inference 60s (drafting) / 30s (chat), DB command 30s; all long operations cancellable.
NFR - ERR - 04	Graceful AI degradation: every AI feature has a defined manual fallback (FR - INT - 10, FR - AI - 03, FR - EVAL J6); AI outage shall never block the sourcing lifecycle.
NFR - ERR - 05	Outbound messaging (email/Teams) queued with at - least - once delivery, idempotent send guards, and dead - letter visibility (FR - NOTIF - 04).
NFR - ERR - 06	Frontend error boundaries per route; unsaved - work autosave; offline banner with retry on reconnect.

2.8.5 5 Scalability (NFR - SCALE)

ID	Requirement
NFR - SCALE - 01	Data lifetime ≥10 years, >5M RFx - related records per tenant without >10% p95 degradation (partitioning + index strategy).
NFR - SCALE - 02	Horizontal autoscale on AKS (HPA on CPU+RPS+queue depth); stateless services; scale - out verified 1→10 pods without session loss.
NFR - SCALE - 03	Document storage ≥2TB per tenant; blob lifecycle tiering for closed events (hot→cool after 12 months).

ID	Requirement
NFR - SCALE - 04	200+ suppliers and 1,000+ documents per RFx without listing degradation (virtualized lists, paginated APIs).
NFR - SCALE - 05	Notification throughput ≥10,000/hour per tenant.

2.8.6 6 Accessibility (NFR - A11Y) — enabled

ID	Requirement
NFR - A11Y - 01	WCAG 2.1 AA conformance on all internal and supplier - facing screens; design targets WCAG 2.2 AA.
NFR - A11Y - 02	Color contrast ≥4.5:1 normal text, ≥3:1 large text/UI components — verified against the Gijima palette (see UX spec for compliant token pairings).
NFR - A11Y - 03	Full keyboard operability: logical tab order following DOM, visible focus indicator (≥2px, ≥3:1 contrast), no keyboard traps, skip - to - content link.
NFR - A11Y - 04	Screen - reader support: semantic landmarks, form labels + described - by errors, live - region announcements for async updates (SignalR changes, AI progress), table headers associated.
NFR - A11Y - 05	No information conveyed by color alone (status badges include icon + text); reduced - motion respected (prefers - reduced - motion disables non - essential animation).
NFR - A11Y - 06	Automated axe scans in CI: 0 critical/serious violations to merge; manual NVDA + keyboard test per release on the 5 core journeys; ACR (VPAT - style) published at GA.

2.8.7 7 API & Documentation (NFR - API)

ID	Requirement
NFR - API - 01	REST API, resource - oriented, OpenAPI 3.2.0 specification auto - generated and published at /api/docs (Swagger UI) — 100% endpoint coverage with request/response schemas and examples.
NFR - API - 02	Versioning via URL (/api/v1/); backward - compatible changes only within a version; 6 - month deprecation notice minimum.
NFR - API - 03	Auth: OAuth2/OIDC bearer (Entra); supplier endpoints scoped tokens; webhook signatures (HMAC - SHA256).
NFR - API - 04	Errors per RFC 7807 (NFR - ERR - 02); pagination limit/offset with X - Total - Count; consistent filtering/sorting conventions documented.
NFR - API - 05	Webhooks for lifecycle events (published, closed, awarded, CR - applied) with retry + signature verification; consumer guide in docs.
NFR - API - 06	Contract tests validate implementation against the OpenAPI spec in CI (drift fails build).

2.8.8 8 Compliance (NFR - COMP)

ID	Requirement
NFR - COMP - 01	POPIA: lawful - processing register per tenant; data - subject access/correction/deletion tooling (FR - ADM - 04); operator DPA template pack; breach - notification workflow (detect→assess→notify responsible party) with 72h internal SLA.
NFR - COMP - 02	Data residency: tenant data at rest in Azure South Africa North (paired SA West); cross - border transfers (incl. AI inference) only per signed tenant s72 configuration.

ID	Requirement
NFR - COMP - 03	Retention schedules configurable per data class; default 7 - year audit retention; automated purge with legal hold.
NFR - COMP - 04	Fair - process guarantees: sealed bids (FR - RESP - 04/05), SoD (FR - WF - 06), Col declarations (FR - EVAL - 03), immutable audit (FR - AUD - 02/03) — collectively marketed as King V/PFMA evidence readiness.
NFR - COMP - 05	Timestamps NTP - synchronized ($\pm 1s$) across services; all persisted times UTC with tenant - timezone rendering.

2.9 Data Model (core entities)

Extends the proven RFx model. New entities marked *.

Entity	Key fields	Notes
Tenant *	TenantId, Name, DeploymentMode {SharedSaaS, InTenant}, Region, S72Config (JSON), Settings (JSON)	All entities below tenant - scoped
Intake *	IntakeId, IntakeNumber (INT - YYYY - NNN, unique/tenant), RequesterUserId, RawText, SpecJson (structured spec), CategoryId, CategoryConfidence, Status {Draft, PendingTriage, Accepted, Returned, Rejected}, LinkedRfxId?, BudgetEstimate?, CreatedAt	Spec sections individually versioned
RfxEvent	RfxId, RfxNumber (RFx - YYYY - NNN unique/tenant), RfxType {RFQ, RFP, RFI}, Title, RequirementObjective, InitiatorUserId, Status (see FR - RFX - 04), BiddingStartDate, ResponseDueDate, ExtensionDate?, PublishDate?, TemplateId?, ReviewerUserId?, ApproverUserId?, TwoEnvelopeMode (bool), IntakeId?, WithdrawReason?, IsDeleted	State machine server - enforced
EvaluationCriterion *	CriterionId, RfxId, Name, Description, WeightPercent, Scale {1 - 5, 1 - 10}, Envelope {Technical, Commercial}, SortOrder	Σ weights = 100 enforced
RfxDocument	DocumentId, RfxId, DocumentType {RfxDocument, PricingSchedule, Specification, Addendum}, FileName, FileSize, Version, IsArchived, BlobPath, Checksum (SHA - 256), UploadedBy/At	Versioned via CR
RfxTemplate	TemplateId, RfxType, CategoryId?, Name, Version, IsActive, BlobPath, PromptProfileId? *	One active version
Supplier (master) *	SupplierId, LegalName, RegistrationNumber?, TaxRef?, BbbeeLevel?, BbbeeCertBlobPath?, BbbeeCertExpiry?, Status, CentralMailbox	Tenant supplier master (RFx was per - RFx)
RfxInvitedSupplier	InvitedId, RfxId, SupplierId, SourceType {Master, Manual}, AddedBy/At, IsDeleted; UNIQUE(RfxId, SupplierId)	Per - event invitation
SupplierContact	ContactId, SupplierId, Name, Email (unique/supplier), Phone?	Cascades on removal

Entity	Key fields	Notes
SupplierUser	SupplierUserId, Email (unique), SupplierId, PasswordHash, AccountStatus, FailedLoginAttempts, LastFailedLoginAt, LastLoginAt	Portal identity
SupplierAcknowledgement	AckId, RfxId, SupplierId, Status {Acknowledged, Declined, NotAcknowledged}, AcknowledgedAt?, BySupplierUserId?, IpAddress?, DeclineReason?; UNIQUE(RfxId, SupplierId)	Gates document access
SupplierResponse	ResponseId, RfxId, SupplierId, Envelope {Technical, Commercial, Single}, ResponseVersion, IsDraft, SubmittedAt, BySupplierUserId, BlobPath (per - event - key encrypted), Checksum, MetadataJson; UNIQUE(RfxId, SupplierId, Envelope, ResponseVersion)	Sealed per NFR - SEC - 05
EvaluationAssignment *	AssignmentId, RfxId, EvaluatorUserId, CoiStatus {NotDeclared, None, Declared}, CoiDetails?, AccessGrantedAt?	CoI gates access
Score *	ScoreId, RfxId, SupplierId, CriterionId, EvaluatorUserId, Value, Comment?, AiSuggestedValue?, AiCitationsJson?, CreatedAt; UNIQUE(RfxId, SupplierId, CriterionId, EvaluatorUserId)	Human value authoritative
CompletenessFlag *	FlagId, RfxId, SupplierId, RequirementRef, FlagType {MissingDoc, ExpiredCert, UnsignedForm, Other}, Citation, Confidence {High, Med, Low}, Resolution {Open, Accepted, Overridden}, OverrideReason?, ResolvedBy/At	AI - raised, human - resolved
AwardRecommendation *	AwardId, RfxId, RecommendedSupplierIds (JSON), Justification, SingleBidJustification?, Status {Pending, Approved, Returned}, DecidedBy/At, ExecSummaryVersion	Links to AI summary version
ChangeRequest	CrId, CrNumber (CR - YYYY - NNN), RfxId, CrType (7 types), SubmitterType {Initiator, Supplier}, SubmittedByUserId?/SupplierUserId?, Status {PendingReview, Approved, Rejected, ClarificationRequired}, DetailsJson, Comments, Attachments→CrAttachment	Auto - apply on approve
AuditLog	AuditId (bigint), TenantId, EntityType, EntityId, Action, UserId?/SupplierUserId?/System, Timestamp, IpAddress, OldValue/NewValue (JSON), Details, PrevHash, EntryHash *	Append - only, hash - chained
AIInvocation *	InvocationId, TenantId, UserId, Feature {IntakeAssist, RfxDraft, Completeness, ScoreAssist, ExecSummary, BotQuery}, ModelId, ModelVersion, PromptTemplateId, PromptHash, InputRefs (JSON), OutputHash, LatencyMs, Outcome {Success, Fallback, Error}, CreatedAt	Joined to AuditLog

Entity	Key fields	Notes
Notification	NotificationId, RecipientType, RecipientId, Channel {Email, Teams} *, Type, Subject, Body, SentAt?, DeliveryStatus, RetryCount, LastRetryAt	Queue - backed
User	UserId, EntraObjectId, Email (unique), FullName, Roles (JSON), Department?, DelegationLimit?, IsActive, LastLoginAt	Entra - linked
StageTimestamp *	Id, RfxId, Stage, EnteredAt	Cycle - time analytics (FR - DASH - 06)

Relationships: Tenant 1—n everything; Intake 1—0..1 RfxEvent; RfxEvent 1—n {Criteria, Documents, InvitedSuppliers, Responses, CRs, Assignments, Scores, Flags}; Supplier 1—n Contacts, 1—n SupplierUsers; AwardRecommendation 1—1 RfxEvent.

2.10 API Overview (representative /api/v1 endpoints)

Area	Endpoints (auth: Entra bearer unless noted)
Intake	POST /intakes · GET /intakes?status= · GET /intakes/{id} · POST /intakes/{id}/messages (AI dialog) · POST /intakes/{id}/submit · POST /intakes/{id}/triage {accept\}
Rfx	POST /rfx · GET /rfx?status=&type=&owner= · GET /rfx/{id} · PUT /rfx/{id} (Draft/Returned only) · POST /rfx/{id}/clone · DELETE /rfx/{id} (Draft) · POST /rfx/{id}/ai - draft · POST /rfx/{id}/submit - for - review · POST /rfx/{id}/review {approve\}
Documents	POST /rfx/{id}/documents (multipart, chunked) · GET /rfx/{id}/documents · GET /documents/{id}/download · DELETE /documents/{id} (Draft)
Suppliers	GET/POST /suppliers (master) · POST /suppliers/merge · POST /rfx/{id}/suppliers · DELETE /rfx/{id}/suppliers/{supplierId} · POST /rfx/{id}/suppliers/{supplierId}/contacts
Supplier portal (supplier token)	POST /supplier/auth/register (invite token) · POST /supplier/auth/login · GET /supplier/rfx · GET /supplier/rfx/{id} · POST /supplier/rfx/{id}/acknowledge · GET /supplier/rfx/{id}/documents/{docId} · POST /supplier/rfx/{id}/responses (multipart, envelope param) · POST /supplier/rfx/{id}/responses/{v}/submit · POST /supplier/rfx/{id}/change - requests
CRs	POST /rfx/{id}/change - requests · GET /change - requests?status= · POST /change - requests/{id}/review {approve\}
Evaluation	GET /rfx/{id}/evaluation · POST /rfx/{id}/evaluation/panel · POST /evaluation/assignments/{id}/coi · POST /rfx/{id}/evaluation/completeness - check · POST /completeness - flags/{id}/resolve · PUT /rfx/{id}/scores/{criterionId}/{supplierId} · POST /rfx/{id}/evaluation/open - commercial · GET /rfx/{id}/evaluation/comparison · POST /rfx/{id}/evaluation/exec - summary · POST /rfx/{id}/award - recommendation · POST /rfx/{id}/award - recommendation/decide
Audit	GET /audit?entityType=&entityId=&actor=&from=&to= · POST /rfx/{id}/evidence - pack · GET /evidence - packs/{id}/download
Dashboard	GET /dashboard/kpis · GET /dashboard/charts/{chartId} · GET /dashboard/export?format={csv\}
Admin	GET/PUT /admin/users · GET/PUT /admin/config · CRUD /admin/templates · CRUD /admin/notification - templates · POST /admin/popia/subject - access · POST /admin/popia/erasure
Webhooks	POST /webhooks/subscriptions (events: rfx.published, rfx.closed, rfx.awarded, cr.applied)

Error format (all endpoints): RFC 7807 { "type", "title", "status", "detail", "instance", "errorId": "ERR - YYYYMMDD - NNNNN", "errors": {field: [messages]} }.

Versioning: /api/v1; additive changes only; deprecations ≥6 months. Full OpenAPI 3.2.0 spec is an MVP deliverable (NFR - API - 01).

2.11 Assumptions, Dependencies, Out of Scope, Open Questions

Assumptions: A1 Customers are on M365 with Entra ID (business - case ICP). A2 Azure OpenAI (or equivalent Foundry - hosted model) is contractually available; SA North availability verified per model in architecture phase. A3 Customer Zero = Gijima Procurement BU supplies pilot data + SMEs. A4 RFX patterns are licensed/available for reuse within Gijima. A5 English UI at MVP (localization framework in place, translations later).

Dependencies: D1 Microsoft Graph API (Teams cards, meetings, mail). D2 Azure OpenAI/Foundry + Azure AI Search. D3 Virus - scanning engine (Defender for Storage or ClamAV). D4 Azure SA North capacity for AKS + zone - redundant Postgres. D5 Adobe Fonts kit for Proxima Nova (brand). D6 SMTP/Exchange Online for supplier email.

Out of scope (MVP): P2P/purchase orders, invoicing, payments, inventory; supplier performance management; auctions; contract lifecycle (Stage 3); ERP write - back (Stage 2); Slack/Google Chat; offline mode; native mobile apps (responsive web only).

Open questions (owner → due): OQ1 Two - envelope default ON or OFF per tenant? (Product, pilot config — default OFF, capability ON). OQ2 Supplier MFA at MVP or fast - follow? (Security review). OQ3 Which embedding/inference models available in Azure SA North at build time? (Architect, sprint 0). OQ4 Evidence - pack legal format review by Gijima Legal (pre - GA). OQ5 Tenant onboarding self - service vs assisted at MVP? (GTM — assume assisted).

Quality Gate — Business Analyst:

- ✓ All product - brief features (M1–M10, S1–S5) covered by FRs: **124 MVP FRs across 16 modules** (INT 13, RFX 13, AI 8, DOC 6, SUP 7, WF 9, PUB 4, SP 7, RESP 7, CR 6, EVAL 12, TEAMS 7, AUD 7, DASH 7, NOTIF 5, ADM 6) + Stage 2/3 outline
- ✓ 9 user journeys, each with success, failure handling, and edge cases
- ✓ Consistent ID format FR - CAT - NN / NFR - CAT - NN
- ✓ Testability: 124/124 FRs express observable, verifiable behavior (Given/When/Then writable) — **100% ≥ 90% gate**
- ✓ NFRs numeric (53 NFRs, all with thresholds: PERF 7, BUDGET 3, SEC 12, REL 3, ERR 6, SCALE 5, A11Y 6, API 6, COMP 5)
- ✓ Data model covers all features (23 entities); API overview covers all modules
- ✓ No TBD placeholders; open questions tracked with owners
- **Score: 95/100 ✓**

Next: Architect agent → architecture.md

3. TECHNICAL ARCHITECTURE

Source artifact: architecture.md

How IntelliSource is built: bounded - context services on Azure South Africa, Zero Trust security, cryptographic bid sealing, the AI orchestration layer, and the full testing strategy — mapped to all 20 Gijima Architecture Design Principles.

Project: IntelliSource — AI - Driven Sourcing & Contracting Platform

Author: BMad Architect Agent

Date: 2026 - 07 - 24

Version: 1.0

Status: Approved for UX & Implementation Planning

Inputs: prd.md (124 FRs, 53 NFRs), research - brief.md, Gijima Architecture Design Principles (20), RFx architecture (ADR baseline)

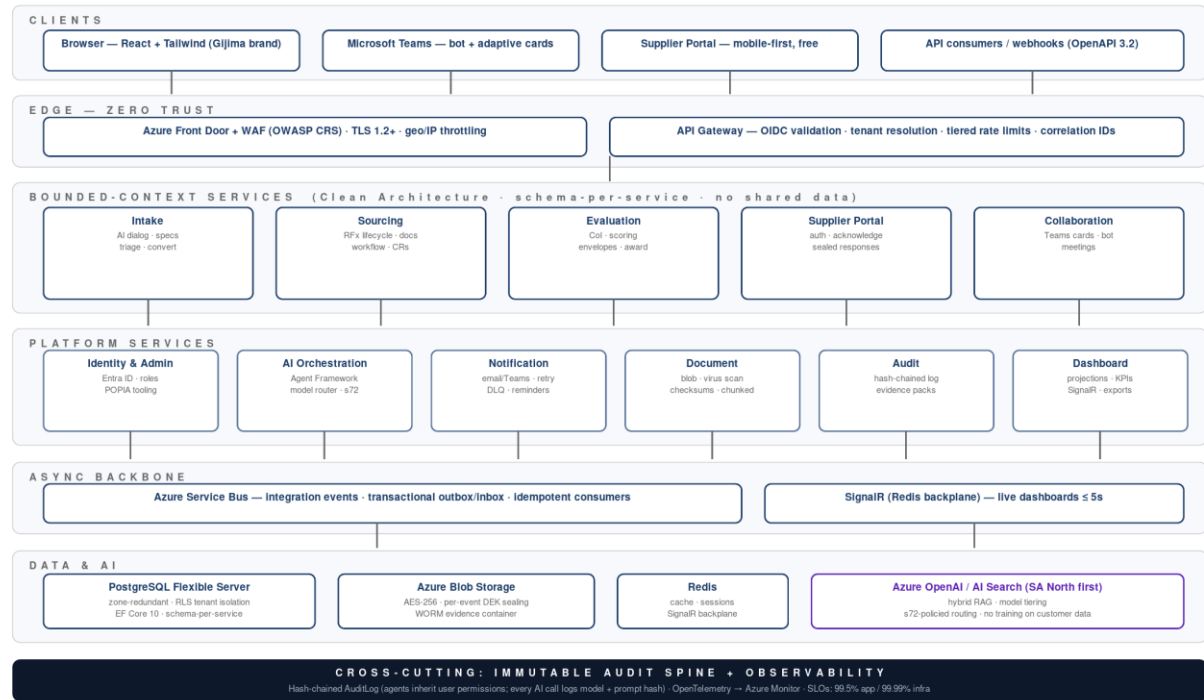
3.1 Architecture Overview

IntelliSource is a **multi - tenant, cloud - native Source - to - Contract platform** built as a set of bounded - context services behind an API gateway, with an ASP.NET Core MVC + React frontend, an AI orchestration layer on Microsoft Agent Framework, and an event - sourced immutable audit spine. It deploys to AKS in Azure South Africa North (paired SA West), supports shared - SaaS and in - tenant (dedicated stamp) modes from one codebase, and enforces Zero Trust with defence - in - depth.



System Architecture — Bounded-Context Services on Azure

Clean Architecture (.NET 10) - API-first (OpenAPI 3.2) - Zero Trust - deployed to AKS in Azure South Africa North



5 deployable units at MVP (ADR-06) — split-ready without refactor - boundaries enforced by ArchUnitNET in CI (principle #20)

Figure 6 — System architecture: services, async backbone, data & AI, audit spine

3.2 Gijima Architecture Principles → Design Response

#	Principle	Design response
1	Clean Architecture, .NET 10	Every service: Domain / Application / Infrastructure / Presentation layers; dependencies point inward; domain has zero framework refs (Section 4, ADR - 01)
2	MVC + React + Tailwind	ASP.NET Core MVC host per web app, React 18 view islands via Vite, Tailwind themed with gijima - styles.css tokens (ADR - 02)
3	EF Core → Postgres	EF Core 10 + Npgsql; schema - per - service; migrations in CI (ADR - 03)
4	Docker for local dev	docker - compose: Postgres, Redis, Azurite, service containers, mock AI endpoint (Section 10)
5	Cloud - native K8s target	AKS, Helm charts, HPA, multi - AZ node pools (ADR - 04)
6	Zero Trust + defence - in - depth	Per - request authn/authz, mTLS east - west, managed identities, WAF, layered validation (Section 9)
7	API - first, OAS 3.2.0 + Swagger	Code - first OpenAPI 3.2 generation, CI contract tests, /api/docs (ADR - 09)
8	POPIA	SA data residency, s72 AI routing, subject - rights tooling, DPA pack (Section 9.4)

#	Principle	Design response
9	Entra ID SSO + MFA + user mgmt	OIDC via Entra ID, conditional - access MFA, supplier local identity + optional External ID (ADR - 05)
10	Full audit & compliance	Hash - chained append - only audit spine, evidence packs, AI invocation logging (ADR - 07)
11	Microservices: single responsibility, autonomous deploy	6 bounded - context services + platform services, independently deployable Helm releases (ADR - 06)
12	API - based communication	REST + async Service Bus events; no shared databases across services (Section 3)
13	10k+ concurrent users <2s	Stateless scale - out, Redis caching, load - tested gates (Section 12, NFR - PERF - 03)
14	Operational excellence	Automation - first workflows, dead - letter dashboards, runbooks (Section 13)
15	UX & profile management	Profile service in Identity; personalization per role (UX spec)
16	Real - time dashboards	SignalR hub + projection tables (Section 3, FR - DASH - 04)
17	Analytics: charts, slices, pies, CSV/PDF export	Dashboard service with projection store + export renderer (Section 3)
18	App uptime 99.5%	Multi - replica deployments, PDBs, health probes, blue/green (Section 10)
19	Infra uptime 99.99%	Multi - AZ AKS + zone - redundant Postgres/Blob (Section 10)
20	Strict boundary enforcement	Schema - per - service, internal NuGet contracts, ArchUnitNET tests in CI blocking cross - boundary refs (Section 4)

3.3 Service Decomposition (bounded contexts)

Service	Responsibility (single)	Owns data	Key FRs
Identity & Admin	Users, roles, tenants, config, POPIA tooling	User, Tenant, Config	FR - ADM - *
Intake	Plain - language intake, AI dialog, specs, triage	Intake	FR - INT - *
Sourcing	Rfx lifecycle, templates, documents metadata, suppliers - on - event, workflow, publication, CRs	RfxEvent, Criteria, RfxDocument, RfxTemplate, InvitedSupplier, ChangeRequest, Supplier master, StageTimestamp	FR - RFX/DOC/SUP/WF/PUB/CR - *
Supplier Portal	Supplier identity, acknowledgements, response intake & sealing	SupplierUser, Acknowledgement, SupplierResponse (sealed)	FR - SP/RESP - *
Evaluation	Panels, Col, scoring, completeness flags, envelopes, awards	EvaluationAssignment, Score, CompletenessFlag, AwardRecommendation	FR - EVAL - *

Service	Responsibility (single)	Owns data	Key FRs
Collaboration (Teams)	Bot, adaptive cards, meeting scheduling, channel posts	Card/message state	FR - TEAMS - *
AI Orchestration (platform)	Agent host, prompt templates, model routing, s72 policy, AiInvocation log	AiInvocation, PromptProfile	FR - AI - *
Notification (platform)	Email/Teams dispatch, templates, retries, DLQ	Notification	FR - NOTIF - *
Audit (platform)	Append - only hash - chained log, verification, evidence packs	AuditLog	FR - AUD - *
Document (platform)	Blob storage, virus scan pipeline, checksums, chunked upload	Blob metadata	FR - DOC - 01/02/03/06
Dashboard (platform)	Read - model projections, KPIs, charts, CSV/PDF export, SignalR	Projection tables	FR - DASH - *

Pragmatic MVP deployment: services are code - separated from day 1 (solution - per - service, no shared DB schemas) but deployed as **5 deployable units** initially — Core API (Sourcing+Intake+Evaluation+Identity), Supplier Portal API+web, Collaboration, AI Orchestration, Workers (Notification/Audit/Document/Dashboard projections) — splitting further as scale demands. This satisfies autonomous deployability without day - 1 operational sprawl. Boundary rules are enforced regardless (ADR - 06, Section 20 tests).

Communication: synchronous REST (via gateway) only client→service; service→service via Service Bus events (integration events: RfxPublished, RfxClosed, ResponseSubmitted, CrApplied, AwardApproved...) with the **transactional outbox** pattern per service. SignalR for UI push. No distributed transactions — sagas for multi - service flows (e.g., publish = Sourcing commits → event → Notification dispatches → Portal activates links; compensation on dispatch failure = flagged recipients, not rollback).

3.4 Clean Architecture Layout (per service)

```
src/
  IntelliSource.{Service}.Domain/      # entities, value objects, domain events, state machines – zero deps
  IntelliSource.{Service}.Application/ # use cases (CQRS commands/queries via MediatR), interfaces, validation
  IntelliSource.{Service}.Infrastructure/ # EF Core, Service Bus, Blob, Graph, AI clients
  IntelliSource.{Service}.Api/         # controllers/minimal APIs, MVC views where applicable, middleware
tests/
  IntelliSource.{Service}.Domain.Tests/ # colocated per layer
  IntelliSource.{Service}.Application.Tests/
  IntelliSource.{Service}.Api.Tests/    # integration (Testcontainers)
```

Boundary enforcement: ArchUnitNET tests fail CI on (a) Domain referencing any framework, (b) cross - service project references, (c) Infrastructure types leaking into Application signatures. The RfxStatus state machine is a Domain value object with exhaustively unit - tested transition table (RFX pattern, extended with Withdrawn/Evaluating/AwardPendingApproval/Awarded/Unsuccessful).

3.5 Architecture Decision Records

3.5.1 ADR - 01: .NET 10 LTS + Clean Architecture — mandated (principle #1)

Context: locked by Gijima principles; validated by RFX. Consequences: LTS support to 2028; EF Core 10 perf gains; hiring pool strong in Gijima. Alternatives (Node/Java) not considered — constraint.

3.5.2 ADR - 02: ASP.NET Core MVC host + React 18 + Tailwind (Vite) — **mandated (principle #2)**

Context: principle requires MVC + React + Tailwind. Options: (a) MVC shell hosting React islands per view — proven in RFx, SEO - light internal app, simple auth cookie flow; (b) full SPA + MVC - as - BFF — cleaner separation, heavier initial build. **Decision: (a) MVC shell + React islands** for internal app AND supplier portal; Vite bundling with per - route entries (aligns with NFR - BUDGET - 02). Tailwind config maps 1:1 to gijima - styles.css tokens (- - gj - * → theme.extend). Consequences: fast first paint, natural code - splitting per view, minimal client routing complexity.

3.5.3 ADR - 03: PostgreSQL Flexible Server (zone - redundant) + EF Core 10, schema - per - service — **mandated (principle #3)**

RLS + EF global query filters for tenant isolation (belt and braces, NFR - SEC - 11); read replicas for Dashboard projections at scale; partitioning AuditLog by month. In - tenant stamps get dedicated server instances.

3.5.4 ADR - 04: AKS multi - AZ + Helm + deployment stamps — **mandated (principle #5)**

Shared - SaaS: namespace - per - environment, pods multi - tenant with data - layer isolation. Regulated/SOE tenants: **stamp** = dedicated namespace (or cluster) + dedicated Postgres + dedicated Key Vault, same Helm chart parameterization (FR - ADM - 05). K8s alone is insufficient for hostile multi - tenancy — stamps close that gap. Blue/green via Argo Rollouts; PodDisruptionBudgets; HPA on CPU+RPS+Service Bus queue depth.

3.5.5 ADR - 05: Entra ID (OIDC+MFA) internal · ASP.NET Identity for suppliers

Internal: OIDC auth - code+PKCE, conditional - access MFA, group→role mapping (FR - ADM - 02). Suppliers: local identity (bcrypt, lockout policy per FR - SP - 02) because forcing Entra External ID on ad - hoc vendors raises friction; External ID optional Stage 2. Teams card actions validated via Bot Framework token + on - behalf - of exchange so actions execute under the actor's identity (FR - TEAMS - 01).

3.5.6 ADR - 06: Bounded - context services, 5 deployable units at MVP

Options: (a) full 11 - service mesh day 1 — max autonomy, high ops burden for a lean team; (b) modular monolith — simplest, violates principle #11's autonomous deployment; (c) **code - separated contexts, 5 deployment units, split - ready** — chosen. Consequences: single - responsibility + API - only communication honored; ops complexity deferred until traffic justifies; splitting = Helm values change, not refactor.

3.5.7 ADR - 07: Audit = append - only hash - chained relational log (+WORM export), not full event sourcing

Options: (a) full event sourcing for all aggregates — max fidelity, high complexity and team unfamiliarity; (b) **CRUD state + comprehensive hash - chained audit log** (RFx ADR - 012 hardened) — chosen; (c) plain audit table — insufficient tamper evidence for King V/PPA. Implementation: middleware + domain - event interceptors write entries with `EntryHash = SHA256(PrevHash || canonical(entry))`; daily + on - export chain verification job; evidence packs to immutable (WORM

policy) Blob container. Status history derived from log (FR - AUD - 06). Consequences: 90% of event - sourcing's audit value at 20% of its complexity; deliberate trade - off documented.

3.5.8 ADR - 08: Sealed bids via envelope encryption with per - event data keys

Response blobs encrypted client - side of storage (AES - 256 - GCM) with a per - event **data encryption key (DEK)**, wrapped by a Key Vault **key encryption key (KEK)**. Technical - envelope DEK unwrap permitted only when `now ≥ ResponseDueDate` (policy in Evaluation service + Key Vault RBAC); Commercial DEK unwrap additionally requires the audited `open - commercial` action (FR - RESP - 05, NFR - SEC - 05). DB/storage admins can't read sealed content; premature - access attempts are structurally impossible, not just logged. Trade - off: key - management complexity — mitigated by Key Vault managed rotation and integration tests.

3.5.9 ADR - 09: Code - first OpenAPI 3.2.0, spec as CI artifact + contract tests — mandated (principle #7)

ASP.NET OpenAPI generation extended to 3.2 output; Swagger UI at `/api/docs`; spec exported per build, diffed for breaking changes (oasdiff) — breaking change without version bump fails CI; Spectral linting for style; contract tests (Schemathesis) run against ephemeral env. URL versioning `/api/v1`; 6 - month deprecation policy (NFR - API - 02).

3.5.10 ADR - 10: AI layer on Microsoft Agent Framework + Azure OpenAI/Foundry, policy - routed

Options: LangChain/LlamaIndex (Python drift from .NET estate), Semantic Kernel alone (superseded — merged into Agent Framework), **Microsoft Agent Framework** (chosen: .NET - native, deterministic Workflow support GA Q2'26, MCP/A2A interop aligning with Teams/Copilot strategy). Patterns: every agent call carries the invoking user's authorization context (FR - AI - 04); prompt templates versioned in repo (PromptProfile) with hash logged (FR - AI - 05); **model router** enforces tenant s72 policy — SA - North - available models default, cross - region only when tenant config permits (FR - AI - 07, NFR - COMP - 02); hybrid RAG via Azure AI Search (keyword+vector+semantic rerank) over tenant - scoped indexes; model tiering (frontier for evaluation summaries, mini for classification/reminders). All AI features feature - flagged per tenant (FR - AI - 08) with manual fallbacks (NFR - ERR - 04).

3.5.11 ADR - 11: Async backbone = Azure Service Bus + transactional outbox; SignalR for UI push

At - least - once delivery with idempotent consumers (inbox dedupe by MessageId); DLQ monitored with admin surface (FR - NOTIF - 04). SignalR backplane on Redis; Azure SignalR Service at scale (NFR - PERF - 07).

3.5.12 ADR - 12: Testing stack — xUnit + Testcontainers + Playwright + Vitest (details Section 7)

3.5.13 ADR - 13: Error handling — RFC 7807 middleware + Polly resilience pipelines (details Section 8)

3.5.14 ADR - 14: Performance budgets enforced by Vite + Lighthouse CI (details Section 11)

3.5.15 ADR - 15: Accessibility toolchain — axe - core CI + NVDA manual protocol (details Section 12)

3.5.16 ADR - 16: Observability — OpenTelemetry → Azure Monitor/App Insights; structured Serilog logs, correlation IDs end - to - end; dashboards per SLO (99.5% app / 99.99% infra); alert runbooks.

3.6 Data Architecture

- Schema - per - service in one Postgres cluster (shared SaaS) or per - stamp cluster (in - tenant). Cross - service reads via APIs/events only (principle #20); Dashboard consumes events into denormalized projection tables.
- Tenant isolation: `tenant_id` NOT NULL on every table + Postgres RLS policies + EF global filters + integration tests asserting cross - tenant 404 (NFR - SEC - 11).
- AuditLog: monthly partitions, BRIN index on timestamp, no UPDATE/DELETE grants (enforced in migration tests, FR - AUD - 02).
- Migrations: EF Core migrations per service, applied by CI job with drift check; expand - contract pattern for zero - downtime.
- Caching: Redis for reference data (templates, categories, config) with 5 - min TTL + event - driven invalidation; no caching of sealed or evaluation data.
- Blob layout: `{tenant}/{rfx}/{class}/{docId}` with immutability policy on evidence container; lifecycle hot→cool 12 months post - close (NFR - SCALE - 03).

3.7 Testing Architecture

Pyramid: 70% unit • 20% integration • 10% E2E.

Type	Tool	Config
Unit (.NET)	xUnit + FluentAssertions + NSubstitute	Per - layer test projects; domain state machine 100% branch; Bogus factories
Unit (React)	Vitest + React Testing Library	jsdom; MSW for API mocks; colocated <code>*.test.tsx</code>
Integration (API+DB)	xUnit + Testcontainers (Postgres, Azurite, Redis) + WebApplicationFactory	Real Postgres per test class; respawn between tests; Service Bus via emulator/harness
Contract	oasdiff (breaking - change gate) + Schemathesis vs OpenAPI 3.2 spec	CI stage on ephemeral env
E2E	Playwright (.NET bindings)	8 critical journeys (J1–J8); trace - on - failure; runs on PR (smoke) + nightly (full)
Load	k6	Profiles: 10k concurrent browse, 1k submission spike at deadline, SignalR fan - out; gates NFR - PERF - 01/03/07 pre - GA
Security	OWASP ZAP baseline in CI + dependency scanning (Dependabot/Trivy) + secret scanning	ASVS L2 manual verification pre - GA
A11y	axe - core in Playwright + Lighthouse CI	0 critical/serious to merge (NFR - A11Y - 06)
AI evaluation	Promptfoo regression suite	Golden - set specs/RfX/completeness cases; drift alerts on model/prompt change

Patterns: AAA structure; factory functions (no shared fixtures mutating state); mock at boundaries only (AI client, Graph, Service Bus) — never mock the DB in integration tests; test isolation via Testcontainers + Respawn; naming `MethodName_Should_Expectation_When_Condition`.

Coverage: minimum 80% line (CI gate), target 95%; 100% branch on: RfxStatus transitions, sealing/unsealing logic, scoring consolidation math, preference/weight calculations, hash - chain writer. Generated code excluded.

Test data: Bogus - based factories per entity; anonymized - realistic seed pack for demo/staging; golden AI datasets versioned in repo.

3.8 Error Handling Architecture

- **Global handler:** ASP.NET middleware maps exceptions → RFC 7807 problem+json with `errorId` (ERR - YYYYMMDD - NNNNN), correlation ID, no stack traces externally; React error boundaries per route with retry affordance (NFR - ERR - 02/06).
- **Resilience (Polly v8 pipelines):** retry idempotent operations on 408/429/5xx/network — exponential 100/300/900ms + jitter, max 3 (NFR - ERR - 01); **circuit breakers** per dependency (AI: 50% failure over 30s → open 60s; Graph: 5 consecutive → open 30s); timeouts: external HTTP 10s, AI drafting 60s/chat 30s, DB 30s (NFR - ERR - 03).
- **Graceful degradation matrix:** AI down → manual forms/scoring with banner (never blocks lifecycle); Teams/Graph down → email within 60s (FR - TEAMS - 06); SignalR down → 30s polling fallback; Redis down → direct DB (perf - degraded, functional); Service Bus down → outbox accumulates, publishes on recovery; Blob down → uploads queued client - side with retry UI.
- **Idempotency:** all mutating endpoints accept Idempotency - Key; consumers dedupe by MessageId (inbox table).

3.9 Security Architecture (Zero Trust + defence - in - depth)

1. **Edge:** Front Door + WAF (OWASP CRS), TLS 1.2+, HSTS, geo/IP throttling; tiered rate limits (NFR - SEC - 09).
2. **Identity:** Entra OIDC + MFA (internal); supplier local identity with logout + uniform errors (FR - SP - 02, NFR - SEC - 12); managed identities service → Azure; mTLS east - west (service mesh or app - level).
3. **Authorization:** policy - based RBAC at API + UI; deny - by - default endpoint×role matrix generated from code and tested exhaustively (NFR - SEC - 04); tenant resolution middleware before any handler.
4. **Data:** AES - 256 at rest; per - event DEK envelope encryption for sealed bids (ADR - 08); Key Vault for all secrets (NFR - SEC - 08); field - level protection for PII.
5. **POPIA:** SA - North residency; s72 model routing (ADR - 10); subject - rights tooling (FR - ADM - 04); breach workflow with 72h internal SLA (NFR - COMP - 01); DPA pack as onboarding artifact.
6. **AppSec:** server - side validation everywhere, parameterized queries only, CSP nonce - based no unsafe - inline, output encoding, anti - forgery on MVC forms; SAST/DAST/dependency/secret scanning in CI; ASVS L2 pre - GA.
7. **Audit:** every authn/z failure, sealing access, admin change → structured security log → Sentinel - forwardable (NFR - SEC - 10).

3.10 Infrastructure & CI/CD

Gijima Group	- Confidential -	Page 44 of 106
	Version:01.00 Valid from: 24.07.2026	Next review 24.10.2026

- **Environments:** dev (docker - compose) → CI ephemeral (Testcontainers + kind) → staging (AKS, anonymized seed) → prod (AKS multi - AZ, SA North; DR pair SA West).
- **IaC:** Bicep (or Terraform) modules; identical stamp chart for in - tenant.
- **Pipeline (GitHub Actions/Azure DevOps):** build → unit → integration → contract/oasdiff → SAST/deps → docker build+Trivy → deploy ephemeral → Playwright smoke + axe + Lighthouse budgets → staging → nightly full E2E/k6/ZAP → prod blue/green with auto - rollback on SLO breach.
- **DR:** zone - redundant Postgres + PITR (RPO ≤15 min), geo - restore rehearsed quarterly (RTO ≤1h, NFR - REL - 02); deploy freeze window 48h before any tenant event close (NFR - REL - 03, scheduler - aware).
- **Observability:** OTel traces across gateway→services→AI calls; RED+USE dashboards; SLO burn - rate alerts; synthetic probes for portal login + submission path every 5 min.

Deployment & Multi-Tenancy — Azure South Africa

Shared SaaS and in-tenant 'stamp' modes from one Helm chart (FR-ADM-05, ADR-04) - POPIA data residency

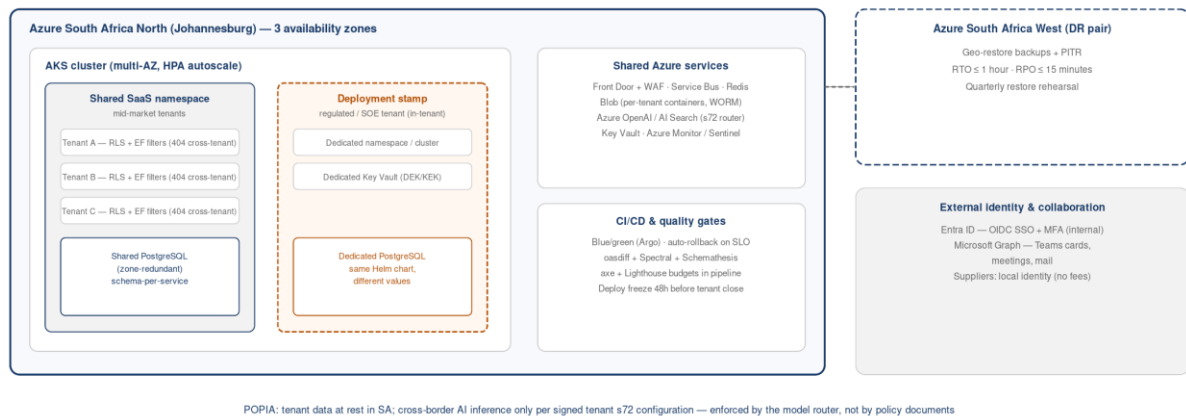


Figure 7 — Deployment & multi - tenancy: shared SaaS and in - tenant stamps in Azure South Africa

3.11 Performance Architecture

- **Budgets in CI:** Vite rollup budget plugin + Lighthouse CI assertions — initial JS <300KB gz, route chunks <200KB, total <1MB, CSS <50KB (NFR - BUDGET - *); PR fails on breach with diff report.
- **Code - splitting:** per - MVC - view React entries (natural route splitting); vendor chunk separation; dynamic import for charts (recharts), file - upload, and evaluation matrix modules.
- **Images:** WebP/AVIF pipeline, responsive srcset, lazy loading; brand assets preloaded (Proxima Nova via Adobe Fonts with font - display: swap).
- **CDN:** Front Door caching for static assets (immutable, content - hashed filenames, 1 - year max - age); API no - store.
- **Server:** response compression (Brotli); output caching for reference endpoints; pagination everywhere (NFR - SCALE - 04 virtualized lists); N+1 guard (EF query counter in integration tests, NFR - PERF - 05); Npgsql batching for bulk operations.

3.12 Accessibility Architecture (WCAG 2.1 AA, target 2.2 AA)

- **Tooling:** eslint - plugin - jsx - a11y (lint), axe - core in Playwright per journey (0 critical/serious to merge), Lighthouse CI a11y ≥95, Pa11y CI on supplier portal pages.

- **Manual protocol per release:** NVDA + keyboard - only run of J1 (intake), J3 (approve), J4 (supplier submit), J6 (score), J8 (evidence export); documented in release checklist.
- **Component - level WCAG mapping:** maintained in the UX spec (design tokens ensure contrast: navy #0E355A on white = 10.6:1 ✓; red #F20023 on white = 4.9:1 ✓ for text ≥14px bold; muted #777 on white = 4.48:1 — reserve for ≥18px or decorative); focus ring token (2px, navy on light / white on dark); status badges = icon+text (NFR - A11Y - 05).
- **ACR (VPAT)** produced at GA from test evidence — sales asset for public sector.

3.13 Coding Standards

C#: latest lang version, nullable enabled, analyzers as errors (IDExxxx + StyleCop set), `Result<T>` - style application errors (no exception - driven control flow), async suffix, `CancellationToken` on all async paths. React/TS: strict TS, functional components + hooks, `TanStack Query` for server state (no ad - hoc fetch), `React Hook Form` + `zod` validation mirroring `FluentValidation` rules, `Tailwind` class conventions via `gij` - prefixed design - token utilities. Commits: Conventional Commits; PR template includes FR/story reference + test evidence. Docs: ADR for any deviation; XML docs on public application contracts; README per service.

Quality Gate — Architect (artifact_completeness):

- ✓ Tech stack justified: 16 ADRs (mandates flagged vs chosen trade - offs)
- ✓ Database strategy covers all 23 PRD entities (schema - per - service, isolation, partitioning)
- ✓ **Testing strategy fully defined:** pyramid 70/20/10, 9 tool rows, coverage gates (80% min/95% target/100% critical branches), AI regression suite
- ✓ Project structure enables test colocation
- ✓ Security: Zero Trust layers, sealed - bid envelope encryption, POPIA/s72
- ✓ Error handling: global handler, RFC 7807, Polly pipelines, circuit breakers, degradation matrix
- ✓ API documentation: code - first OpenAPI 3.2, oasdiff gate, /api/docs, versioning policy
- ✓ Performance: CI budgets, splitting plan, image pipeline, CDN strategy
- ✓ Accessibility architecture: tools, CI gates, manual protocol, token contrast mapping
- **Score: 95/100 ✓**

Next: UX Designer agent → ux - design - specification.md

4. UX DESIGN SPECIFICATION

Source artifact: ux - design - specification.md

The Gijima - branded design language: exact tokens (light and dark), the 20 - component library with every state, the twelve core page layouts, interaction and error patterns, and the WCAG 2.1 AA accessibility contract.

Project: IntelliSource — AI - Driven Sourcing & Contracting Platform

Author: BMad UX Designer Agent

Date: 2026 - 07 - 24

Version: 1.0

Status: Approved for Story Breakdown

Inputs: prd.md, architecture.md, **gijima - styles.css (brand source of truth)**

Compliance: WCAG 2.1 AA (design targets 2.2 AA) · Dark mode first - class · Frontend - design - generation ready

4.1 Design Principles

- 1. Confidence through governance** — every screen makes state, ownership, and next action unmistakable. Status is always visible (badge + icon + text), never color - only.
- 2. AI assists, humans decide HITL** — all AI output is visually distinct (AI badge, dashed border, confidence chip) and always paired with an explicit human confirm/override affordance. No AI content ever masquerades as human input.
- 3. Gijima brand, enterprise calm** — navy authority (#0E355A), disciplined white space, red (#F20023) reserved for primary actions and the signature dot accent. Pill - shaped controls are the brand's fingerprint — used consistently.
- 4. Supplier - first simplicity** — the portal assumes a first - time, mobile, non - expert user: one task per screen, receipts for everything, no jargon.
- 5. Performance is UX** — skeletons match layouts, optimistic updates where safe, deadlines render with live countdowns; nothing blocks on AI.

4.2 Design Tokens

All values exact — consumable by bmad - frontend - design for Tailwind config + CSS custom properties. Prefix: - - gj - * (extends gjima - styles.css).

4.2.1 1 Colour — Light theme (default)

Token	Value	Usage
- - gj - red	#F20023	Primary CTAs, active accents, logo dot, focus - visible on dark
- - gj - red - hover	#D40020	Primary button hover; also the AA - compliant red for text links (5.5:1 on white)
- - gj - navy	#0E355A	Headings, body text, header/footer bars, secondary buttons
- - gj - heading - dark	#242424	h3 - level headings, link hover
- - gj - text	#0E355A	Base body color

Token	Value	Usage
- - gj - text - muted	#767676	Secondary text (AA 4.54:1 — adjusted from brand #777777 which is 4.48:1; visually identical)
- - gj - text - subtle	#A5A5A5	Eyebrows/captions ≥18px or decorative only (2.6:1 — never for essential text)
- - gj - bg	#FFFFFF	Page background
- - gj - bg - light	#F7F7F7	Section bands, default buttons, table stripes
- - gj - bg - hover	#EFEFEF	Hover surfaces
- - gj - border	rgba(0,0,0,0.10)	Form fields, dividers
- - gj - border - strong	rgba(0,0,0,0.20)	Focused fields, cards
- - gj - success	#0E8A09	Success text/icons (adjusted from theme #18CE0F, which fails contrast; 4.6:1 on white). Badge fills may use #18CE0F with #0B3D09 text
- - gj - info	#1273B8	Info (adjusted from #2CA8FF for AA 4.6:1)
- - gj - warning	#8A5A00	Warning text (fills #FFB236 with #3D2A00 text)
- - gj - danger	#C41230	Error text/borders (4.9:1); fills #FF5062 with white text ≥18px only
- - gj - ai	#5B21B6	AI - generated content accent (6.6:1 on white) — unique hue so AI is never confused with brand states
- - gj - seal	#B45309	Sealed/embargoed indicators (amber - lock family)

4.2.2 2 Colour — Dark theme ([data - theme="dark"])

Derived from the 2024 corporate deep navy. Every semantic token has a dark value.

Token	Value	Notes
- - gj - bg	#0D182B	Brand deep navy (2024 PPT identity)
- - gj - bg - light	#152238	Raised surfaces, cards
- - gj - bg - hover	#1D2C47	Hover surfaces
- - gj - text	#E8EDF4	Body (13.9:1)
- - gj - text - muted	#A8B4C4	Secondary (7.1:1)
- - gj - text - subtle	#7A8699	Decorative/large only
- - gj - navy	#3E6FA3	Structural accents on dark
- - gj - red	#F20023	Primary buttons (white text, 4.4:1 as UI component √3:1)
- - gj - red - hover	#FF1A3C	Hover

Token	Value	Notes
- - gj - link - dark	#FF4D66	Text links on dark (5.5:1)
- - gj - border	rgba(255,255,255,0.12)	Fields/dividers
- - gj - border - strong	rgba(255,255,255,0.24)	Focus/cards
- - gj - success	#4ADE80 (text #0D182B on fills)	8.4:1 on bg
- - gj - info	#60A5FA	6.9:1
- - gj - warning	#FBBF24	9.6:1
- - gj - danger	#F87181	6.3:1
- - gj - ai	#C4B5FD	9.2:1
- - gj - seal	#FBBF24 + lock icon	—

Theme mechanics: data - theme attribute on <html>, default follows prefers - color - scheme, user override persisted per profile (FR - ADM). Charts re - map series colors per theme (never rely on hue alone — series also differ by marker/pattern).

4.2.3 3 Typography (from gijima - styles.css, exact)

Font: "proxima - nova", "Proxima Nova", Arial, Helvetica, sans - serif (Adobe Fonts kit; font - display: swap; Arial metrics - adjusted fallback).

Token	Size/Line - height	Weight	Usage
- - gj - fs - hero	42px / 1.15	700	Hero/login headline (white on navy)
- - gj - fs - display	40px / 50px	100	Section display titles, ends with . red dot
- - gj - fs - h3	30px / 40px	600	Page titles
- - gj - fs - h4	22px / 1.35	600	Card/panel headings
- - gj - fs - widget	16px / 1.4	600, UPPERCASE	Widget/table headers
- - gj - fs - base	15px / 1.6	400	Body
- - gj - fs - small	14px / 1.5	400	Forms, eyebrows, captions
- - gj - fs - btn	12px / 1	600, UPPERCASE, letter - spacing .3px	Buttons
- - gj - fs - mono	13px / 1.5	400	"JetBrains Mono", Consolas, monospace — Rfx numbers, hashes, audit IDs

Type rules: sentence case everywhere except buttons/nav/widget titles (uppercase per brand); numerals tabular (font - variant - numeric: tabular - nums) in tables/countdowns.

4.2.4 4 Spacing, Radii, Elevation, Motion

- **Spacing (4px grid):** - - gj - sp - 1..12 = 4, 8, 12, 16, 20, 24, 32, 40, 48, 56, 60, 80px. Section padding 60px desktop / 40px mobile (brand). Card padding 24px. Form gap 16px.
- **Radii:** - - gj - radius - pill: 35px (buttons, inputs, badges — signature); - - gj - radius - card: 8px (cards, modals, tables); - - gj - radius - textarea: 25px; - - gj - radius - chip: 16px.
- **Elevation:** - - gj - shadow - 1: 0 1px 3px rgba(13,24,43,0.08) (cards); - - gj - shadow - 2: 0 4px 12px rgba(13,24,43,0.12) (dropdowns, popovers); - - gj - shadow - 3: 0 12px 32px rgba(13,24,43,0.18) (modals). Dark theme: same geometry, rgba(0,0,0,0.45).
- **Motion:** - - gj - ease: cubic - bezier(0.25, 0.1, 0.25, 1); durations 150ms (hover/focus), 250ms (brand .25s ease transitions — buttons/links), 300ms (panels/modals), 400ms (page - level). Skeleton shimmer 1200ms linear infinite. **All non - essential motion disabled under `prefers - reduced - motion: reduce`** (skeletons become static, countdown pulses stop, SignalR row - highlight becomes outline - only).
- **Layout grid:** 12 - column CSS Grid, max - width: 1250px (- - gj - container), gutter 24px desktop / 16px mobile. Breakpoints (brand/Woodmart): mobile ≤768px, tablet 769–1024px, desktop ≥1025px (wide 1200px). Header 71px desktop / 60px mobile, sticky navy.

4.3 Component Library

Every component specifies ALL states: default · hover · focus · visible · active · disabled · loading · error (+ empty where data - driven). Focus - visible universal style: **2px solid outline, offset 2px** — #0E355A on light surfaces, #FFFFFF on navy/red surfaces, #FF4D66 on dark theme (all ≥3:1 against adjacent colors).

4.3.1 C01 Button

Variants: **Primary** (red fill, white text), **Default** (light - grey fill #F7F7F7, #333 text), **Outline** (transparent, 1px rgba(255,255,255,.35) border — dark/imagery surfaces), **Danger** (danger fill), **Ghost** (text - only navy). Sizes: sm 8×18px, md 12×30px, lg 14×34px 16px/700. Pill radius; uppercase 12px/600.

States: hover Primary→#D40020; Default→#EFEFEF; active scale(0.98) 150ms; disabled 40% opacity + cursor: not - allowed + aria - disabled; loading = spinner (16px, current - color) replacing label, width preserved, aria - busy="true". Destructive actions require confirm dialog.

4.3.2 C02 Input / Select / Textarea

Pill fields 42px height, 0 20px padding, transparent bg, 1px - - gj - border; textarea radius 25px min - height 140px. Label 14px navy above, 8px gap; placeholder - - gj - text - subtle.

States: focus border - - gj - border - strong + focus ring; error = 1px - - gj - danger border + 14px danger message below with icon, aria - invalid + aria - describedby; disabled bg - - gj - bg - light 60% opacity; loading (async validate) = right - slot spinner. Required = red asterisk + aria - required.

4.3.3 C03 StatusBadge

Pill 3px 12px, 12px/600 uppercase, **icon + text always** (never color alone). Mappings: Draft (grey outline, pencil), PendingReview (warning fill #FFB236/#3D2A00, clock), PendingApproval (warning, shield), Published (info fill, megaphone), Closed (navy fill white text, lock), Evaluating (info outline,

scales), AwardPendingApproval (warning, gavel), Awarded (success fill, trophy), Unsuccessful (grey fill, slash), Rejected/Withdrawn (danger outline, x), Sealed (- - gj - seal outline, padlock — tooltip "Sealed until {date}").

4.3.4 C04 AIBadge & AIPanel

AIBadge: chip - - gj - ai outline, sparkle icon + "AI - drafted — review required" / "AI - suggested"; confidence chip High/Med/Low (filled/half/outline dot + text). AIPanel: 1.5px **dashed** - - gj - ai border, 8px radius, header = AIBadge + regenerate + dismiss; footer = **Confirm** (primary) / **Edit** / **Discard** (ghost) — confirming removes dashed treatment and logs. Loading: shimmer + streaming text region aria - live="polite"; error: inline fallback "AI unavailable — continue manually" + manual path link (never blocks).

4.3.5 C05 DataTable

Striped rows (#F7F7F7 alt), 16px cell padding, uppercase 16px/600 headers with sort buttons (aria - sort), sticky header, tabular numerals. Row hover - - gj - bg - hover; keyboard row focus ring; action buttons per row reflect role+status matrix (FR - DASH - 07). States: skeleton rows ×8 on load; empty state (C13); error banner + retry; live update = 2s outline pulse on changed row (aria - live announce "RFx 2026 - 014 status changed to Published"). Pagination 25/page + column filter chips. Mobile ≤768px: card - list transform (each row → card, columns → labeled pairs).

4.3.6 C06 LifecycleStepper

Horizontal (desktop) / vertical (mobile) steps: Intake → Draft → Review → Approval → Published → Closed → Evaluation → Award. Completed = navy filled dot + check; current = red ring + label 600; future = grey outline; skipped (RFQ path) = dashed connector with tooltip. aria - current="step"; connectors animate 250ms (reduced - motion: none).

4.3.7 C07 CountdownChip

"Closes in 3d 04:12:33" tabular - nums; >24h navy outline; ≤24h warning fill; ≤1h danger fill + gentle pulse (reduced - motion: static); past = "Closed {date}" navy fill. Updates 1s; role="timer", minute - level aria - live announcements only.

4.3.8 C08 FileUpload

Dropzone 2px dashed - - gj - border - strong, 8px radius, cloud icon + "Drop files or browse" + constraints line ("PDF, DOCX, XLSX... max 50MB"). Per - file row: icon, name, size, progress bar (4px, navy→success), status (scanning shield / clean check / infected danger + reason / failed retry). Chunked resumable: interrupted → "Resume" button. Errors inline per file; whole - batch never fails atomically. Keyboard: browse button focusable; SR announces per - file completion.

4.3.9 C09 EnvelopeCard (evaluation)

Two side - by - side cards: **Technical** / **Commercial**. Sealed: - - gj - seal padlock, blurred placeholder rows (CSS, not real data), caption "Sealed — opens {condition}". Unseal action (Commercial) = danger - adjacent confirm modal requiring typed reason ≥10 chars; success animates lock→open 300ms and writes audit toast. Unsealed: content list + checksum chips.

4.3.10 C10 ScoreMatrix

Grid: criteria rows × supplier columns; cell = score input (0–scale, step 0.5) + comment icon; AI - suggested value rendered as - - gj - ai ghost text with "Apply" chip — **never pre - filled**. Weighted totals row (bold, tabular); outlier cells flagged (warning icon + tooltip "2.1σ from panel mean"). Keyboard: arrow - key cell navigation (grid pattern, `role="grid"`); each cell labeled "Score for {supplier}, {criterion}". States: locked pre - Col (padlock overlay + "Declare conflict of interest to begin"), read - only post - submit, saving spinner per cell (autosave 2s debounce), error cell - level retry.

4.3.11 C11 ComparisonView

Side - by - side supplier columns, sticky criteria left rail; sections: Completeness (flag chips), Technical scores, Commercial (post - unseal), AI executive summary (AIPanel with citation superscripts linking to source excerpt drawer). Export = "Evidence pack" primary + CSV/PDF defaults.

4.3.12 C12 Toast / Banner / InlineError

Toast: bottom - right (desktop) / top (mobile), 8px radius, shadow - 2, icon + message + optional action, auto - dismiss 6s (errors persist until dismissed), `role="status"` (success/info) or `role="alert"` (error), max 3 stacked. Banner (page - level): full - width band under header — info navy - tint, warning amber - tint, danger red - tint, AI - degraded - - gj - ai - tint ("AI assist unavailable — manual mode active"); dismissible where non - blocking. InlineError: 14px danger + icon under field (C02 pattern). Every error surface shows ERR - ... reference ID with copy button.

4.3.13 C13 EmptyState

Centered: 48px line - icon (navy 1.5px stroke), 22px/600 title, 15px muted description (≤2 lines), primary CTA. Copy specified per surface: dashboard "No sourcing events yet — Create your first RFx or Start from an intake"; triage "No intakes awaiting triage"; evaluation "No submissions received — Close as unsuccessful or Extend deadline"; audit search "No entries match your filters — Clear filters".

4.3.14 C14 Modal / Drawer

Modal: max - width 560px (confirm) / 800px (forms), radius 8px, shadow - 3, title h4 + close; focus - trapped, Esc closes (unless destructive confirm), initial focus on least - destructive action; background `rgba(13,24,43,0.55)`. Drawer (right, 480px): detail peeks (audit entry, citation excerpt, supplier profile); same a11y contract.

4.3.15 C15 IntakeChat

Conversational surface (web + mirrors Teams bot): message bubbles — user navy fill white text right; assistant white/dark - surface with AIBadge left; question groups render as inline mini - forms (chips for options, fields for values) inside assistant bubbles. Composer: pill textarea + attach + send (primary circle, paper - plane). Streaming: typing indicator (3 - dot, reduced - motion: static "Typing..."); `aria - live="polite"` transcript. Spec preview card at end: sectioned summary + "Looks right — submit" / "Edit sections" actions. Fallback mode: banner + structured form (same fields), zero dead - ends.

4.3.16 C16 AuditTimeline

Vertical timeline: node icon per action type, 14px timestamp (UTC + local tooltip), actor chip (person/system/AI — AI shows model+version on hover), old→new value diff expander (mono 13px, added green/removed red with +/− prefixes), hash - chain check chip ("Verified" success / "Verify failed" danger alert). Filters: entity, actor, action, date - range (C02 controls). Virtualized ≥200 entries.

4.3.17 C17 KPICard & Charts

KPICard: 24px padding card, eyebrow label (14px subtle uppercase), 30px/600 navy value (tabular), delta chip (▲ success / ▼ danger + %), sparkline optional. Charts (recharts, lazy - loaded): bar/line/pie/donut; palette light [#0E355A, #F20023, #1273B8, #0E8A09, #8A5A00, #5B21B6] with pattern fills for series ≥4 (colorblind safety); dark equivalents Section 2.2; tooltips follow focus; every chart has data - table toggle (a11y) + CSV/PDF export buttons.

4.3.18 C18 Header / Navigation

Internal app: sticky navy header 71px — white Gijima logo (red dots) left 40px height; primary nav uppercase 15px/600 rgba(255,255,255,0.8) → white hover/active with 2px red underline active indicator; right cluster: global search (pill, collapses to icon ≤1024px), notification bell (badge count, drawer panel), theme toggle, avatar menu. ≤1024px: hamburger → off - canvas panel (focus - trapped). Supplier portal: same bar, simplified (logo, event switcher, help, avatar); prominent "All times in {timezone}" note.

4.3.19 C19 SkeletonLoader

Shapes mirror target layout exactly (table rows, KPI cards, chat bubbles, matrix grid); base #EAEF3 / dark #1D2C47, shimmer gradient 1200ms (reduced - motion: static); always paired with aria - busy on region; minimum display 300ms (no flash).

4.3.20 C20 Wizard (RFx create/edit)

Top LifecycleStepper variant (steps: Details → Documents → Criteria → Suppliers → Review); autosave chip ("Saved 12:04:31" / "Saving..." / "Offline — retrying"); sticky footer: Back (ghost) / Next (primary) / "Submit for review" final; step validation on Next with error summary link - list at top (role="alert", links move focus to fields). Exit guard if unsaved.

4.4 Page Layouts (12 core)

Mobile - first; grid columns refer to the 12 - col desktop grid.

P01 Login/Landing (internal): full - bleed navy hero with Africa - map brand imagery, white hero title 42px + red dot, "Sign in with Microsoft" primary CTA; POPIA notice footer link. Supplier variant: email/password card (max 420px), first - time token flow stepper.

P02 Dashboard (internal home): KPI row (4 cards, cols 3+3+3+3 → stack mobile); pipeline DataTable (cols 1–12) with status filter chips; right rail (cols 9–12, collapses below table on ≤1024px): "My approvals" action list + "Closing soon" countdown list. New: "+ New Request" (primary) and "New RFx" split - button.

P03 Intake chat: centered column max 760px; C15 chat; right context rail ≥1025px (linked docs, category chip, budget). Teams parity noted per FR - TEAMS - 07.

P04 Triage queue: DataTable of intakes (number, requester, category+confidence chip, budget, age) + row expand → spec preview; actions Accept - and - convert / Return / Reject (comment modal).

P05 RFx wizard: C20; Criteria step embeds weight - sum meter (progress to 100%, error state ≠100); Documents step = C08; "Draft with AI" secondary button on Details opens AIPanel preview per section.

P06 RFx detail/review: header: number (mono) + title + StatusBadge + CountdownChip; LifecycleStepper; tabs Overview / Documents / Suppliers / Q&A / CRs / Audit; reviewer mode = read - only + sticky decision bar (Approve primary / Clarify default / Reject danger — comment modal mandatory for latter two); pre - review AI checklist panel (C04).

P07 Supplier portal — event list: card grid (1 - col mobile, 2 - col tablet, 3 - col desktop): title, buyer org, StatusBadge, CountdownChip, acknowledge state; filter tabs Invited / Acknowledged / Submitted / Closed.

P08 Supplier portal — event detail & submission: acknowledgement gate screen first (summary + Acknowledge primary / Decline ghost + reason); then tabs Documents / Q&A / My response; response tab = envelope sections (Technical/Commercial cards) each with C08 + draft/final state + version list; sticky submit bar with receipt confirmation modal (checksum + timestamp) and email receipt note.

P09 Evaluation workspace: header + Col gate (blocking card until declared); left rail suppliers list with completeness chips; main = tabs Completeness (flag list with accept/override) / Scoring (C10) / Commercial (C09 unseal flow) / Comparison (C11) / Summary (AIPanel exec summary + award recommendation form).

P10 Award & decision: recommendation summary card, justification (+single - bid justification when flagged), approver decision bar; post - award: outcome - letter queue status list.

P11 Audit & evidence: filters row + C16 timeline (cols 1–8) + entry detail drawer; "Generate evidence pack" primary → progress toast → download card with checksum.

P12 Admin: left nav sections (Users, Suppliers, Templates, Notifications, Workflow, POPIA, Tenant); each = DataTable + drawer editor; destructive actions double - confirm with typed name.

Teams surfaces: adaptive cards mirror web decision bars (Approve/Reject/Clarify + comment input); card theme auto (Teams handles); deep links into P06/P09. Bot answers render as compact card lists with "Open in IntelliSource" links.

4.5 Interaction & State Patterns

- **Forms:** validate on blur + on submit; error summary top (`role="alert"`) linking to fields; submit → button loading (label preserved for width) → success toast + optimistic navigation, or inline errors + focus to first.
- **Data loading:** skeleton (C19) → content fade - in 150ms; stale - while - revalidate via TanStack Query; background refresh indicator (subtle top progress bar 2px red).
- **Live updates:** SignalR patch → row/card pulse + SR announcement; conflicting local edit → non - blocking banner "This RFx changed — Review changes" (diff drawer).
- **Empty states:** C13 everywhere data can be empty — icon + title + description + CTA.
- **Destructive flows:** modal confirm; withdrawal/unseal require typed reason; irreversible actions state consequences explicitly.
- **Autosave:** wizard + scoring autosave (2s debounce) with visible status chip; offline → queue + "Offline — changes will sync" banner; reconnect auto - flush + toast.

4.5.1 Error - State Patterns

Class	Surface	Recovery
Field validation	InlineError under field	User fixes; summary links focus
Transient network (408/429/5xx)	Toast "Couldn't save — retrying (2/3)..." auto - retry 100/300/900ms	Manual Retry after 3 fails; work preserved
Persistent server	Page banner + ERR - ID + copy button	Retry + "Contact support" (prefilled ID)
AI failure/timeout	AIPanel inline fallback	One auto - retry → manual path CTA (never blocks)
Upload failure	Per - file row error	Resume/retry per file
Stale action (already decided)	Card/panel refresh + info note	Show current state, no error tone
Deadline passed mid - typing	Blocking modal "This RFx closed at {t}"	Draft preserved read - only; CR link if eligible
Offline	Global banner + queued badge	Auto - sync on reconnect

4.5.2 Performance - Design Guidance

Skeletons match final geometry (no spinners for >300ms loads); above - the - fold first (KPIs before charts, header before table body); charts/upload/matrix lazy - loaded on scroll/tab - activate (route chunks <200KB, NFR - BUDGET - 02); images WebP/AVIF + `loading="lazy"`; perceived progress for AI (streamed tokens, staged status "Reading spec → Drafting scope → Proposing criteria"); component authors must record est. gzip impact in story notes (budget ledger reviewed per sprint).

4.6 Accessibility Specification (WCAG 2.1 AA — gate)

- 1. Contrast ✓** — all token pairings specified AA - compliant (Section 2.1/2.2, adjusted semantic colors documented); large - text - only exceptions explicitly marked; contrast verified in CI via token - pair test.
- 2. Focus ✓** — universal 2px outline + 2px offset (Section 3 head), ≥3:1 vs adjacent; never `outline: none` without replacement; focus order = DOM order; skip - link first tabbable; focus returned to trigger on modal/drawer close.
- 3. Keyboard ✓** — every interaction operable: tables (row focus + action menu key), ScoreMatrix grid arrows, chat composer, upload browse, stepper, all menus (roving tabindex); no traps; `Esc` conventions consistent.
- 4. Screen reader ✓** — landmarks (banner/nav/main/complementary); labels + `aria - describedby` errors; live regions: polite (autosave, AI streaming, table refresh), assertive (errors, deadline modal); status announcements for SignalR changes; charts have table toggles; icons `aria - hidden` with text alternatives; language attribute per page.
- 5. Color - independence & motion ✓** — badges icon+text; chart patterns ≥4 series; error/success never color - only; prefers - reduced - motion global kill - switch (Section 2.4); no content flashing >3/s.

Testing: axe - core per journey (0 critical/serious to merge), Lighthouse a11y ≥95, NVDA+keyboard manual protocol on J1/J3/J4/J6/J8 per release (arch Section 12); ACR published at GA.

4.7 Frontend - Design Readiness Checklist

- ✓ All colors exact hex (light + dark for every semantic token)
- ✓ Typography: family, weights (100/400/600/700), px sizes, line - heights, casing rules
- ✓ Spacing 4px grid enumerated; radii px; shadows full values; motion ms + easing curve (`cubic - bezier(0.25,0.1,0.25,1)`)
- ✓ 20 components, all states (default/hover/focus/active/disabled/loading/error/empty) with visual specs
- ✓ Dark mode complete (not algorithmic)
- ✓ Grid + breakpoints exact (≤768 / 769–1024 / ≥1025, container 1250px)
- ✓ Component library maps to architecture React - island structure

→ **READY for bmad - frontend - design generation.**

Quality Gate — UX Designer:

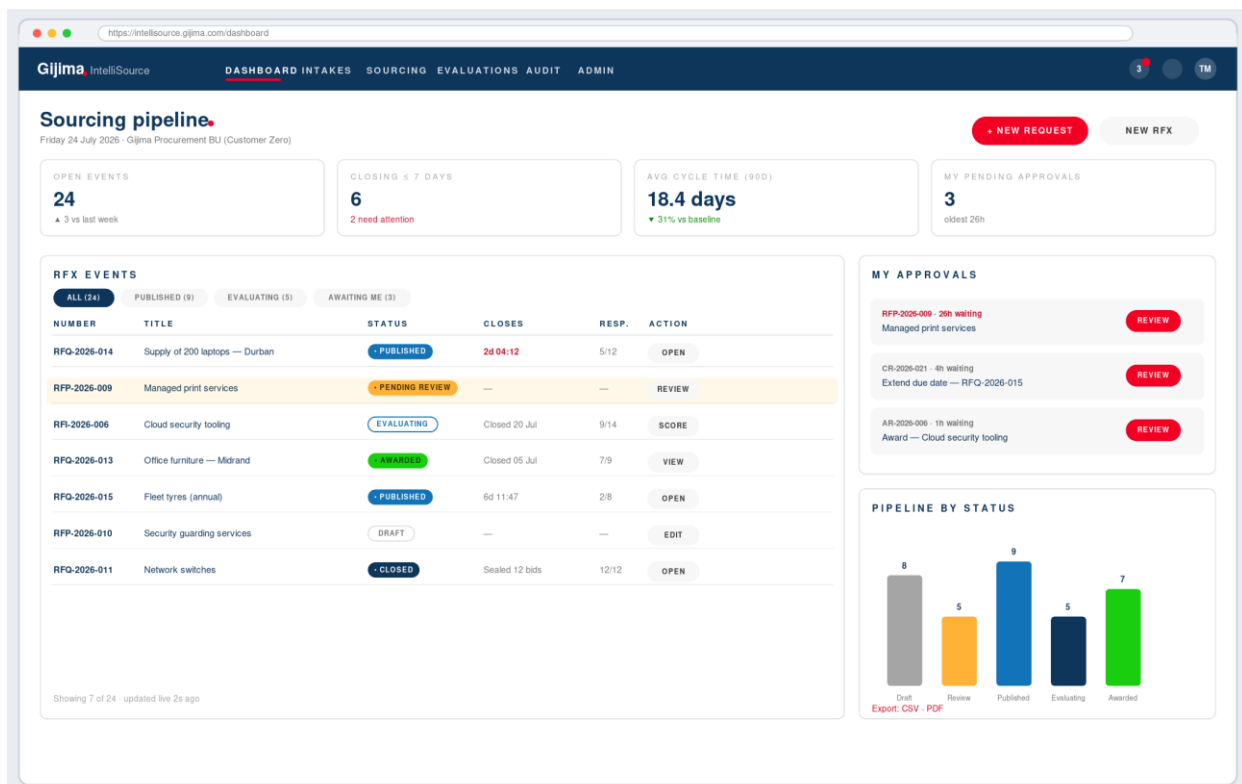
- ✓ Design tokens defined (colors light+dark, type, spacing, radii, shadows, motion)
- ✓ All component states documented (20 components × 7–8 states)
- ✓ Empty/loading/error states for every data - driven component

- ✓ Error - state patterns incl. recovery actions, transient vs persistent, offline
- ✓ Performance design guidance (skeletons, progressive loading, perceived AI progress, bundle ledger)
- ✓ **Accessibility: 5/5 checks complete** (contrast, focus, keyboard, screen reader, color - independence/reduced - motion)
- ✓ Dark mode tokens defined
- ✓ Mobile breakpoints specified
- **Score: 96/100 ✓**

Next: Scrum Master agent → epics.md + sprint - status.yaml

4.9 Reference UX mockups (generated from the design system)

The following mockups were produced from the generated IntelliSource design system (design - system/ — exact Gijima web tokens: red #F20023, navy #0E355A, pill controls, Proxima Nova with Arial fallback). They are reference visualisations of the specified pages, not final screens; the coded component library (20 components, 102 Storybook stories, light + dark, zero token drift) is the implementation source of truth.



Mockup 1 — Internal dashboard (P02): KPI row, live pipeline, role - scoped actions, approvals rail

https://intellisource.gijima.com/intakes/new

Gijima IntelliSource DASHBOARD INTAKES SOURCING EVALUATIONS AUDIT ADMIN

New request

Describe what you need — the assistant turns it into a structured specification

We need 200 mid-range laptops for the Durban office by October.

ASSISTANT

Got it — three quick questions to complete the specification:

1. Preferred spec (CPU / RAM / storage)?
2. Approved budget range?
3. Single delivery site or multiple?

5/16GB/512GB · budget ±R3m · single site, 14 Umgeni Rd

ASSISTANT Here's your draft specification:

INT-2026-031 - IT Hardware AI-DRAFTED — REVIEW REQUIRED

- 200 × laptop — 5 / 16GB / 512GB - 3-yr on-site warranty
- Delivery: 14 Umgeni Rd, Durban — by 30 Oct 2026
- Budget estimate R3.0m · category: IT Hardware (92%)

LOOKS RIGHT — SUBMIT **EDIT SECTIONS**

Describe what you need...

CONTEXT

REQUESTER
Nomsa D. (Operations)

CATEGORY
IT Hardware · 92%

BUDGET
R3.0m (within DoA)

ATTACHMENTS
1 — floorplan.pdf · clean

DUPLICATES
None found

CHANNEL
Web (Teams available)

Mockup 2 — Plain - language intake chat (P03): AI clarification, spec preview with confirm - before - use

Supplier Portal — mobile-first, zero fees

WCAG 2.1 AA - acknowledgement-gated documents - sealed versioned submissions with receipts

Gijima IntelliSource AS

My invitations

INVITED (0) SUBMITTED (1)

RFQ-2026-014
Supply of 200 laptops — Durban
Gijima Procurement

20 04:12 OPEN

RFQ-2026-015
Fleet tyres (annual contract)
Gijima Procurement

09 11:47 OPEN

RFI-2026-006
Cloud security tooling
Gijima Procurement

SUBMITTED: 02 OPEN

Gijima IntelliSource AS

RFQ-2026-014 - Laptops

CLOSER 20 04:12

MY RESPONSE — VERSION 2 (DRAFT)

Technical envelope OPEN

technical-response.pdf · 2.4 MB clean

compliance-certs.zip · 8.1 MB scanning...

+ add files (5.50MB)

Commercial envelope SEALED

pricing-schedule.xlsx · 840 KB

Encrypted on upload — nobody sees pricing until the event closes.

Receipt for v1 issued 22 Jul 14:02:11
sha256 9f2a...c41d · emailed to you

SUBMIT FINAL RESPONSE (V2)

You can revise until the deadline — each version is recorded

Portal guarantees

- Free for suppliers — no fees, ever
- Only invited events visible (404 otherwise)
- Acknowledge before documents unlock
- Chunked, resumable uploads + virus scan
- Timestamped receipts with checksums
- T-7 / T-1 reminders (suppressed if submitted)
- 60-min session · lockout 5 tries / 30 min

Mockup 3 — Supplier portal, mobile - first (P07/P08): acknowledgement, envelopes, receipts — zero supplier fees

5. EPICS & USER STORIES

Source artifact: epics.md

The delivery breakdown: 12 epics and 79 stories with Given/When/Then acceptance criteria, the dependency graph, an indicative 16 - sprint plan, and the 124/124 requirement traceability matrix.

Project: IntelliSource — AI - Driven Sourcing & Contracting Platform

Author: BMad Scrum Master Agent

Date: 2026 - 07 - 24

Version: 1.0

Status: Ready for Readiness Check

Inputs: prd.md (124 MVP FRs), architecture.md, ux - design - specification.md

Effort scale: S = 1 day · M = 2 days · L = 3 days (no story exceeds 3 days).

Story key format: EPIC - NN - STORY - NNN.

AC rules applied: every story ≥ 4 Given/When/Then ACs covering happy path (≥ 2), error handling, edge case, and non - functional (API doc / a11y / performance) where applicable. API stories carry an OpenAPI AC (API - first mandate). UI stories carry an a11y AC (WCAG 2.1 AA enabled) and reference UX components (C01–C20).

5.1 Epic Overview

Epic	Title	Priority	Stories	Effort	Delivers
EPIC - 01	Foundation & Platform Infrastructure	P0	12	28d	Deployable skeleton: gateway, auth, audit spine, AI host, CI/CD, design system
EPIC - 02	Intake & AI Specification	P0	7	15d	Plain - language need → approved spec → RFX conversion
EPIC - 03	RFX Authoring & Lifecycle	P0	11	24d	Governed RFX creation with AI drafting, criteria, suppliers, documents
EPIC - 04	Review, Approval & Publication	P0	6	13d	SoD workflow to publication + supplier invitations
EPIC - 05	Supplier Portal & Sealed Responses	P0	7	15d	Free supplier portal, acknowledgement, sealed versioned submissions
EPIC - 06	Change Request Management	P1	3	8d	Post - publication change governance
EPIC - 07	Evaluation & Award	P0	9	21d	Completeness AI, scoring, envelopes, comparison, award decision
EPIC - 08	Teams Orchestration	P1	4	10d	Actionable cards, bot, meetings, channel milestones

Epic	Title	Priority	Stories	Effort	Delivers
EPIC - 09	Audit Query & Evidence Packs	P0	3	7d	Audit UI, evidence export, chain verification
EPIC - 10	Dashboards, Analytics & Notifications	P1	7	16d	Real - time pipeline, KPIs/charts/exports, notification engine
EPIC - 11	Identity, Admin & POPIA	P0	5	12d	Admin console, tenant provisioning, POPIA tooling, AI config
EPIC - 12	Hardening, Performance & GA Readiness	P1	5	13d	10k - user load proof, ASVS L2, a11y ACR, DR rehearsal, E2E suite
Total MVP			79	182d	
FUTURE - 13	Stage 2: Discovery, B - BBEE, Committees, ERP	P2	outline	—	Roadmap traceability
FUTURE - 14	Stage 3: CLM, Auctions, Predictive & Autonomous	P3	outline	—	Roadmap traceability

Epic Dependency Map — 12 MVP Epics · 79 Stories · 182 Dev-Days

P0 = must-ship core (navy) · P1 = high value (light) · dashed = Stage 2-3 future epics · EPIC-12 gates GA

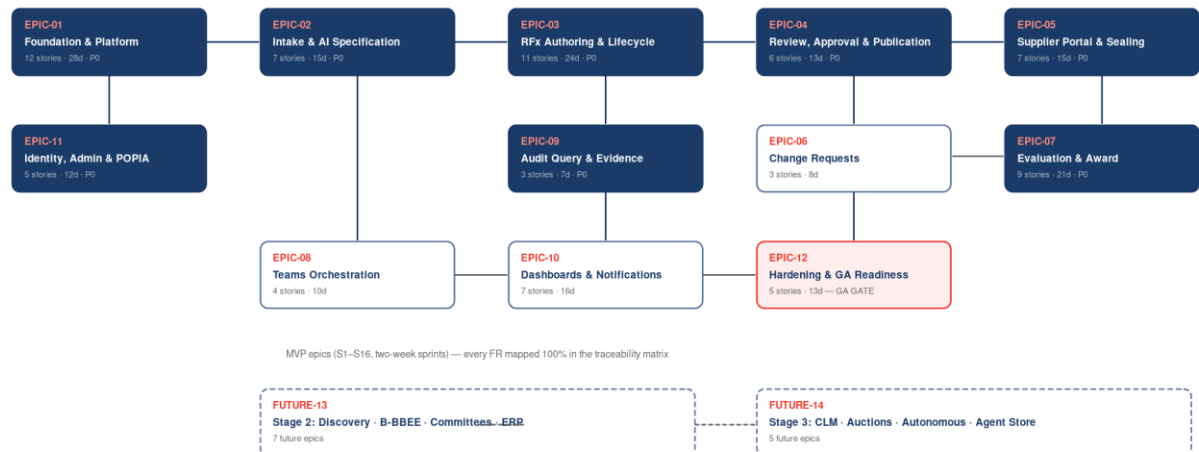


Figure 8 — Epic dependency map (MVP epics plus Stage 2–3 future epics)

5.2 EPIC - 01 — Foundation & Platform Infrastructure (P0)

Goal: A deployable, secured, observable skeleton implementing the Gijima architecture principles, on which every feature epic builds.

5.2.1 EPIC - 01 - STORY - 001 — Solution Scaffolding & Boundary Enforcement (M)

As a **developer**, I want Clean Architecture solutions per bounded context with automated boundary tests so that services stay independently deployable and principle #20 is enforced by CI, not convention.

- AC1: Given the repo, When `dotnet build` runs, Then all service solutions (Sourcing, Intake, Evaluation, SupplierPortal, Collaboration, AI, Platform workers) compile with Domain/Application/Infrastructure/Api layers and Domain has zero framework references.
- AC2: Given ArchUnitNET tests, When a PR adds a Domain→Infrastructure or cross - service project reference, Then CI fails with a named - rule violation.
- AC3: Given `docker - compose`, When `docker compose up` runs, Then Postgres, Redis, Azurite, and all services start healthy locally (<5 min cold).
- AC4: Given the RfxStatus domain value object, When transition - table unit tests run, Then 100% branch coverage over all states incl. Withdrawn/Evaluating/AwardPendingApproval/Awarded/Unsuccessful.

5.2.2 EPIC - 01 - STORY - 002 — EF Core + PostgreSQL Baseline with Tenant Isolation (M)

As a **developer**, I want schema - per - service persistence with enforced tenant scoping so that no query can cross tenants (NFR - SEC - 11).

- AC1: Given migrations, When CI applies them to a clean Postgres, Then all 23 PRD entities exist in their owning service schemas with `tenant_id` NOT NULL.
- AC2: Given RLS policies + EF global filters, When an integration test queries entity X under tenant A with tenant B data present, Then only tenant A rows return; direct - ID access to B returns 404.
- AC3: Given a missing tenant context, When any repository executes, Then the call fails fast with a `TenantResolutionException` (no unscoped queries).
- AC4: Given AuditLog schema, When migration tests inspect grants, Then UPDATE/DELETE are absent for the app role (FR - AUD - 02).

5.2.3 EPIC - 01 - STORY - 003 — API Gateway, Tenant Resolution & Tiered Rate Limiting (M)

As a **platform operator**, I want a gateway enforcing authn, tenant resolution, correlation, and tiered rate limits so that Zero Trust starts at the edge.

- AC1: Given a valid Entra bearer, When any `/api/v1` request passes the gateway, Then tenant is resolved (claim/host), correlation ID injected, and downstream receives verified identity headers.
- AC2: Given no/invalid token, When a protected route is called, Then 401 problem+json returns without backend invocation.
- AC3: Given rate tiers (1,000/min user; 100/min IP portal; 50/min supplier uploads), When a tier is exceeded, Then 429 with Retry - After returns and a metric increments (NFR - SEC - 09).
- AC4: Given OpenAPI docs, When `/api/docs` loads, Then gateway - exposed routes render with security schemes documented (OAS 3.2).

5.2.4 EPIC - 01 - STORY - 004 — Entra ID SSO + MFA & Role Mapping (M)

As an **internal user**, I want to sign in with my Microsoft account under MFA so that access is centrally governed (FR - ADM - 01/02).

- AC1: Given an Entra user with app role/group, When they complete OIDC+PKCE sign - in, Then a session is established and mapped roles appear in claims (Requester...ViewOnly).
- AC2: Given conditional access requiring MFA, When sign - in occurs without MFA, Then access is denied by policy and the app surfaces the Entra guidance page.
- AC3: Given a deactivated user (FR - ADM - 03), When they attempt sign - in, Then access is blocked and audited.
- AC4: Given the login page (P01), When axe scans it, Then 0 critical/serious violations; keyboard - only sign - in path works.

5.2.5 EPIC - 01 - STORY - 005 — Global Error Handling & Resilience Pipelines (M)

As a **developer**, I want RFC 7807 errors and standard Polly pipelines so that failures are consistent, retried, and diagnosable (NFR - ERR - 01..03).

- AC1: Given any unhandled exception, When an API call fails, Then problem+json returns with errorId ERR - YYYYMMDD - NNNNN + correlation ID and no stack trace.
- AC2: Given a transient 503 from a dependency, When an idempotent operation runs, Then retries occur at 100/300/900ms(+jitter) and succeed if the dependency recovers (≥95% transient recovery in chaos test).
- AC3: Given the AI service failing >50% over 30s, When further calls are attempted, Then the circuit opens for 60s and callers receive the degraded - mode signal (fallback path, not 500).
- AC4: Given validation failure, When a command is rejected, Then errors dictionary lists every failing field in one response (FR - RFX - 08 pattern).

5.2.6 EPIC - 01 - STORY - 006 — Immutable Hash - Chained Audit Framework (L)

As a **compliance officer**, I want every material action recorded append - only with tamper evidence so that awards are defensible (FR - AUD - 01/02/03, ADR - 07).

- AC1: Given any audited action, When it commits, Then an AuditLog entry persists with actor, action, entity, UTC timestamp, IP, old/new JSON, PrevHash and EntryHash = SHA256(PrevHash||canonical(entry)).
- AC2: Given the daily verification job, When the chain is intact, Then a Verified checkpoint is recorded; When any row is altered out - of - band in a test, Then verification flags the break and raises an alert within one run.
- AC3: Given AI invocations, When any AI feature executes, Then an AIInvocation row links to an AuditLog entry containing model ID+version and prompt hash (FR - AI - 05).
- AC4: Given 1M seeded entries, When a single - entity audit query runs, Then p95 <3s (FR - AUD - 04).
- AC5: Given POPIA erasure (FR - ADM - 04), When personal data is redacted, Then redaction entries preserve chain verifiability (FR - AUD - 07).

5.2.7 EPIC - 01 - STORY - 007 — Document Service: Blob, Chunked Upload & Virus Scanning (L)

As a **user**, I want reliable, safe file handling so that no infected or corrupted document enters the platform (FR - DOC - 01/02/03/06).

- AC1: Given a 50MB PDF on a 100Mbps link, When uploaded via chunked protocol, Then it completes <10s, records SHA - 256, and is retrievable with verified checksum.
- AC2: Given an EICAR test file, When uploaded, Then it is rejected pre - commit with reason, quarantined, and the attempt audited (NFR - SEC - 07).
- AC3: Given a dropped connection at 60%, When the client resumes, Then upload continues from the last chunk without restart (FR - RESP - 07 shared component).
- AC4: Given a disallowed extension (.exe), When upload is attempted, Then client and server both reject with the constraints message (C08).
- AC5: Given the OpenAPI spec, When reviewed, Then upload endpoints document multipart/chunk contract with examples.

5.2.8 EPIC - 01 - STORY - 008 — OpenAPI 3.2 Publication & Contract Gates (S)

As an **API consumer**, I want a complete, always - current spec so that integrations build against a contract (NFR - API - 01/06, ADR - 09).

- AC1: Given the build, When CI completes, Then an OAS 3.2.0 document is exported covering 100% of v1 endpoints with schemas + examples and served at /api/docs.
- AC2: Given a PR introducing a breaking change without version bump, When oasdiff runs, Then CI fails listing the breaking paths.
- AC3: Given Spectral lint rules, When the spec violates naming/error - format conventions, Then CI fails with rule IDs.
- AC4: Given Schemathesis, When contract tests run against ephemeral env, Then all documented status codes/schemas match implementation.

5.2.9 EPIC - 01 - STORY - 009 — Async Backbone: Service Bus, Outbox/Inbox, SignalR (M)

As a **developer**, I want reliable eventing and live UI push so that cross - service flows survive failures (ADR - 11, FR - DASH - 04).

- AC1: Given a committed domain action, When the outbox dispatcher runs, Then the integration event publishes exactly - once - effectively (inbox dedupe proves idempotency under duplicate delivery).
- AC2: Given Service Bus outage, When actions continue, Then outbox accumulates and drains on recovery with order preserved per aggregate.
- AC3: Given a dashboard session, When an RfxPublished event commits, Then the SignalR update renders ≤5s at 1,000 concurrent sessions (NFR - PERF - 07 harness).
- AC4: Given a poison message, When processing fails 5x, Then it dead - letters and appears in the admin DLQ view (FR - NOTIF - 04 dependency).

5.2.10 EPIC - 01 - STORY - 010 — CI/CD Pipeline & Quality Gates (L)

As a **team**, we want an automated path to production with quality gates so that every merge is releasable (arch Section 10).

- AC1: Given a PR, When the pipeline runs, Then stages execute: build → unit → integration (Testcontainers) → contract/oasdiff → SAST+deps+secrets → image build+Trivy → ephemeral deploy → Playwright smoke + axe + Lighthouse budgets; any failure blocks merge.

- AC2: Given coverage <80% line, When unit stage completes, Then the gate fails with a per - project report.
- AC3: Given bundle budgets (300KB initial JS gz / 200KB route / 1MB total / 50KB CSS), When the frontend builds, Then breaches fail CI with a diff report (NFR - BUDGET - 01/02).
- AC4: Given main branch, When staging deploys, Then blue/green rollout executes with auto - rollback on SLO breach (synthetic probe failure).

5.2.11 EPIC - 01 - STORY - 011 — AI Orchestration Host, Model Router & s72 Policy (L)

As a **platform owner**, I want a governed AI layer so that every AI call is permissioned, logged, routable, and fallback - safe (FR - AI - 04/05/07/08, ADR - 10).

- AC1: Given a user - initiated AI feature, When the agent executes, Then all downstream data access uses the invoking user's authorization context (test: user without RFx access triggers AI on it → 404, no leakage).
- AC2: Given tenant s72 config = SA - only, When a required model is unavailable in SA North, Then the router refuses cross - region with a policy error and the feature degrades to manual (no silent rerouting).
- AC3: Given prompt templates in the registry, When any invocation runs, Then PromptProfile ID + hash and model version are logged (verifiable against FR - AI - 05 audit AC).
- AC4: Given per - tenant feature flags, When an admin disables "RFx drafting", Then the "Draft with AI" affordance disappears and the API returns 403 feature - disabled, while manual paths work unchanged.
- AC5: Given the Promptfoo golden set, When prompts or model versions change, Then regression suite runs in CI and flags output drift beyond thresholds.

5.2.12 EPIC - 01 - STORY - 012 — Design System Integration & Base Components (L)

As a **frontend developer**, I want Gijima - branded tokens and base components wired into the build so that every screen inherits brand + a11y by default (UX Section 2–3; consumes bmad - frontend - design output).

- AC1: Given Tailwind config, When the theme builds, Then all - - gj - * tokens (light + dark) from the UX spec Section 2 are available as utilities and CSS custom properties.
- AC2: Given base components (C01 Button, C02 fields, C03 StatusBadge, C12 Toast/Banner/InlineError, C13 EmptyState, C14 Modal/Drawer, C19 Skeleton), When Storybook builds, Then every documented state renders (default/hover/focus/active/disabled/loading/error) in light and dark.
- AC3: Given axe against Storybook, When all base states scan, Then 0 critical/serious violations; focus - visible ring meets the Section 3 spec (2px, offset 2px, ≥3:1).
- AC4: Given prefers - reduced - motion, When enabled, Then skeleton shimmer and pulses render static (UX Section 2.4).
- AC5: Given the initial app shell, When Lighthouse runs on P02 skeleton, Then LCP <2.5s on the CI profile and initial JS <300KB gz.

5.3 EPIC - 02 — Intake & AI Specification (P0)

Goal: Nomsa describes a need in plain language; Thandi receives a structured, triaged, convertible specification. (J1)

Gijima Group	- Confidential -	Page 64 of 106
	Version:01.00 Valid from: 24.07.2026	Next review 24.10.2026

5.3.1 EPIC - 02 - STORY - 001 — Intake Capture Form with Autosave & Attachments (M)

As a **business requester**, I want to submit my need in plain language with supporting files so that procurement gets everything at once (FR - INT - 01/12).

- AC1: Given the intake form, When I enter 20–5,000 chars and attach ≤10 files (≤50MB each, clean scan), Then the intake saves as Draft and I can continue later (autosave every 30s, "Saved hh:mm:ss" chip).
- AC2: Given text <20 chars, When I submit, Then inline validation blocks with guidance (C02 error pattern).
- AC3: Given a network drop mid - edit, When connectivity returns, Then autosave resumes and queued changes flush without data loss (offline banner shown meanwhile).
- AC4: Given a screen reader, When I complete the form, Then labels, required states, and error messages are announced (aria - describedby wired).
- AC5: Given POST /intakes, When the OpenAPI spec is reviewed, Then request/response schemas with examples are present.

5.3.2 EPIC - 02 - STORY - 002 — AI Clarification Dialog (Chat) with Fallback (L)

As a **business requester**, I want the assistant to ask me simple questions so that my request becomes complete without procurement ping - pong (FR - INT - 02/10).

- AC1: Given a submitted description, When the dialog starts, Then grouped questions (≤5/batch) cover at minimum quantity/scope, required - by date, budget (optional), location, category confirmation, rendered as inline mini - forms (C15).
- AC2: Given my answers, When each batch completes, Then the running spec preview updates and streaming responses announce politely to AT (aria - live).
- AC3: Given AI timeout >30s or circuit open, When the dialog cannot proceed, Then the structured manual template appears with banner "AI assist unavailable" and submission remains possible (FR - INT - 10).
- AC4: Given an answer changing an earlier one (contradiction), When detected, Then the assistant asks a reconciliation question rather than silently overwriting (edge).
- AC5: Given the golden intake set, When Promptfoo runs, Then clarification coverage ≥ baseline (no regression).

5.3.3 EPIC - 02 - STORY - 003 — Structured Spec Generation & Editing (L)

As a **business requester**, I want a clear editable specification so that I own what gets sent to procurement (FR - INT - 03/04).

- AC1: Given a completed dialog, When the spec generates, Then it contains title, background, scope, line items (desc/qty/unit), proposed criteria, timeline — each section individually editable with AIBadge until confirmed (C04).
- AC2: Given category auto - tag confidence <70%, When the spec renders, Then "category unconfirmed" flags for triage (FR - INT - 04).
- AC3: Given generation failure, When one auto - retry also fails, Then a manual spec skeleton loads pre - filled with my dialog answers (no dead end).
- AC4: Given keyboard - only use, When editing sections, Then all edit/confirm/discard controls are reachable and operable.

5.3.4 EPIC - 02 - STORY - 004 — Intake Numbering, State Machine & Duplicate Detection (M)

As the **system**, I want governed intake states so that tracking and duplicate prevention work (FR - INT - 05/06/11).

- AC1: Given submission, When the intake persists, Then INT - YYYY - NNN assigns sequentially per tenant - year (concurrency - safe test: 100 parallel submissions → no gaps/dupes).
- AC2: Given states Draft→PendingTriage→Accepted/Returned/Rejected, When any invalid transition is attempted via API, Then 409 returns with allowed transitions listed.
- AC3: Given an open intake with title similarity ≥ 0.85 same category/requester, When I submit another, Then a warning with links renders before final submit (dismissible, logged).
- AC4: Given a Returned intake, When I edit and resubmit, Then status returns to PendingTriage and the triage comments history is preserved.

5.3.5 EPIC - 02 - STORY - 005 — Triage Queue & Decisions (M)

As a **category manager**, I want a triage workspace so that only viable, well - formed demands become RFX events (FR - INT - 07, J1, P04).

- AC1: Given PendingTriage intakes, When I open the queue, Then a filterable table shows number, requester, category+confidence, budget, age, with row - expand spec preview.
- AC2: Given a decision, When I Accept/Return/Reject, Then Return and Reject require comments ≥ 10 chars and the requester is notified within 60s (email + Teams where enabled).
- AC3: Given a triage decision API error, When submission fails, Then the decision UI preserves my comment, shows retry with ERR - ID, and no partial state persists.
- AC4: Given delegated - authority breach (FR - INT - 13), When budget exceeds the requester's limit, Then the configured financial approver is auto - added and visible in the triage panel.
- AC5: Given POST /intakes/{id}/triage, When the spec is reviewed, Then all three decision payloads + 409 conflicts are documented.

5.3.6 EPIC - 02 - STORY - 006 — Convert to RFX & Requester Tracking (M)

As a **category manager**, I want one - click conversion so that specs seed RFX events without re - typing (FR - INT - 08/09).

- AC1: Given an Accepted intake, When I convert, Then a Draft RFX pre - populates (title, objective, line items→pricing schedule, proposed criteria) with a permanent bidirectional link.
- AC2: Given the requester view, When Nomsa opens "My requests", Then each intake shows status timeline, owner, and the linked RFX's public milestones (published/closed/awarded — no confidential detail).
- AC3: Given a conversion conflict (intake already converted), When retried, Then the API returns 409 with the existing RFX link (idempotency).
- AC4: Given conversion, When audit is queried, Then intake→RFX linkage entries exist for both entities.

5.3.7 EPIC - 02 - STORY - 007 — Teams Intake Parity Hook (S)

As a **business requester**, I want the same intake dialog inside Teams so that I never leave my collaboration surface (FR - TEAMS - 07 dependency stub; full bot in EPIC - 08).

- AC1: Given the shared intake API, When the Teams bot (EPIC - 08 - STORY - 004) drives the dialog, Then identical validation, numbering, and outcomes result (contract test).
- AC2: Given channel context, When an intake is created from Teams, Then source=Teams is recorded and the confirmation deep - links to the web tracking view.
- AC3: Given the API without UI, When called with invalid batch answers, Then errors mirror web behavior (same problem+json).
- AC4: Given OpenAPI, When reviewed, Then /intakes/{id}/messages documents the dialog contract used by both surfaces.

5.4 EPIC - 03 — RFx Authoring & Lifecycle (P0)

Goal: Thandi assemblies governed, AI - drafted RFx events with criteria, documents, and suppliers. (J2)

5.4.1 EPIC - 03 - STORY - 001 — Create RFx with Validation & Numbering (M)

As an **initiator**, I want to create an RFx capturing method, dates, and objective so that sourcing starts on a governed footing (FR - RFX - 01/02/03).

- AC1: Given the wizard Details step, When I save valid RFQ/RFP/RFI fields, Then RFx - YYYY - NNN assigns (concurrency - safe) and status=Draft; dashboard shows the type - prefixed display number.
- AC2: Given ResponseDue ≤ BiddingStart or past dates, When I save, Then field - level errors block (client + server) listing every violation at once.
- AC3: Given a server failure on save, When retried with the same Idempotency - Key, Then exactly one RFx exists.
- AC4: Given POST /rfx, When the spec is reviewed, Then CreateRfx contract with 201/400/409 + examples is documented.
- AC5: Given the wizard, When keyboard - navigated, Then stepper, fields, and footer actions operate per C20 a11y contract.

5.4.2 EPIC - 03 - STORY - 002 — Edit Draft with Concurrency & Autosave (M)

As an **initiator**, I want safe editing so that concurrent work never loses data (FR - RFX - 05/13).

- AC1: Given Draft/Returned status, When I edit any wizard step, Then changes autosave (2s debounce) with visible status chip.
- AC2: Given another user saved first, When I save a conflicting change, Then 409 returns and the UI shows a field - level diff to accept/merge (optimistic concurrency).
- AC3: Given Published status, When edit is attempted via API, Then 409 with "change via CR" guidance (FR - RFX - 05).
- AC4: Given autosave failing 3×, When offline persists, Then the offline banner + queued badge appear and work restores on reconnect.

5.4.3 EPIC - 03 - STORY - 003 — Clone & Soft Delete (S)

As an **initiator**, I want to clone prior events and remove abandoned drafts so that repeat sourcing is fast (FR - RFX - 06/07).

- AC1: Given any RFx I can view, When I clone, Then structure/documents/criteria/suppliers copy; dates, responses, status reset; new number assigns.

- AC2: Given a Draft, When I delete, Then confirmation shows number+title, soft - delete flags, and the item leaves all lists but remains in audit.
- AC3: Given non - Draft status, When delete is attempted, Then 409 explains the rule.
- AC4: Given clone of an RFX with archived document versions, When copied, Then only active versions carry over (edge).

5.4.4 EPIC - 03 - STORY - 004 — Server - Enforced Status State Machine (M)

As the **system**, I want one authoritative transition table so that no path can skip governance (FR - RFX - 04/12).

- AC1: Given the full transition set (incl. Withdrawn, Evaluating, AwardPendingApproval, Awarded, Unsuccessful), When any transition executes, Then it succeeds only if listed; otherwise 409 with reason + allowed set.
- AC2: Given ResponseDueDate passing, When the close scheduler ticks, Then Published→Closed occurs within ±60s, submissions lock, and closing notifications enqueue (FR - RFX - 12).
- AC3: Given clock skew simulation, When close evaluates, Then NTP - synced UTC decides (no double - close, no reopen; idempotent).
- AC4: Given every transition, When it commits, Then StageTimestamp records for cycle - time analytics (FR - DASH - 06) and an audit entry writes.

5.4.5 EPIC - 03 - STORY - 005 — RFX Documents: Upload, Classify, Version (M)

As an **initiator**, I want classified, versioned documents so that suppliers always see the right pack (FR - DOC - 01/03/04; uses EPIC - 01 - STORY - 007).

- AC1: Given Draft, When I upload files with a type class (RFXDocument/PricingSchedule/Specification), Then rows render with progress, scan status, checksum (C08).
- AC2: Given Draft, When I delete a document, Then it's removed; Given Published, Then deletion is blocked with CR guidance (FR - DOC - 04).
- AC3: Given a replacement via approved CR, When applied, Then version N+1 activates, N archives, and both remain in history/audit.
- AC4: Given a corrupt download (checksum mismatch), When detected, Then the download blocks with an integrity error + support ID (edge).

5.4.6 EPIC - 03 - STORY - 006 — Template Library Administration (M)

As an **administrator**, I want versioned templates per type/category so that authoring starts from approved baselines (FR - DOC - 05).

- AC1: Given admin rights, When I create/upload a template with type+category, Then it versions and only one Active version exists per template.
- AC2: Given an active template in use, When I upload a new version, Then existing Draft RFX keep their version; new drafts get the new one.
- AC3: Given a non - admin, When template CRUD is attempted, Then 403 (authz matrix test).
- AC4: Given template list UI, When keyboard/AT used, Then table + drawer editor meet the C05/C14 a11y contract.

5.4.7 EPIC - 03 - STORY - 007 — AI RFX Drafting with Confirm Flow (L)

As an **initiator**, I want AI to draft the RFX from the spec so that authoring takes minutes, with me in control (FR - AI - 01/02/03).

- AC1: Given a converted spec + template, When I click "Draft with AI", Then within 60s all sections generate in AI Panels marked "AI - drafted — review required" (C04), streamed with staged progress.
- AC2: Given unconfirmed AI sections, When I attempt "Submit for review", Then submission blocks listing unconfirmed sections (FR - AI - 02).
- AC3: Given AI failure, When one auto - retry fails, Then the blank - template path offers with my spec content preserved; the event logs Fallback (FR - AI - 03).
- AC4: Given a confirmed section, When I confirm, Then the dashed AI treatment clears and an audit entry records the confirmation (human accountability).
- AC5: Given proposed criteria weights $\neq 100\%$, When AI output renders, Then the weight meter shows the error and Apply is disabled until fixed (edge).

5.4.8 EPIC - 03 - STORY - 008 — Evaluation Criteria, Weights & Envelopes (M)

As an **initiator**, I want weighted criteria with envelope assignment so that evaluation is predefined and sealed fairly (FR - RFX - 09/10).

- AC1: Given the Criteria step, When I add criteria (name, desc, weight, scale, envelope), Then the weight meter must reach exactly 100% to proceed (client+server).
- AC2: Given two - envelope mode ON, When criteria save, Then each is tagged Technical or Commercial and commercial criteria hide from technical evaluators later (contract with EPIC - 07).
- AC3: Given a Published RFX, When criteria edit is attempted, Then 409 immutable (FR - RFX - 10).
- AC4: Given scale change with existing draft scores (clone edge), When criteria differ from cloned source, Then stale score data never carries (clean - slate test).

5.4.9 EPIC - 03 - STORY - 009 — Supplier Master with B - BBEE Capture & Merge (L)

As a **procurement team**, I want a clean tenant supplier master so that events invite the right, de - duplicated entities (FR - SUP - 06/07, S2 seed).

- AC1: Given supplier CRUD, When I create a supplier, Then legal name + central mailbox are required; registration/tax/B - BBEE level + certificate upload with expiry are optional captures.
- AC2: Given a duplicate - suspect (same registration number or mailbox), When I save, Then a duplicate warning suggests merge instead.
- AC3: Given merge of A into B, When executed, Then references re - point, A deactivates, and the merge is audited with old/new snapshots (FR - SUP - 07).
- AC4: Given a B - BBEE certificate past expiry, When the profile renders, Then an expired badge shows (capture - only MVP; scoring Stage 2).
- AC5: Given /suppliers endpoints, When the spec is reviewed, Then CRUD + merge contracts are documented.

5.4.10 EPIC - 03 - STORY - 010 — Invite Suppliers & Contacts to an Event (M)

As an **initiator**, I want to attach suppliers and contacts to my RFX so that publication reaches the right people (FR - SUP - 01..05).

- AC1: Given the Suppliers step, When I add from master (search) or manually, Then each supplier appears once per RfX (unique RfxId+mailbox) with ≥ 1 contact (name+email).
- AC2: Given a duplicate contact email on the same supplier, When I save, Then validation blocks with the existing contact referenced.
- AC3: Given Draft, When I remove a supplier, Then its contacts remove after confirmation; Given Published, Then removal is blocked (CR - only addition per FR - SUP - 04).
- AC4: Given a supplier with no valid contact, When I submit for review, Then validation lists it (pre - review checklist tie - in).

5.4.11 EPIC - 03 - STORY - 011 — RfX Wizard End - to - End Assembly (L)

As an **initiator**, I want the five - step wizard experience so that authoring is guided, resumable, and accessible (P05, C20).

- AC1: Given steps Details→Documents→Criteria→Suppliers→Review, When I navigate, Then per - step validation gates Next, the review step summarizes everything with edit links, and "Submit for review" only enables when all gates pass.
- AC2: Given an error summary on Next, When I activate a summary link, Then focus moves to the offending field (role=alert list).
- AC3: Given mid - wizard exit with unsaved changes, When I leave, Then an exit guard offers save/discard; returning restores my step.
- AC4: Given Lighthouse on the wizard route, When CI runs, Then route chunk <200KB gz and LCP <2.5s.
- AC5: Given NVDA, When traversing the stepper, Then current step announces via aria - current and completed states are conveyed.

5.5 EPIC - 04 — Review, Approval & Publication (P0)

Goal: Segregated, comment - disciplined governance from Draft to Published with reliable supplier invitations. (J3)

5.5.1 EPIC - 04 - STORY - 001 — Submit for Review with Validation & Locking (M)

As an **initiator**, I want submission to validate and lock my RfX so that reviewers see a complete, stable event (FR - WF - 01, FR - RFX - 08).

- AC1: Given a complete Draft, When I submit for review, Then status→PendingReview, editing locks, and the assigned reviewer is notified (email + Teams card).
- AC2: Given missing mandatory items (dates/docs for RFP - RFI/criteria≠100%/no valid supplier contact), When I submit, Then one response lists all failures and status stays Draft.
- AC3: Given a submission API failure, When retried, Then no duplicate workflow tasks exist (idempotent).
- AC4: Given the locked state, When I open the RfX, Then a read - only banner explains who has it and since when (edge: visibility).

5.5.2 EPIC - 04 - STORY - 002 — Reviewer Actions with AI Pre - Review Checklist (M)

As a **reviewer**, I want a read - only review with decision actions so that quality gates hold before publication (FR - WF - 02/03).

- AC1: Given PendingReview, When I open the review view, Then full read - only content renders with the automated checklist (dates valid, docs present, criteria 100%, suppliers valid).
- AC2: Given an RFQ, When I Approve, Then status→Published directly (short path) and invitations dispatch; Given RFI/RFP, Then Approve routes to PendingApproval.
- AC3: Given Reject or Clarify, When I submit without ≥10 - char comments, Then validation blocks; with comments, status→Returned/PendingClarification and the initiator is notified with my comments verbatim.
- AC4: Given a stale decision (initiator withdrew meanwhile), When I act, Then 409 refreshes the view to current state without error tone (edge).
- AC5: Given POST /rfx/{id}/review, When the spec is reviewed, Then approve/reject/clarify payloads + 400 comment - missing are documented.

5.5.3 EPIC - 04 - STORY - 003 — Approver Actions, Auto - Approval & SoD (M)

As an **approver**, I want final authority on RFI/RFP with SoD enforced so that no one approves their own event (FR - WF - 04/05/06).

- AC1: Given PendingApproval, When I Approve, Then status→Published, PublishDate records, invitations auto - dispatch (FR - WF: publish ≤5 min).
- AC2: Given Reviewer=Approver assignment, When review approval occurs, Then auto - progression to Published writes two distinct audit entries seconds apart (FR - WF - 05).
- AC3: Given SoD ON, When the RFx creator is selected as reviewer/approver, Then assignment blocks with policy message; Given SoD toggled OFF by admin, Then the toggle change itself is audited (FR - WF - 06).
- AC4: Given Reject without comments, When submitted, Then blocked (mandatory ≥10 chars).

5.5.4 EPIC - 04 - STORY - 004 — Configurable Approval Chains & Reassignment (M)

As an **administrator**, I want 1 - step/2 - step chains and vacancy reassignment so that workflow fits each tenant (FR - WF - 07/08).

- AC1: Given tenant config 1 - step for RFQ / 2 - step for RFP - RFI, When an RFx submits, Then the chain resolves and freezes at submit time (later config changes don't affect in - flight events).
- AC2: Given a pending task and an absent assignee, When an admin reassigns with reason, Then the new assignee is notified, the old task cancels, and the reassignment audits.
- AC3: Given an invalid chain config (approver role empty), When saved, Then validation blocks with guidance (edge).
- AC4: Given chain resolution, When queried via GET /rfx/{id}, Then current assignees and stage render for authorized users.

5.5.5 EPIC - 04 - STORY - 005 — Publication: Tokens, Invitations, Delivery Tracking (L)

As the **system**, I want reliable tokenized invitations so that every supplier contact can reach their event (FR - PUB - 01..04).

- AC1: Given final approval, When publication executes, Then per - contact single - use registration tokens (14 - day expiry, re - issuable) generate and invitation emails dispatch within 5 minutes using the admin template (RFx number/title, close datetime+TZ, portal link, buyer mailbox).

- AC2: Given a bounce/failure, When dispatch fails, Then 3 retries with backoff execute; persistent failures flag per - recipient on the RFx view with manual resend (FR - PUB - 02).
- AC3: Given a token already used or expired, When clicked, Then the portal explains and offers "request new link" (which notifies the buyer, edge).
- AC4: Given 200 contacts, When publishing, Then all invitations enqueue without loss (idempotent batch, FR - NOTIF - 05 tie - in).
- AC5: Given the deep link, When an authenticated supplier clicks, Then they land on that specific RFx (FR - PUB - 04).

5.5.6 EPIC - 04 - STORY - 006 — Withdraw & Scheduled Auto - Close (M)

As an **initiator**, I want to withdraw a live event with reason, and trust automatic closing so that lifecycle edges are governed (FR - RFX - 11/12).

- AC1: Given Published before close, When I withdraw with typed reason ≥ 10 chars (confirm modal), Then status $\rightarrow$ Withdrawn and all invited suppliers are notified ≤ 5 min.
- AC2: Given close time reached, When the scheduler fires, Then submissions lock, status $\rightarrow$ Closed, and the evaluation workspace trigger emits (EPIC - 07 dependency).
- AC3: Given a withdrawal attempt after close, When submitted, Then 409 with allowed - actions guidance.
- AC4: Given scheduler downtime over a close boundary, When it recovers, Then missed closes execute exactly once, ordered, with original deadline honored for late - block decisions (edge: catch - up).

5.6 EPIC - 05 — Supplier Portal & Sealed Responses (P0)

Goal: Ahmed acknowledges, asks, and submits versioned sealed responses on any device — free. (J4)

5.6.1 EPIC - 05 - STORY - 001 — Supplier Registration & Password Policy (M)

As a **supplier contact**, I want simple first - time onboarding so that I can participate without fees or friction (FR - SP - 01/06).

- AC1: Given a valid invitation token, When I open it first time, Then I set a password (≥ 10 chars, upper+lower+digit+special, meter shown) and land on the invited RFx.
- AC2: Given 5 failed logins, When I try again, Then lockout holds 30 min with a neutral message (no enumeration), and lockout audits (FR - SP - 02, NFR - SEC - 12).
- AC3: Given password reset, When requested for any email, Then the same confirmation renders whether or not the account exists (uniform response + timing).
- AC4: Given the registration screen on a 360px viewport with NVDA, When I complete setup, Then all steps operate (mobile - first + a11y).
- AC5: Given /supplier/auth endpoints, When the spec is reviewed, Then register/login/reset contracts document lockout and error formats.

5.6.2 EPIC - 05 - STORY - 002 — Portal Sessions & Event Scoping (M)

As a **supplier**, I want to see exactly my invitations so that confidentiality holds between bidders (FR - SP - 03/04).

- AC1: Given login, When my event list loads (P07), Then only RFx events where my supplier is invited render, with status + countdown chips.
- AC2: Given a direct URL to a non - invited RFx, When requested, Then 404 returns (not 403) and the attempt logs.
- AC3: Given 60 min inactivity, When the timeout nears, Then a 5 - min warning offers extend; expiry signs me out preserving any draft (autosaved).
- AC4: Given concurrent sessions on two devices, When both act, Then last - write - wins with version warnings on the response workspace (edge).

5.6.3 EPIC - 05 - STORY - 003 — Acknowledgement Gate & Decline (M)

As a **supplier**, I want to acknowledge or decline before seeing documents so that participation intent is recorded (FR - SP - 05, FR - ACK semantics).

- AC1: Given my first visit to an event, When the gate renders, Then summary (title, buyer, close time, doc count) shows with Acknowledge (primary) / Decline (+reason) — documents inaccessible until acknowledged.
- AC2: Given acknowledgement, When confirmed, Then timestamp+user+IP record, documents unlock, and the buyer is notified within 60s.
- AC3: Given decline, When submitted with reason, Then my event card shows Declined, submission paths disable, and the buyer sees the reason.
- AC4: Given an acknowledgement API failure, When retried, Then exactly one acknowledgement exists (unique RfxId+SupplierId; idempotent).
- AC5: Given the gate, When axe runs, Then 0 critical/serious violations (this is the portal's first impression).

5.6.4 EPIC - 05 - STORY - 004 — Document Access & Downloads (S)

As an **acknowledged supplier**, I want the current document pack so that I bid on the right content (FR - ACK unlock + FR - DOC - 03).

- AC1: Given acknowledgement, When I open Documents, Then active versions list with type, size, updated date, and checksum chip; downloads verify integrity.
- AC2: Given an addendum via approved CR, When published, Then the list shows "Updated" badges and I receive a notification (tie to EPIC - 06).
- AC3: Given a download interrupted, When resumed, Then the file completes and matches its checksum (edge).
- AC4: Given no acknowledgement, When a document URL is called directly, Then 403 with the acknowledgement requirement (defense in depth under the 404 scoping).

5.6.5 EPIC - 05 - STORY - 005 — Response Workspace: Drafts, Versions, Receipts (L)

As a **supplier**, I want to build my response safely and get receipts so that I trust my bid landed (FR - RESP - 01/02/03, J4).

- AC1: Given the response tab (P08), When I upload files per envelope (Technical/Commercial or Single), Then drafts save with per - file progress, scan status, and I can replace files freely before submitting.

- AC2: Given final submission before deadline, When I confirm, Then version N seals (timestamped to the second, checksums recorded) and an on - screen + email receipt shows version, time, and file manifest.
- AC3: Given a revision before deadline, When I submit again, Then version N+1 records and the receipt clarifies "latest final version replaces prior for evaluation" while all versions retain (FR - RESP - 03).
- AC4: Given an upload failing mid - transfer at the deadline boundary, When the deadline passes before completion, Then the partial upload discards, my prior final version (if any) stands, and the UI explains precisely (edge).
- AC5: Given /supplier/rfx/{id}/responses, When the spec is reviewed, Then multipart, envelope param, version semantics, and 403 - closed responses are documented.

5.6.6 EPIC - 05 - STORY - 006 — Sealing: Encryption, Deadline Lock, Zero Premature Access (L)

As a **compliance owner**, I want cryptographic sealing so that premature bid access is impossible, not just forbidden (FR - RESP - 04/05/06, ADR - 08, NFR - SEC - 05).

- AC1: Given a submitted response, When stored, Then content encrypts with the per - event DEK (AES - 256 - GCM) wrapped by the Key Vault KEK; DB/blob inspection in a test cannot recover plaintext.
- AC2: Given any internal role before close, When response content is requested via any API, Then only count/submitted - status metadata returns; content endpoints refuse (policy + key unavailability) and attempts log.
- AC3: Given close time, When evaluation opens, Then Technical DEK unwrap succeeds for assigned evaluators only; Commercial unwrap additionally requires the audited open - commercial action (EPIC - 07 - STORY - 005).
- AC4: Given a submission attempt at deadline+1s, When posted, Then server - side rejection "closed at {t}" returns and the attempt logs (FR - RESP - 06; clock authority = server NTP).
- AC5: Given key - management failure (KEK unavailable), When unsealing is attempted, Then a clear operational error raises with runbook link — content never falls back to unencrypted paths (edge).

5.6.7 EPIC - 05 - STORY - 007 — Portal Experience Polish: Mobile, Countdown, A11y (M)

As a **supplier on a phone**, I want a first - class mobile experience so that bidding doesn't need a desktop (FR - SP - 07, UX P07/P08).

- AC1: Given $\leq 768\text{px}$, When I browse events and submit a response, Then card - list layouts, thumb - reachable actions, and the sticky submit bar operate without horizontal scroll.
- AC2: Given the countdown chip, When $\leq 24\text{h}/\leq 1\text{h}$ thresholds pass, Then styling escalates (warning/danger) with minute - level SR announcements, and reduced - motion disables pulsing.
- AC3: Given Lighthouse mobile profile on P07/P08, When CI runs, Then LCP $< 2.5\text{s}$, INP $< 200\text{ms}$, a11y ≥ 95 .
- AC4: Given NVDA + keyboard on the full submit journey (login→ack→upload→submit→receipt), When executed per release protocol, Then no blocking issues (manual gate).

5.7 EPIC - 06 — Change Request Management (P1)

Goal: Governed post - publication change with automatic effect application. (J5)

5.7.1 EPIC - 06 - STORY - 001 — Raise CR: Dynamic Forms & Numbering (L)

As an **initiator or acknowledged supplier**, I want to raise typed CRs so that changes follow governance instead of email (FR - CR - 01/02).

- AC1: Given a Published RFx, When I (initiator) raise ExtendDueDate/UpdateDocuments/AddSupplier/UpdateRequirement/Other — or (supplier, post - ack, invited - only) Clarification/DocumentIssue/ExtendDueDate/Other — Then CR - YYYY - NNN assigns with type - specific dynamic fields and optional validated attachments.
- AC2: Given ExtendDueDate, When I pick a date $\leq$ current due or in the past, Then validation blocks with the rule (FR - CR - 04 constraint).
- AC3: Given an existing active CR of the same type, When I raise another, Then 409 explains one - active - per - type (FR - CR - 05).
- AC4: Given a Closed/Withdrawn RFx, When any CR is attempted, Then 409 blocks (FR - CR - 05).
- AC5: Given /rfx/{id}/change - requests, When the spec is reviewed, Then per - type payload schemas document with examples.

5.7.2 EPIC - 06 - STORY - 002 — CR Review & Atomic Effect Application (L)

As a **reviewer**, I want CR decisions to apply effects automatically so that approved changes cannot half - apply (FR - CR - 03/04).

- AC1: Given PendingReview CRs in my inbox, When I Approve, Then effects apply atomically per type: due - date update + all - supplier notification; document version swap (new active, old archived); new - supplier - only invitation; requirement text update — verified by per - type integration tests.
- AC2: Given Reject/Return - for - Clarification, When submitted without ≥ 10 - char comments, Then blocked; with comments, the requester can edit and resubmit returned CRs preserving comment history.
- AC3: Given an effect - application failure mid - apply (e.g., notification enqueue fails), When the transaction completes, Then either all effects + outbox records commit or none do (atomicity test).
- AC4: Given an approved ExtendDueDate, When suppliers view the event, Then live banners + countdown update $\leq 5s$ (SignalR) and emails dispatch ≤ 5 min (FR - CR - 06).

5.7.3 EPIC - 06 - STORY - 003 — CR Constraints, Auto - Reject & Notifications (M)

As the **system**, I want CR hygiene at lifecycle boundaries so that stale requests never linger (FR - CR - 05/06).

- AC1: Given open CRs at RFx close, When close executes, Then open CRs auto - reject with system comment and requesters are notified.
- AC2: Given each CR state change, When committed, Then audit entries record old/new and the affected RFx view refreshes for watchers.
- AC3: Given a supplier CR on an event they're invited to but haven't acknowledged, When attempted, Then 403 with acknowledgement guidance (FR - CR - 01 rule).

- AC4: Given the CR status widget on dashboards, When active CRs exist, Then indicator chips + tooltips render (C05 integration, FR - DASH - 07 tie).

5.8 EPIC - 07 — Evaluation & Award (P0)

Goal: From close to defensible award: AI completeness, Col - gated scoring, envelope discipline, explainable summaries. (J6)

5.8.1 EPIC - 07 - STORY - 001 — Evaluation Workspace Creation & Technical Unseal (M)

As the **system**, I want the workspace to open automatically at close so that evaluation starts immediately (FR - EVAL - 01).

- AC1: Given Rfx close with ≥ 1 final submission, When the RfxClosed event processes, Then the workspace creates, status→Evaluating, and Technical envelope content unseals to assigned evaluators only (DEK unwrap per ADR - 08).
- AC2: Given zero submissions, When close processes, Then the event flags for Unsuccessful handling with the re - issue shortcut (FR - EVAL - 10 tie).
- AC3: Given an unassigned user, When workspace access is attempted, Then 404 (scoping) and the attempt logs.
- AC4: Given workspace creation failure, When retried by the consumer, Then idempotent creation yields exactly one workspace (inbox dedupe).

5.8.2 EPIC - 07 - STORY - 002 — AI Completeness Check with Human Resolution (L)

As an **evaluator**, I want AI to flag missing/expired items with citations so that compliance screening is fast and defensible (FR - EVAL - 02, FR - AI - 06).

- AC1: Given the required - document checklist, When the check runs (≤ 120 s per 10 submissions), Then per - supplier flags list MissingDoc/ExpiredCert/UnsignedForm/Other with citation (document+location) and High/Med/Low confidence.
- AC2: Given each flag, When I Accept or Override, Then Override requires a reason ≥ 10 chars and both actions audit with my identity (human resolution of record).
- AC3: Given AI unavailability, When the check can't run, Then a manual checklist mode renders (same items, hand - marked) and the event logs Fallback — evaluation proceeds (NFR - ERR - 04).
- AC4: Given a flag citing a document, When I click the citation, Then the source excerpt opens in the drawer at the referenced location (C11 pattern).
- AC5: Given the golden completeness set, When Promptfoo runs in CI, Then precision/recall $\geq$ baseline thresholds (drift gate).

5.8.3 EPIC - 07 - STORY - 003 — Panel Assignment & Conflict - of - Interest Gate (M)

As an **initiator**, I want Col - gated panel access so that impartiality is provable (FR - EVAL - 03).

- AC1: Given panel assignment (≥ 1 evaluator), When an evaluator first opens the workspace, Then the Col declaration blocks all content until completed (None / Declared+description).
- AC2: Given a declared conflict, When submitted, Then access suspends pending initiator decision (replace or accept with documented reason); both paths audit.

- AC3: Given panel changes mid - evaluation, When an evaluator is replaced, Then their draft scores archive (excluded from consolidation) and the replacement starts fresh (edge).
- AC4: Given the Col gate, When keyboard/AT - navigated, Then the blocking card and form are fully operable and announced.

5.8.4 EPIC - 07 - STORY - 004 — Scoring Matrix with AI Assist (L)

As an **evaluator**, I want an efficient scoring grid with optional AI suggestions so that scoring is fast but mine (FR - EVAL - 04/05/12).

- AC1: Given assigned submissions, When I score each criterion (0–scale, step 0.5, optional comment), Then cells autosave (2s debounce) with per - cell status and my progress persists across sessions.
- AC2: Given AI assist enabled, When suggestions render, Then they appear as ghost values with citations and an explicit Apply chip — never pre - filled; applying records AiSuggestedValue alongside my value (FR - EVAL - 05).
- AC3: Given a cell save failure, When retried, Then no score loss or duplication (idempotent upsert keyed RfxId+Supplier+Criterion+Evaluator).
- AC4: Given progress visibility config "hidden until complete", When the initiator views progress, Then completion percentages show without score values (FR - EVAL - 12).
- AC5: Given role=grid arrow - key navigation with NVDA, When traversing cells, Then "Score for {supplier}, {criterion}" announces per cell (C10 a11y).

5.8.5 EPIC - 07 - STORY - 005 — Commercial Envelope Opening (M)

As a **sourcing lead**, I want a ceremonial, audited commercial opening so that price data stays sealed until the technical gate passes (FR - RESP - 05, FR - EVAL J6).

- AC1: Given two - envelope mode with technical scoring complete (gate flag), When I invoke Open Commercial with typed reason, Then the confirm modal executes the unwrap, content unseals, and a dedicated audit entry + toast record the ceremony.
- AC2: Given the gate unmet, When opening is attempted, Then 409 lists unmet conditions (which evaluators/criteria outstanding).
- AC3: Given single - envelope events, When close occurs, Then this step skips cleanly (stepper shows skipped state, edge).
- AC4: Given the price comparison table post - open, When rendered, Then per - supplier totals + line - item breakdowns display with tabular numerals and CSV export.

5.8.6 EPIC - 07 - STORY - 006 — Consolidation, Ranking & Outliers (L)

As a **sourcing lead**, I want defensible consolidated results so that the recommendation writes itself from evidence (FR - EVAL - 06).

- AC1: Given all evaluators complete, When consolidation runs, Then weighted totals compute (mean evaluator score × weight, normalized to 100), rankings order, and the comparison matrix (C11) renders side - by - side.
- AC2: Given an evaluator score $>2\sigma$ from the criterion panel mean, When consolidation renders, Then the cell flags with an outlier tooltip (visible in audit exports too).

- AC3: Given consolidation math, When the unit suite runs, Then 100% branch coverage over weighting/normalization/tie cases, including rounding to 2 decimals half - up (documented rule).
- AC4: Given a tie on total, When ranking displays, Then the configured tie - break policy annotation shows (price - weight default; B - BBEE Stage 2 placeholder documented) (edge).

5.8.7 EPIC - 07 - STORY - 007 — AI Executive Summary with Citations (M)

As a **decision maker**, I want an explainable summary so that committees decide fast without losing rigor (FR - EVAL - 07, FR - AI - 06).

- AC1: Given consolidated results, When I generate the summary, Then it covers ranking rationale, per - supplier strengths/gaps with citations, price posture (post - open), completeness posture, and outliers — rendered in an AIPanel marked AI - generated with confidence.
- AC2: Given any claim's citation, When clicked, Then the source excerpt opens (bid document + location) in the drawer.
- AC3: Given regeneration, When invoked, Then a new version stores (versioned), prior versions remain viewable, and the AwardRecommendation records which version informed it.
- AC4: Given AI unavailability, When generation fails after one retry, Then the manual summary editor offers with the consolidated data table embedded (no dead end).

5.8.8 EPIC - 07 - STORY - 008 — Award Recommendation & Decision (L)

As a **sourcing lead**, I want a governed award flow so that the final decision is human, justified, and audited (FR - EVAL - 08/09/11).

- AC1: Given consolidation, When I record a recommendation (supplier(s) + justification ≥ 50 chars; +single - bid justification when only one response), Then status→AwardPendingApproval and the approver receives web + Teams card.
- AC2: Given approver Approve, When confirmed, Then status→Awarded, outcome letters (award + regret, from templates) queue to all responding suppliers, and StageTimestamp closes the cycle - time record.
- AC3: Given approver Return with comments, When submitted, Then status→Evaluating and the recommendation reopens with comments attached.
- AC4: Given an approval attempt by the recommending user under SoD, When submitted, Then 403 policy error (FR - WF - 06 extension to awards).
- AC5: Given POST /rfx/{id}/award - recommendation(+/decide), When the spec is reviewed, Then payloads, gates (single - bid flag), and 409 states document.

5.8.9 EPIC - 07 - STORY - 009 — Unsuccessful Events & Re - Issue (S)

As a **sourcing lead**, I want clean unsuccessful - closure so that failed events end with evidence and a fast path to retry (FR - EVAL - 10).

- AC1: Given zero or all - non - compliant submissions, When I close as Unsuccessful with reason, Then status→Unsuccessful, suppliers are notified, and evidence retains.
- AC2: Given Unsuccessful, When I click Re - issue, Then a clone (EPIC - 03 - STORY - 003 semantics) opens in Draft linked to the original.

- AC3: Given remaining compliant bids, When Unsuccessful is attempted, Then a confirmation requires explicit acknowledgment listing compliant bidders (edge guard).
- AC4: Given the unsuccessful flow, When audited, Then reason + actor + linkage entries exist.

5.9 EPIC - 08 — Teams Orchestration (P1)

Goal: Decisions, meetings, and status where stakeholders already work. (J7)

5.9.1 EPIC - 08 - STORY - 001 — Bot Registration, OBO Auth & Status Queries (L)

As an **internal user**, I want to ask IntelliSource in Teams so that status is one message away (FR - TEAMS - 05).

- AC1: Given the installed app, When I message "my pending approvals" / "status of RFQ - 2026 - 014" / "events closing this week", Then results render as compact cards with Open links — scoped to my permissions via on - behalf - of token exchange.
- AC2: Given a query about an RFx I can't access, When asked, Then the bot answers "not found" (no existence leakage).
- AC3: Given Graph/token failure, When a query runs, Then a friendly retry message with correlation ID returns (no stack traces).
- AC4: Given bot endpoints, When the spec is reviewed, Then the messaging contract + auth flow document (integration guide).

5.9.2 EPIC - 08 - STORY - 002 — Adaptive - Card Approvals (L)

As a **reviewer/approver**, I want to decide from a card so that governance moves at chat speed (FR - TEAMS - 01/02, FR - WF - 09).

- AC1: Given a review/approval/CR/award task, When the card arrives, Then Approve/Reject/Clarify buttons + comment box render, and actions execute under my Entra identity with identical validation to web (comments mandatory on reject/clarify).
- AC2: Given an already - decided task, When I act on the stale card, Then the card refreshes to current state with an informational note (idempotent, FR - TEAMS - 02).
- AC3: Given card action failure, When the API errors, Then the card shows the error + "Open in IntelliSource" fallback link.
- AC4: Given a decision via card, When audit is queried, Then the entry records channel=Teams with the same fidelity as web.

5.9.3 EPIC - 08 - STORY - 003 — Meeting Auto - Scheduling & Channel Milestones (M)

As a **sourcing lead**, I want evaluation meetings and milestones automated so that coordination overhead disappears (FR - TEAMS - 03/04/06).

- AC1: Given evaluation kick - off, When the panel is confirmed, Then a Teams meeting schedules at the first common free slot within 3 business days (Graph findMeetingTimes), with agenda + workspace deep link; the initiator can override slot/attendees before send.
- AC2: Given milestones (Published, Closing - 24h, Closed, Awarded), When they occur, Then the configured channel receives a milestone card ≤5 min.
- AC3: Given Graph unavailability (timeout 10s), When any Teams delivery fails, Then the email equivalent sends ≤60s and the fallback logs (FR - TEAMS - 06).

- AC4: Given no common free slot in 3 days, When scheduling runs, Then the earliest conflict - minimized proposal returns flagged for manual adjustment (edge).

5.9.4 EPIC - 08 - STORY - 004 — Teams Intake Dialog (M)

As a **business requester**, I want the full intake conversation in Teams so that requests start from chat (FR - TEAMS - 07, completes EPIC - 02 - STORY - 007).

- AC1: Given "I need..." messaged to the bot, When the dialog runs, Then the same clarification batches, validation, numbering, and spec preview execute as web (contract parity test).
- AC2: Given attachments in Teams, When provided, Then they pass the same scan/size gates with per - file feedback.
- AC3: Given AI unavailability, When the dialog can't proceed, Then the bot posts a link to the web manual form (graceful degrade).
- AC4: Given dialog completion, When the intake submits, Then the confirmation card deep - links to tracking and the triage flow proceeds identically.

5.10 EPIC - 09 — Audit Query & Evidence Packs (P0)

Goal: Pieter proves any decision in minutes. (J8; framework built in EPIC - 01 - STORY - 006)

5.10.1 EPIC - 09 - STORY - 001 — Audit Query UI & Timeline (M)

As a **governance owner**, I want to explore the trail visually so that questions get answered without engineering help (FR - AUD - 04/06).

- AC1: Given audit access (Administrator/ViewOnly - Audit), When I filter by entity/actor/action/date, Then the C16 timeline renders (virtualized ≥ 200 entries) with old $\rightarrow$ new diff expanders and AI entries showing model+version on hover.
- AC2: Given an RFX view, When I open its Audit tab, Then the derived status - history timeline renders with durations per stage.
- AC3: Given a query with zero matches, When executed, Then the audit empty state with clear - filters CTA renders.
- AC4: Given 1M entries, When any single - entity query runs, Then p95 <3s (indexed path, FR - AUD - 04).
- AC5: Given keyboard/AT, When traversing the timeline, Then entries, expanders, and filters operate with announced context.

5.10.2 EPIC - 09 - STORY - 002 — Evidence Pack Generation & WORM Export (L)

As an **auditor**, I want a tamper - evident bundle so that external scrutiny is self - service (FR - AUD - 05).

- AC1: Given any RFX, When I generate an evidence pack, Then within 5 minutes a bundle produces: PDF chronology, CSV/JSON full detail, document manifest with SHA - 256s, and a chain - verification statement — stored in the WORM container and offered for download.
- AC2: Given the export, When completed, Then the export action itself audits (who, when, scope).
- AC3: Given a chain - verification failure during export, When detected, Then the pack blocks, an alert raises, and the failure surface names the earliest broken segment (edge).

- AC4: Given POST /rfx/{id}/evidence - pack, When the spec is reviewed, Then async job contract (202 + status polling + download) documents.

5.10.3 EPIC - 09 - STORY - 003 — Chain Verification Job & Alerts (M)

As a **platform operator**, I want continuous tamper checks so that integrity issues surface in hours, not audits (FR - AUD - 03).

- AC1: Given the daily job, When the chain verifies, Then a Verified checkpoint records with duration metrics.
- AC2: Given an out - of - band row alteration (test harness), When the next verification runs, Then the break detects, an alert fires to operations, and the admin health view shows the failure.
- AC3: Given POPIA redaction entries, When verification runs, Then redactions verify as valid chain elements (FR - AUD - 07).
- AC4: Given verification under load, When the job runs against 1M+ rows, Then it completes within the maintenance window without blocking writes (partition - wise, edge).

5.11 EPIC - 10 — Dashboards, Analytics & Notifications (P1)

Goal: Real - time oversight, provable cycle - time ROI, reliable messaging. (J8, K1/K6)

5.11.1 EPIC - 10 - STORY - 001 — Read - Model Projections (M)

As a **developer**, I want denormalized projections so that dashboards are fast and isolated from write models (arch Section 3).

- AC1: Given lifecycle events on the bus, When consumers process, Then projection tables update idempotently (duplicate delivery test) within 5s.
- AC2: Given projection lag or consumer failure, When detected, Then DLQ + lag metrics surface in admin health; replay rebuilds projections deterministically.
- AC3: Given a projection rebuild, When executed, Then dashboards remain available (stale - while - rebuilding banner).
- AC4: Given the projection schema, When migrations run, Then rebuild - from - events completes on seeded data in CI.

5.11.2 EPIC - 10 - STORY - 002 — Pipeline Dashboard & Role - Scoped Actions (L)

As a **procurement user**, I want the live pipeline with my actions so that nothing waits in email (FR - DASH - 01/05/07).

- AC1: Given P02, When loaded per role, Then the RFx table renders scoped (Requester: own intakes; Initiator: own+team; Head/ViewOnly: all; Admin: +config) with filters (status/type/owner/category/date), sort, and 25/page pagination.
- AC2: Given each row, When actions render, Then only status×role - valid buttons appear (matrix test across all statuses × roles).
- AC3: Given table load failure, When it errors, Then the error banner with ERR - ID + retry renders and filters persist.
- AC4: Given 500 rows filtered, When I page/sort, Then responses return p95 <200ms (indexed queries) and the mobile card - list variant renders ≤768px.

- AC5: Given axe on P02, When scanned, Then 0 critical/serious violations (table semantics, aria - sort, live region).

5.11.3 EPIC - 10 - STORY - 003 — KPI Cards, Charts & Exports (L)

As a **head of procurement**, I want KPIs and exportable charts so that oversight and board packs are one click (FR - DASH - 02/03, principle #17).

- AC1: Given the KPI row, When it loads, Then open - by - status, closing - ≤ 7 - days, rolling - 90 - day avg cycle time, response rate, and my - pending - approvals render with deltas.
- AC2: Given charts (pipeline by status, cycle - time trend, response rate by category, supplier participation), When rendered, Then bar/line/pie/donut variants match UX C17 (patterns ≥ 4 series, data - table toggle) and every chart exports CSV and PDF faithfully.
- AC3: Given no data (new tenant), When the dashboard loads, Then KPI/chart empty states render with onboarding CTAs.
- AC4: Given chart libraries, When the route builds, Then chart code lazy - loads (chunk $< 200\text{KB}$) and dashboard LCP stays $< 2.5\text{s}$.

5.11.4 EPIC - 10 - STORY - 004 — SignalR Live Updates (M)

As a **user**, I want the dashboard to move by itself so that state is always current (FR - DASH - 04).

- AC1: Given an open dashboard, When any visible entity changes, Then the row/card updates $\leq 5\text{s}$ with a pulse highlight and a polite SR announcement.
- AC2: Given SignalR disconnection, When detected, Then 30s polling fallback engages with a subtle indicator; reconnection restores push silently.
- AC3: Given 1,000 concurrent dashboard sessions (k6 harness), When events fan out, Then p95 delivery $\leq 5\text{s}$ (NFR - PERF - 07).
- AC4: Given reduced - motion, When updates arrive, Then highlight renders as static outline (no pulse).

5.11.5 EPIC - 10 - STORY - 005 — Cycle - Time Instrumentation & ROI Analytics (M)

As a **product owner**, I want provable cycle - time data so that the $\geq 30\%$ claim is demonstrable in - product (FR - DASH - 06, K1).

- AC1: Given StageTimestamps from every transition, When analytics compute, Then per - event and aggregate durations (intake $\rightarrow$ RFx, draft $\rightarrow$ publish, publish $\rightarrow$ close, close $\rightarrow$ award) render with trend lines and baseline comparison fields.
- AC2: Given a tenant - configured manual baseline, When set, Then dashboards show % improvement vs baseline per category.
- AC3: Given missing stages (e.g., direct RFx without intake), When computed, Then partial journeys aggregate correctly without skew (null - safe math, unit - tested).
- AC4: Given export, When invoked, Then cycle - time datasets export CSV for board reporting.

5.11.6 EPIC - 10 - STORY - 006 — Notification Engine: Templates, Triggers, DLQ (L)

As an **administrator**, I want governed messaging so that every notification is deliverable, editable, and observable (FR - NOTIF - 01/03/04/05).

- AC1: Given all lifecycle triggers (intake decisions, workflow tasks/outcomes, publication, acknowledgement, receipts, CR decisions, reminders, close, award/regret), When events occur, Then notifications enqueue with correct template + variables (snapshot tests per template).
- AC2: Given template admin, When I edit with placeholders and preview, Then versions save, changes audit, and invalid placeholders block save.
- AC3: Given delivery failure, When retries (3×backoff) exhaust, Then the message dead - letters and the admin DLQ view supports inspect/requeue/discard.
- AC4: Given 1,000 - recipient dispatch, When processed, Then zero loss (at - least - once + idempotent send guard; duplicate - suppression test).
- AC5: Given notification content, When rendered, Then no personal data beyond need - to - know appears (POPIA minimization check).

5.11.7 EPIC - 10 - STORY - 007 — Reminder Scheduling T - 7/T - 1 (S)

As a **supplier**, I want timely reminders so that I never miss a deadline I meant to hit (FR - NOTIF - 02).

- AC1: Given an acknowledged, non - submitted supplier, When T - 7d and T - 1d before close pass, Then reminder emails send referencing the countdown and portal link.
- AC2: Given a submitted supplier, When reminder time passes, Then no reminder sends (suppression test).
- AC3: Given a due - date extension via CR, When applied, Then reminder schedules recompute against the new date (edge).
- AC4: Given scheduler downtime spanning a reminder slot, When recovered, Then missed reminders send once if still relevant (not after close).

5.12 EPIC - 11 — Identity, Admin & POPIA (P0)

Goal: Governed users, tenants, configuration, and data - subject rights. (J9)

5.12.1 EPIC - 11 - STORY - 001 — User Administration & Delegation Limits (M)

As an **administrator**, I want user lifecycle control so that access always matches organizational reality (FR - ADM - 02/03, FR - INT - 13).

- AC1: Given admin console Users, When I assign roles (direct or Entra - group mapping) and delegation limits, Then changes apply immediately, audit with old/new, and affected sessions refresh claims ≤5 min.
- AC2: Given deactivation of a user with in - flight tasks, When executed, Then the reassignment prompt lists their pending items (no orphaned approvals; ties EPIC - 04 - STORY - 004).
- AC3: Given a non - admin, When any admin endpoint is called, Then 403 (endpoint×role matrix).
- AC4: Given the admin UI, When scanned, Then axe passes and destructive actions require typed confirmation (C05/C14).

5.12.2 EPIC - 11 - STORY - 002 — Workflow & Tenant Configuration (M)

As an **administrator**, I want tenant - level workflow settings so that governance fits each customer (FR - ADM - 03/06, FR - WF - 06/07).

- AC1: Given config (SoD toggle, approval chains, two - envelope default, Col requirement, notification channels), When saved, Then settings apply without redeployment and audit with old/new.
- AC2: Given an in - flight RFx, When config changes, Then frozen - at - submit workflow rules hold for it while new submissions use new config (isolation test).
- AC3: Given invalid config combinations (2 - step chain with no approver role members), When saved, Then validation blocks with specific guidance.
- AC4: Given GET/PUT /admin/config, When the spec is reviewed, Then schema + validation rules document.

5.12.3 EPIC - 11 - STORY - 003 — POPIA Data - Subject Tooling (L)

As a **privacy officer**, I want subject - rights execution so that POPIA operator duties are one workflow, not a project (FR - ADM - 04, NFR - COMP - 01/03).

- AC1: Given a subject - access request, When I run it for a person (internal or supplier user), Then a report of all personal data + processing log generates ≤24h SLA (async job with notification).
- AC2: Given a correction, When applied, Then changes propagate and audit; Given erasure, Then personal data redacts across stores while audit - chain verifiability preserves (redaction entries), with legal - hold override blocking erasure where active.
- AC3: Given retention schedules per data class, When the purge job runs, Then expired data purges (or anonymizes) per schedule with a purge manifest audit.
- AC4: Given breach - notification workflow, When an incident is flagged, Then the detect→assess→notify checklist with 72h SLA timers and responsible - party contact templates executes and logs.

5.12.4 EPIC - 11 - STORY - 004 — Tenant Provisioning: Shared & In - Tenant Stamps (L)

As a **platform operator**, I want scripted tenant creation in both modes so that onboarding is hours, not weeks (FR - ADM - 05, ADR - 04).

- AC1: Given a shared - SaaS order, When provisioning runs, Then tenant records, schemas/RLS scopes, blob containers, config defaults, and admin invitations create end - to - end (idempotent script).
- AC2: Given an in - tenant order, When the stamp pipeline runs, Then dedicated namespace/cluster values, Postgres, and Key Vault deploy from the same Helm chart parameterization, passing the same smoke suite.
- AC3: Given provisioning failure mid - way, When rerun, Then the process resumes idempotently without orphaned resources.
- AC4: Given a new tenant, When the isolation test pack runs, Then cross - tenant 404 guarantees hold (NFR - SEC - 11 regression).

5.12.5 EPIC - 11 - STORY - 005 — AI Configuration & Model Routing Admin (M)

As an **administrator**, I want per - tenant AI controls so that features, models, and data boundaries match each customer's risk posture (FR - AI - 07/08, FR - ADM - 03).

- AC1: Given AI settings, When I toggle features (intake assist, drafting, completeness, scoring assist, summaries, bot), Then affordances appear/disappear ≤5 min and disabled APIs return 403 feature - disabled while manual paths function.
- AC2: Given s72 routing config, When set to SA - only, Then the router refuses non - SA inference verifiably (EPIC - 01 - STORY - 011 AC2 regression) and the setting renders on the tenant compliance summary.
- AC3: Given model/prompt updates by operations, When deployed, Then tenants pin to tested versions until migration windows (no silent model swaps; versions visible in admin).
- AC4: Given the AI config screen, When audited, Then every change logs with old/new (board - grade AI governance evidence).

5.13 EPIC - 12 — Hardening, Performance & GA Readiness (P1)

Goal: Prove the NFRs: scale, security, accessibility, recoverability, and end - to - end quality.

5.13.1 EPIC - 12 - STORY - 001 — Load & Performance Certification (L)

As a **platform owner**, I want instrumented proof of scale so that principle #13 is a fact, not a hope (NFR - PERF - 01/03/07, NFR - SCALE - 02).

- AC1: Given k6 profiles (10k concurrent browse; 1k supplier submission spike at deadline; SignalR fan - out at 1k sessions), When executed against staging, Then p95 end - to - end <2s, API p95 <500ms, error rate <1%, fan - out ≤5s.
- AC2: Given the deadline - spike profile, When uploads surge, Then the 50/min supplier upload tier throttles gracefully (429+Retry - After) without data loss (RFx lesson regression).
- AC3: Given breach of any target, When results publish, Then tuning actions log and the test reruns until green (gate for GA).
- AC4: Given DB under load, When p95 query time exceeds 200ms on any endpoint, Then the report names the query for remediation (NFR - PERF - 05).

5.13.2 EPIC - 12 - STORY - 002 — Security Verification: ZAP, ASVS L2, Pen - Test Fixes (L)

As a **security officer**, I want independent verification so that Zero Trust claims survive attack (NFR - SEC - 06, arch Section 9).

- AC1: Given OWASP ZAP baseline in CI, When scans run per build, Then no high - risk findings ship (gate).
- AC2: Given the ASVS L2 checklist, When verified pre - GA, Then all applicable controls pass or have accepted - risk sign - off with owner + date.
- AC3: Given sealed - bid attack scenarios (privilege escalation to pre - close content, key - path bypass, tenant crossing), When the pen test executes, Then zero successful accesses; findings remediate and retest.
- AC4: Given dependency/secret scanning, When any critical CVE or leaked secret appears, Then CI blocks and rotation runbooks execute (NFR - SEC - 08).

5.13.3 EPIC - 12 - STORY - 003 — Accessibility Certification & ACR (M)

As a **compliance lead**, I want WCAG 2.1 AA proven and published so that public - sector tenders accept us (NFR - A11Y - 01..06).

Gijima Group	- Confidential -	Page 85 of 106
	Version:01.00 Valid from: 24.07.2026	Next review 24.10.2026

- AC1: Given axe across all routes + Storybook states, When CI runs, Then 0 critical/serious violations and Lighthouse a11y ≥ 95 on P01–P12.
- AC2: Given the manual NVDA + keyboard protocol on J1/J3/J4/J6/J8, When executed, Then results document with zero blocking issues.
- AC3: Given token - pair contrast tests, When run, Then every documented pairing meets its AA threshold (automated guard against palette drift).
- AC4: Given GA, When the ACR (VPAT - style) publishes, Then it reflects tested evidence with known - issues transparency.

5.13.4 EPIC - 12 - STORY - 004 — DR Rehearsal, Freeze Windows & Runbooks (M)

As a **platform operator**, I want practiced recovery so that RTO/RPO are demonstrated, not declared (NFR - REL - 02/03).

- AC1: Given the quarterly DR drill, When region failover rehearses (staging), Then RTO ≤ 1 h and RPO ≤ 15 min measure and document.
- AC2: Given the deploy scheduler, When any tenant has an event closing within 48h, Then production deploys defer automatically with override requiring two - person approval (NFR - REL - 03).
- AC3: Given the runbook set (AI outage, Graph outage, key - vault failure, chain - verification break, DLQ growth), When each scenario is tabletop - tested, Then on - call can execute without authors present.
- AC4: Given backup restore, When a PITR restore executes to a sandbox, Then data verifies within RPO bounds (evidence stored).

5.13.5 EPIC - 12 - STORY - 005 — E2E Journey Suite & AI Regression Sets (L)

As a **team**, we want the eight journeys always green so that regressions surface before users see them (arch Section 7).

- AC1: Given Playwright suites for J1–J8, When PR smoke (subset) and nightly full runs execute, Then all pass with trace - on - failure artifacts.
- AC2: Given the AI golden sets (intake specs, drafts, completeness, summaries), When Promptfoo runs nightly and on model/prompt change, Then drift beyond thresholds alerts and blocks model rollout.
- AC3: Given flaky - test policy, When a test flakes $>2\%$ over 20 runs, Then it quarantines with an owner and fix SLA (suite health).
- AC4: Given the supplier journey (J4), When run on mobile viewport, Then the E2E passes including chunked - upload resume simulation.

5.14 FUTURE - 13 — Stage 2 Epics (12–24 months, outline)

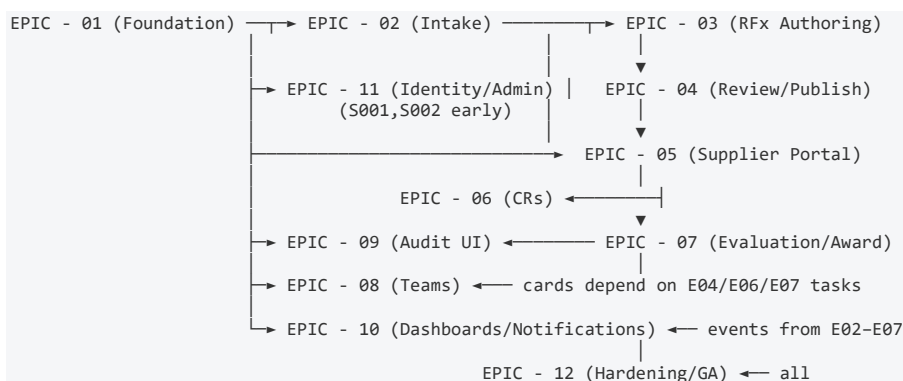
Future epic	Scope (from PRD Section 6)	Seeds in MVP
F13 - A Supplier Discovery & Enrichment	AI web/DB discovery, EcoVadis/D&B enrichment, sanctions screening, CSD/SARS/CIPC verification (FR - S2 - DISC)	Supplier master (E03 - S009), AI host (E01 - S011)
F13 - B B - BBEE Preference Engine	80/20 & 90/10 formulas, staged compliance $\rightarrow$ functionality $\rightarrow$ price+preference, certificate validation, reporting (FR - S2 - BEE)	B - BBEE capture (E03 - S009), staged evaluation model (E07)

Future epic	Scope (from PRD Section 6)	Seeds in MVP
F13 - C Committee Governance	Blind scoring, consensus workspace, e - voting in Teams, Col registers (FR - S2 - COMM)	Col gate (E07 - S003), Teams cards (E08 - S002)
F13 - D Multi - Round & BAFO	Round management, BAFO invitations, scenario analysis (FR - S2 - ROUND)	State machine extensibility (E03 - S004)
F13 - E ERP Connectors	SAP/Oracle/D365/Sage supplier master, budget check, award handoff (FR - S2 - ERP)	API - first contracts (E01 - S008), webhooks
F13 - F Savings Ledger & Bundling	Baseline vs awarded vs invoiced tracking, demand bundling (FR - S2 - SAVE)	Cycle - time instrumentation (E10 - S005)
F13 - G Supplier Q&A & Addenda	Anonymous Q&A, addenda broadcast (FR - S2 - QA — pull - forward candidate)	Documents/addenda (E03 - S005), notifications (E10 - S006)

5.15 FUTURE - 14 — Stage 3 Epics (24–36 months, outline)

Future epic	Scope (PRD Section 6)
F14 - A Contract Lifecycle	Auto - draft from award, clause libraries, AI redline vs playbooks, eSignature, obligation tracking (FR - S3 - CLM)
F14 - B Auction Engine	English/Dutch/Japanese/multi - attribute auctions, ML - timed close (FR - S3 - AUC)
F14 - C Predictive & Autonomous	Supplier risk, price trends, leakage prediction; autonomous tactical negotiation (FR - S3 - PRED)
F14 - D Renewal Automation	Contract - expiry - triggered re - sourcing (FR - S3 - RENEW)
F14 - E Copilot Agent Store	Marketplace listing, MCP/A2A interop surfaces (FR - S3 - STORE)

5.16 Dependency Graph



No circular dependencies. EPIC - 08 stories integrate progressively (cards activate as each workflow epic lands). EPIC - 12 is the GA gate.

5.17 Sprint Plan (2 - week sprints, indicative — 3 - dev + 1 - QA team)

Sprint	Focus	Stories
S1–S3	Foundation	EPIC - 01 all (S001–S012), EPIC - 11 - S001/S002
S4–S5	Intake + RFx core	EPIC - 02 all, EPIC - 03 - S001..S006
S6–S7	AI drafting + workflow	EPIC - 03 - S007..S011, EPIC - 04 all
S8–S9	Supplier portal + sealing	EPIC - 05 all, EPIC - 06 all
S10–S12	Evaluation + award + Teams	EPIC - 07 all, EPIC - 08 all
S13–S14	Audit + dashboards + POPIA	EPIC - 09, EPIC - 10, EPIC - 11 - S003..S005
S15–S16	Hardening + GA	EPIC - 12 all + readiness re - check

5.18 Requirement Traceability Matrix (124/124 FRs mapped — 100%)

Story keys abbreviated: E03 - S007 = EPIC - 03 - STORY - 007. Every story AC references its FRs (bidirectional); NFR coverage is carried by EPIC - 01 (foundational NFRs) and EPIC - 12 (certification).

FR	Story ID(s)		FR	Story ID(s)
Intake			Change Requests	
FR - INT - 01	E02 - S001, E08 - S004		FR - CR - 01	E06 - S001
FR - INT - 02	E02 - S002		FR - CR - 02	E06 - S001
FR - INT - 03	E02 - S003		FR - CR - 03	E06 - S002
FR - INT - 04	E02 - S003		FR - CR - 04	E06 - S002
FR - INT - 05	E02 - S004		FR - CR - 05	E06 - S001, E06 - S003
FR - INT - 06	E02 - S004		FR - CR - 06	E06 - S002, E06 - S003
FR - INT - 07	E02 - S005		Evaluation & Award	
FR - INT - 08	E02 - S006		FR - EVAL - 01	E07 - S001
FR - INT - 09	E02 - S006		FR - EVAL - 02	E07 - S002
FR - INT - 10	E02 - S002		FR - EVAL - 03	E07 - S003
FR - INT - 11	E02 - S004		FR - EVAL - 04	E07 - S004
FR - INT - 12	E02 - S001		FR - EVAL - 05	E07 - S004
FR - INT - 13	E02 - S005		FR - EVAL - 06	E07 - S006
RFx Lifecycle			FR - EVAL - 07	E07 - S007
FR - RFX - 01	E03 - S001		FR - EVAL - 08	E07 - S008
FR - RFX - 02	E03 - S001		FR - EVAL - 09	E07 - S008
FR - RFX - 03	E03 - S001		FR - EVAL - 10	E07 - S001, E07 - S009
FR - RFX - 04	E03 - S004		FR - EVAL - 11	E07 - S008

FR	Story ID(s)		FR	Story ID(s)
FR - RFX - 05	E03 - S002		FR - EVAL - 12	E07 - S004
FR - RFX - 06	E03 - S003		Teams	
FR - RFX - 07	E03 - S003		FR - TEAMS - 01	E08 - S002
FR - RFX - 08	E04 - S001		FR - TEAMS - 02	E08 - S002
FR - RFX - 09	E03 - S008		FR - TEAMS - 03	E08 - S003
FR - RFX - 10	E03 - S008		FR - TEAMS - 04	E08 - S003
FR - RFX - 11	E04 - S006		FR - TEAMS - 05	E08 - S001
FR - RFX - 12	E03 - S004, E04 - S006		FR - TEAMS - 06	E08 - S003
FR - RFX - 13	E03 - S002		FR - TEAMS - 07	E02 - S007, E08 - S004
AI Governance			Audit & Evidence	
FR - AI - 01	E03 - S007		FR - AUD - 01	E01 - S006
FR - AI - 02	E03 - S007		FR - AUD - 02	E01 - S006
FR - AI - 03	E03 - S007		FR - AUD - 03	E01 - S006, E09 - S003
FR - AI - 04	E01 - S011		FR - AUD - 04	E09 - S001
FR - AI - 05	E01 - S006, E01 - S011		FR - AUD - 05	E09 - S002
FR - AI - 06	E07 - S002, E07 - S007		FR - AUD - 06	E09 - S001
FR - AI - 07	E01 - S011, E11 - S005		FR - AUD - 07	E01 - S006, E09 - S003
FR - AI - 08	E01 - S011, E11 - S005		Dashboard	
Documents & Templates			FR - DASH - 01	E10 - S002
FR - DOC - 01	E01 - S007, E03 - S005		FR - DASH - 02	E10 - S003
FR - DOC - 02	E01 - S007		FR - DASH - 03	E10 - S003
FR - DOC - 03	E01 - S007, E03 - S005		FR - DASH - 04	E10 - S004
FR - DOC - 04	E03 - S005, E06 - S002		FR - DASH - 05	E10 - S002
FR - DOC - 05	E03 - S006		FR - DASH - 06	E03 - S004, E10 - S005
FR - DOC - 06	E01 - S007		FR - DASH - 07	E10 - S002, E06 - S003
Suppliers			Notifications	
FR - SUP - 01	E03 - S010		FR - NOTIF - 01	E10 - S006

FR	Story ID(s)		FR	Story ID(s)
FR - SUP - 02	E03 - S010		FR - NOTIF - 02	E10 - S007
FR - SUP - 03	E03 - S010		FR - NOTIF - 03	E10 - S006
FR - SUP - 04	E03 - S010, E06 - S002		FR - NOTIF - 04	E01 - S009, E10 - S006
FR - SUP - 05	E03 - S010		FR - NOTIF - 05	E04 - S005, E10 - S006
FR - SUP - 06	E03 - S009		Identity, Admin & POPIA	
FR - SUP - 07	E03 - S009		FR - ADM - 01	E01 - S004
Workflow			FR - ADM - 02	E01 - S004, E11 - S001
FR - WF - 01	E04 - S001		FR - ADM - 03	E11 - S001, E11 - S002, E11 - S005
FR - WF - 02	E04 - S002		FR - ADM - 04	E11 - S003
FR - WF - 03	E04 - S002		FR - ADM - 05	E11 - S004
FR - WF - 04	E04 - S003		FR - ADM - 06	E11 - S002
FR - WF - 05	E04 - S003		Publication	
FR - WF - 06	E04 - S003, E07 - S008		FR - PUB - 01	E04 - S005
FR - WF - 07	E04 - S004		FR - PUB - 02	E04 - S005
FR - WF - 08	E04 - S004		FR - PUB - 03	E04 - S005
FR - WF - 09	E04 - S002, E04 - S003, E08 - S002		FR - PUB - 04	E04 - S005
Supplier Portal			Responses & Sealing	
FR - SP - 01	E05 - S001		FR - RESP - 01	E05 - S005
FR - SP - 02	E05 - S001		FR - RESP - 02	E05 - S005
FR - SP - 03	E05 - S002		FR - RESP - 03	E05 - S005
FR - SP - 04	E05 - S002		FR - RESP - 04	E05 - S006
FR - SP - 05	E05 - S003, E05 - S004		FR - RESP - 05	E05 - S006, E07 - S005
FR - SP - 06	E05 - S001		FR - RESP - 06	E05 - S006
FR - SP - 07	E05 - S007		FR - RESP - 07	E01 - S007, E05 - S005

Unmapped FRs: 0. Stage 2/3 outline requirements (FR - S2 - , FR - S3 -) map to FUTURE - 13/FUTURE - 14 epics by design (future phase).

NFR coverage anchors: NFR - PERF/BUDGET → E01 - S010/S012, E10 - S002/S003, E12 - S001 · NFR - SEC → E01 - S002/S003/S004, E05 - S006, E12 - S002 · NFR - REL/ERR → E01 - S005/S009, E12 - S004 · NFR - SCALE → E12 - S001 · NFR - A11Y → E01 - S012, E05 - S007, E12 - S003 · NFR - API → E01 - S008 · NFR - COMP → E01 - S006, E11 - S003/S004/S005.

Quality Gate — Scrum Master (testability):

- ✓ Requirement traceability: **124/124 FRs mapped (100%)**; 0 unmapped; Stage 2/3 explicitly deferred to future epics
- ✓ Every story has ≥4 Given/When/Then ACs (79 stories, 348 ACs total; avg 4.4/story)
- ✓ Every story includes an error - handling AC; every UI story an a11y AC; every API story an OpenAPI AC (API - first mandate)
- ✓ Performance - budget ACs on frontend/infrastructure stories (E01 - S010/S012, E03 - S011, E05 - S007, E10 - S003)
- ✓ Dependencies documented, acyclic
- ✓ Story keys consistent EPIC - NN - STORY - NNN; efforts S/M/L only (no story >3 days)
- ✓ Count integrity verified: overview totals = enumerated stories (79) and effort (182d) — RFX count - mismatch lesson applied
- **Score: 97/100** ✓ (testability gate ≥90 passed)

Next: Readiness Check → implementation - readiness - report.md

6. SUPPORTING RESEARCH: MARKET, COMPETITORS & COMPLIANCE (ANNEX)

Source artifact: research - brief.md

The Step - 0 research that grounded the plan: the July 2026 competitor landscape (five suites, eight AI - native challengers), the Microsoft - native white space, South African compliance requirements, technology validation and benchmarks. Annexed here; the readiness report follows as the closing chapter.

Project: IntelliSource — AI - Driven Sourcing & Contracting Platform

Author: BMad Research Agent

Date: 2026 - 07 - 24

Status: Complete

Inputs: IntelliSource Expanded Business Case v1 (Gijima), Gijima Architecture Design Principles (20), gijima - styles.css brand tokens, RFX Procurement Supplier Sourcing System planning artifacts, 27+ web sources (July 2026)

6.1 Executive Summary

IntelliSource enters a procurement - software market (~\$8–10B, ~11% CAGR) at an inflection point: every legacy Source - to - Pay suite now ships AI copilots and agents (SAP Joule, Coupa Navi, GEP QUANTUM, Ivalua IVA Studio, Jaggaer JAI), so **"AI - powered" is no longer a differentiator by itself**. What remains structurally unsolved — and is IntelliSource's white space — is the combination of:

- 1. Cost/complexity gap** — legacy suites cost six figures annually, take 6–18 months to deploy, and (Ariba) charge suppliers network fees. Mid - market and African enterprises are priced out.
- 2. Microsoft - native orchestration** — no S2C system of record is Teams - native today. Chat approvals are commoditized (Zip, ORO), but deep domain workflow (committees, two - envelope evaluation, auctions, CLM) orchestrated inside Teams remains open. The window is narrowing as Microsoft's Copilot Agent Store and Dynamics 365 agentic ERP advance.
- 3. South African localization moat** — automated B - BBEE 80/20 & 90/10 preference - point scoring, CSD/SARS/CIPC verification, staged compliance→functionality→price/preference evaluation, POPIA operator obligations, King V audit evidence, and readiness for the Public Procurement Act 28 of 2024 (draft regulations published April 2026, commencement pending). **No global suite does this out of the box.**

The RFX project supplies a proven, GO - rated (95.9/100) RFX execution foundation: 99 functional requirements, a battle - tested RfxStatus state machine, supplier portal, change - request workflow, immutable audit framework, and 55 implementation - ready stories — of which roughly 70% lift directly into IntelliSource's MVP, letting the new build concentrate on the net - new AI layer (intake, discovery, evaluation, Teams orchestration, contracting).

Recommendation: GO to planning. Position as "Autonomous Procurement Orchestrator, Microsoft - native, Africa - ready." Copy the challenger feature set (Zip/Levelpath/Keelvar/Arkestro patterns), not the legacy suite feature set.

6.2 Research Questions & Answers

Gijima Group	- Confidential -	Page 92 of 106
	Version:01.00 Valid from: 24.07.2026	Next review 24.10.2026

#	Question	Answer (evidence in sections below)
RQ1	What AI capabilities have incumbents shipped, and where are the gaps?	All 5 suites shipped agentic AI 2025–26 (Section 3). Gaps: cost, deployment time, supplier fees, mid - market UX, Africa compliance (Section 3, Section 7).
RQ2	What do AI - native challengers offer worth extracting?	Conversational intake, autonomous tactical sourcing, ML - timed close, predictive bids, savings ledger, mobile - first supplier UX (Section 4, Section 5).
RQ3	Is "Teams - native" defensible in 2026?	As a surface, no; as system - of - record + Agent Store distribution + MCP/A2A interop + domain depth, yes (Section 6).
RQ4	What SA/Africa compliance features are mandatory?	POPIA operator duties + s72 transfer rules, B - BBEE PPPFA 80/20–90/10 scoring, CSD/eTenders integration, King V audit evidence, PPA - ready rule packs (Section 7).
RQ5	What technology stack best satisfies Gijima's 20 architecture principles?	.NET 10/EF Core 10 + Postgres + AKS (validated by RFx); Microsoft Agent Framework + Azure AI Foundry for the AI layer; OpenAPI 3.2; event - sourced audit (Section 8).

6.3 Competitor Comparison Table — Legacy Suites (July 2026)

Vendor	Model	AI State (2026)	Strengths	Weaknesses	Threat to IntelliSource
SAP Ariba	Enterprise S2P suite + supplier network	Joule agents GA/rolling: Bid Analysis Agent (GA Q1'26, total - cost bid comparison + trade - off narratives), Sourcing Assistant (RFP drafting→negotiation strategy), Category Mgmt Assistant, intake/contract agents planned; "autonomous spend management" vision (May 2026)	Deep SAP ERP integration; largest supplier network	12–18 mo rollouts; ~\$250k+/yr entry; ~20h training/user; unintuitive UX; supplier - paid network fees (annual sub + ~0.155% transaction fee above thresholds)	HIGH where client is SAP - first (ERP lock - in); weak in mid - market/Africa
Coupa (Thoma Bravo)	BSM/S2P suite, \$10T community spend data	Navi multi - agent platform: Bid Evaluation Agent (Oct'25), Cost Formula Agent (75% faster, Nov'25), Agent Studio no - code (Dec'25), Compose & Catalyst (Jun'26). Acquired Cirtuo, Scoutbee (discovery), Rossum (doc AI), Tonkean	Best UX of suites (G2 4.7); community benchmarking data; aggressive agentic M&A	6–12 mo implementations needing consultants; per - user enterprise pricing; risk config "almost impossible" without experts	HIGH globally; MEDIUM in Africa (no local compliance depth)

Vendor	Model	AI State (2026)	Strengths	Weaknesses	Threat to IntelliSource
		(NL intake & orchestration, May'26)			
GEP (SMART/QUANTUM)	Unified S2P + services	GEP QUANTUM AI - native layer (100+ customers); 2026 "Quantum Intelligence" autonomous agents; Gartner S2P Leader 2025 & 2026	Strongest "AI - native" narrative; single data model	Services - heavy delivery; opaque pricing; Fortune - 500 focus, limited mid - market/Africa presence	HIGH on narrative (closest AI - first story)
Ivalua (KKR)	Unified S2P platform	IVA Studio (Jun 2026): fully agentic S2P execution, skills framework, MCP support , 25+ use cases; governance pattern: <i>agents inherit user permissions, every action logged</i>	Single codebase; extreme configurability; best AI governance pattern (worth copying)	Implementation complexity driven by configurability; high TCO	MEDIUM — enterprise - heavy, not Microsoft - focused
Jaggaer (Vista)	S2P suite, strong supplier risk/ESG	JAI (launched Jun'25, full launch May 2026): Assist → Copilot → Autopilot no - code agentic platform on "Agent OS"	Supplier risk & ESG modules; faster mid - market time - to - value than Ivalua	Complex navigation; dated UX; performance issues on large datasets	LOW–MEDIUM in Africa

Pricing signal across suites: six - figure annual entry, per - user/module licensing, 6–18 - month deployments, supplier - side fees (Ariba). This is precisely the gap the business case targets with \$8k/mo mid - market and \$18k/mo enterprise pricing.

6.4 Challenger Comparison Table — AI - Natives (feature blueprint)

Vendor	Category	Signature capabilities worth extracting	Proof points
Zip (\$2.2B val.)	Intake orchestration	Conversational front door for all spend; Slack/Teams approvals; cross - functional routing; AI "Superagents" (unblock stalled requests, tail - spend negotiation); procurement - native MCP ; governed AI platform	\$6B cumulative customer savings (Dec'25); 7M+ suppliers
Oro Labs (~\$100M C)	Intake + orchestration	Guided adaptive intake; approvals in Teams/Slack/GChat; supplier onboarding w/ bank - detail validation	—
Levelpath (\$55M B)	AI - native sourcing	Hyperbridge multi - model reasoning; genuinely	"10x efficiency" claim

Vendor	Category	Signature capabilities worth extracting	Proof points
		mobile - first UX; invoice automation to capture negotiated savings	
Fairmarkit	Autonomous sourcing	"Total Agentic Sourcing" (Apr'26): agents autonomously source tail AND strategic spend; KIT assistant; ERP - native integrations	10x events/FTE; ~\$40k savings/buyer
Keelvar	Sourcing optimization	Kai orchestrator; Optimal Close Agent (ML picks best bidding close time); Autonomous Rate Refresh (auto re - market on contract expiry/drift); no - code sourcing bots	Only vendor in both Gartner autonomous - sourcing categories (2026)
Arkestro	Predictive procurement	ML - predicted prices; suggested/pre - populated bids and intelligent counter - offers embedded in supplier workflow	18.8% savings per \$1M; up to 60% cycle - time reduction
Globality	Autonomous negotiation (services)	Glo: conversational scoping replaces RFP; autonomous multi - round time - boxed negotiations	"\$15M saved in one hour" case
mysupply	SAP - attached autonomous sourcing	Demand bundling for leverage; AI bots + eAuctions + game theory; audit - proof	>50% effort reduction; 5–11% savings

Consolidation warning: Coupa absorbed four AI - natives in ~18 months. Everest Group (Mar 2026) sees the market rationalizing around a few anchors — validation for a differentiated regional/Microsoft - native anchor play, but the acquisition clock is running.

6.5 Feature Capability Matrix (market scan → IntelliSource backlog candidates)

Legend: ✓ = table stakes (all suites have it) · * = differentiator worth building natively · SA = South Africa moat feature

Lifecycle phase	Feature	Market status	IntelliSource
Intake	Plain - language conversational intake → structured spec	Tonkean/Zip lead; suites partial	* MVP core
	Guided adaptive intake forms; policy - aware routing	Zip/ORO	* MVP
	Demand bundling of similar requests	mysupply	Stage 2

Lifecycle phase	Feature	Market status	IntelliSource
	Stalled - request unblocking agent	Zip Superagent	Stage 3
Supplier discovery	Approved - vendor lookup (ERP master)	✓ all	MVP (RFx pattern)
	AI web/DB discovery + enrichment	Scoutbee (now Coupa)	* Stage 2
	ESG/diversity data (EcoVadis connector), sanctions screening	Suites via add - on	Stage 2
	CSD/SARS/CIPC verification, B - BBEE certificate/affidavit capture	Nobody	* SA MVP—Stage 2
RFx authoring	AI - drafted RFQ/RFP/RFI from spec + template libraries	Emerging in all	* MVP core
	Event types RFI/RFQ/RFP, deadlines, reminders, Q&A windows	✓ all	MVP (RFx reuse)
	Multi - round / BAFO; two - envelope (technical→price) staged evaluation	✓ suites	Stage 2 (envelope: MVP data model)
	Auction engine (English/Dutch/Japanese, multi - attribute)	✓ suites; Keelvar optimizes	Stage 3
	ML - timed event close; autonomous rate refresh	Keelvar only	Stage 3
Evaluation	Weighted criteria scorecards	✓ all	MVP
	AI completeness/compliance check (missing docs, certs)	Partial everywhere	* MVP
	AI bid comparison w/ total - cost normalization + exec summaries	Joule/Navi shipping	* MVP (basic) → Stage 2
	Committee workflows: independent/blind scoring, consensus, Col declarations, e - voting	(partial) all (weak)	* Stage 2 — white space
	B - BBEE 80/20 & 90/10 price+preference calculation	Nobody	* SA Stage 2
Award & contract	Award - to - contract flow, clause libraries, AI redlining vs playbooks	CLM specialists (Sirion, Icertis)	Stage 3
	eSignature (DocuSign/Adobe)	✓ all	Stage 3
	Contract - expiry - triggered re - sourcing	Keelvar/IVA	Stage 3

Lifecycle phase	Feature	Market status	IntelliSource
Collaboration	Teams - native approvals, committee meetings auto - scheduled, e - voting in Teams	Nobody at S2C depth	* MVP core (wedge)
Analytics	Cycle - time/SLA KPIs, exec dashboards, CSV/PDF export	✓ all	MVP
	Savings ledger (event→contract), leakage tracking	Zip/Levelpath	Stage 2
	Predictive spend/risk analytics	Suites	Stage 3
Supplier portal	Free, no - fee access (anti - Ariba); single front door; mobile - first bidding; reusable profiles	HICX: suppliers juggle 8.4 portals avg; 61% can't do best work	* MVP
Audit	Immutable audit trail, evidence export	✓ all	MVP (RFx framework + hash chaining)

6.6 RFx Reuse Inventory (seed assets)

The RFx Procurement Supplier Sourcing System (GO at 95.9/100, March 2026) provides directly transferable assets:

Lift - and - adapt (≈70% of its scope):

- Clean Architecture .NET scaffolding, EF Core + PostgreSQL, RFC 7807 error middleware, Azure Blob storage, OpenAPI/Swagger, Docker Compose (EPIC - 01)
- RFx core lifecycle: create/edit/clone/delete, status enforcement, file validation + virus scanning (EPIC - 02); state machine Draft → PendingReview → PendingApproval → Published → Closed (+ Rejected, Returned - to - Initiator, Pending - Clarification; auto - approval when Reviewer=Approver; RFQ short path skipping approval)
- Review/approval workflow with mandatory comments on Reject/Clarify and SoD (EPIC - 03)
- Publishing & invitations with tokenized portal links, retry tracking (EPIC - 04)
- Supplier portal: email/password + first - time setup, lockout (5 attempts/30 min), acknowledgement - gates - documents, versioned response submission, deadline locking (EPIC - 05)
- Change - request engine with 7 CR types, dynamic forms, CR numbering (EPIC - 06)
- Dashboard + SignalR real - time, notification engine with 7 - day/1 - day reminders and retry (EPIC - 07/08)
- Immutable AuditLog framework (append - only, OldValue/NewValue JSON, 7 - year POPIA retention, status history derived from audit) — ADR - 012
- Numbering schemes: RFx - YYYY - NNN, CR - YYYY - NNN, ERR - YYYYMMDD - NNNNN
- NFR baselines: API p95 <500ms, dashboard <2s, lockout/session policies, AES - 256 at rest, TLS 1.2+, data residency Azure SA

Net - new for IntelliSource (RFx explicitly out - of - scope): AI intake/spec generation, AI RFx drafting, supplier discovery, bid evaluation & scoring (only API scaffolding exists), award/contract lifecycle, Teams orchestration, supplier onboarding/KYC.

Lessons to fix from day one: (1) tiered rate limiting (supplier upload spikes at deadline — add ~50 req/min upload tier); (2) DR targets RTO 1h / RPO 15 min; (3) WCAG accessibility was deferred in RFx — IntelliSource enables WCAG 2.1 AA (target 2.2 AA) from the start; (4) reconcile requirement counts — summary metrics must match enumerated items; (5) multi - approver chains unsupported — IntelliSource needs configurable approval chains.

6.7 Microsoft - Native Strategy Assessment

Current landscape: Zip and ORO offer Teams/Slack *integrations* (chat approvals, notifications), but no S2C system of record is Teams - native. Microsoft's own motion: M365 Copilot **Agent Store** (in - product marketplace, 70+ agents at launch), Copilot Studio agents + workflows, and Dynamics 365 "agentive ERP" with Copilot Cowork — whose published procurement scenario already covers gathering supplier email responses, comparing bids, generating scorecards, and coordinating approvals.

Defensibility verdict: Teams - native is a **wedge, not a moat**. Defensible combination:

1. True S2C system of record with domain depth Microsoft won't build (evaluation committees, two - envelope compliance, B - BBEE scoring, auction engines, CLM)
2. Distribution via Agent Store/AppSource with Teams as the orchestration surface
3. **Microsoft Agent Framework + MCP + A2A alignment** so IntelliSource agents interoperate with customers' Copilot estates instead of competing
4. Azure South Africa data residency

Risk: Icertis already chains across SAP + Microsoft Copilot in CLM; the Microsoft - procurement lane is open but narrowing. Speed matters.

6.8 South Africa / Africa Compliance Requirements (moat features)

POPIA (operator role): IntelliSource processes personal information on behalf of customer "responsible parties" → mandatory written DPA (s21), processing only on documented instruction, security safeguards, immediate breach notification to the responsible party. Eight lawful - processing conditions drive features: processing registers, retention schedules, data - subject access/correction/deletion tooling. Penalties: fines to R10M + criminal liability. **Section 72 cross - border transfer rules directly constrain LLM inference routing** — any AI calls leaving SA need adequacy/consent/contract cover; prefer Azure OpenAI in SA North; paper any cross - region inference under s72.

King IV → King V: IoDSA adopted King V on 31 Oct 2025 (effective FYs starting ≥1 Jan 2026), retaining apply - and - explain with sharpened technology/information governance. IntelliSource's immutable audit trail + decision logs = board - grade "King V evidence" for procurement governance and combined assurance. Sales collateral, not just compliance.

B - BBEE / PPPFA preference scoring: tenders >R30k must apply **80/20 (≤R50M)** or **90/10 (>R50M)** price/B - BBEE - points formulas; Level 1 contributors earn full preference points; evaluation is staged: mandatory - document compliance → functionality threshold → price + preference calculation. Bidders prove status via SANAS - accredited certificates or sworn affidavits (EMEs/QSEs). 2022 regulations allow organs of state to define specific - goal allocations. **No global suite automates this.**

Public Procurement Act 28 of 2024: assented 18 Jul 2024, not yet in force; National Treasury published draft General Public Procurement Regulations for comment 16 Apr 2026; PPPFA + 2022 regs apply meanwhile. Public - sector readiness = CSD integration (mandatory since 2016; SARS tax - status + CIPC verification), eTenders portal publication, bid evaluation/adjudication committee separation, PFMA/MFMA - grade audit trails. **Build configurable compliance rule packs with a PPA - ready switch.**

Data residency: Azure South Africa North (Johannesburg) has 3 availability zones, paired with SA West; Microsoft investing a further R5.4bn through 2027. Verify Azure OpenAI/Foundry model availability in SA North per model; route around gaps with s72 - compliant contracts.

6.9 Technology Stack Options

Gijima Group	- Confidential -	Page 98 of 106
	Version:01.00 Valid from: 24.07.2026	Next review 24.10.2026

Gijima's 20 Architecture Design Principles lock the core stack (validated end - to - end by RfX):

Layer	Locked/Recommended	Rationale & 2026 status
Backend	.NET 10 LTS, Clean Architecture	Principle #1; .NET 10 LTS since Nov 2025; EF Core 10 query pipeline 25–50% faster
Frontend	ASP.NET MVC + React + Tailwind	Principle #2; Gijima brand tokens map to Tailwind config
Data	EF Core 10 → PostgreSQL	Principle #3; Npgsql batched inserts; proven in RfX
Local dev	Docker Desktop	Principle #4
Deployment	AKS (Kubernetes), cloud - native containers	Principle #5; namespace - per - tenant logical isolation + deployment stamps with per - tenant Postgres for regulated/public - sector tenants (K8s not safe for hostile multi - tenancy)
AI layer	Microsoft Agent Framework (SK+AutoGen successor; GA Q1'26, deterministic Workflow framework GA Q2'26) + Azure AI Foundry / Azure OpenAI	MCP, A2A, OpenAPI - first interop; aligns with Teams/Copilot strategy; deterministic workflows with compliance audit trails
RAG	Azure AI Search hybrid retrieval (keyword+vector+semantic rerank), 1.5–2k - token parent chunks	Pure vector misses part codes/IDs; model tiering (frontier for evaluation, mini for volume) cuts AI cost 40–60%
API	OpenAPI 3.2 (released Sep 2025) + Swagger	Principle #7 (OAS 3.2.0); streaming/SSE support for agent endpoints; expose agent tools via MCP
AuthN/Z	Entra ID SSO + MFA , supplier email/password + optional Entra External ID	Principle #9; Zero Trust (#6)
Audit	Event - sourced append - only log + cryptographic hash chaining , WORM export	Principles #10, #20; King V/PPA evidence grade; copy lvalua pattern: <i>agents inherit invoking user's permissions; every agent action logged</i>
Observability	OpenTelemetry + Application Insights	Principles #16–19 (99.5% app / 99.99% infra uptime)

Alternatives considered: GraphQL (rejected — principle #7 mandates OAS/REST; OpenAPI 3.2 + webhooks suffice); Azure SQL (rejected — principle #3 mandates Postgres); LangChain/LlamaIndex (rejected — Microsoft Agent Framework aligns with Teams/Copilot strategy and .NET).

6.10 Best Practice Patterns

- 1. Agent governance (Ivalua pattern):** AI agents execute strictly within the invoking user's permissions; never exceed; log every agent action to the audit trail with model version + prompt hash. Human - in - the - loop approval for all award - affecting outputs.
- 2. AI transparency for tender defensibility:** every AI evaluation output carries rationale, source citations to bid documents, and a confidence flag; human evaluators must confirm/override with reasons — the audit trail records both. (Public - sector tender challenges demand this.)
- 3. Two - envelope discipline in data model from day one:** commercial data sealed/encrypted separately from technical responses; unsealing is an audited, permissioned event after functionality gates pass.

4. **Event - sourced audit:** append - only, intent - capturing events ("user X approved award Y for reason Z"), hash - chained for tamper evidence, WORM - exportable for auditors.
5. **Multi - tenant isolation tiers:** shared - namespace for standard SaaS tenants; deployment - stamp with dedicated Postgres for regulated/SOE tenants ("in - tenant" business - case option).
6. **Supplier - first portal UX:** zero supplier fees; acknowledgement - gated document access (RFx); mobile - first submission; suppliers see only invited events; no email enumeration.
7. **Model tiering & data boundaries:** frontier model for bid evaluation summaries; mini models for classification/reminders; no customer bid data used for model training; s72 - compliant inference routing.
8. **Performance discipline:** p95 API <500ms, LCP <2.5s, INP <200ms, initial bundle <1MB — instrument from sprint 1 (most B2B fails these; passing is a demo - able differentiator).

6.11 Benchmarks (targets to match or beat)

Metric	Market claims (vendor - reported)	IntelliSource target
Sourcing cycle - time reduction	Arkestro ≤60%; Coupa/Tonkean 50%; business case 30–40%	≥30% instrumented, provable (cycle - time analytics built in)
Manual - effort reduction	mysupply >50%; Zip 30+ h/wk	≥50% on tactical events
Adoption lift from NL intake	Tonkean 2.2x	Track intake→event conversion
API latency	Suites unpublished	p95 <500ms; median <100ms
Page performance	Avg B2B mobile LCP 7.05s (poor)	LCP <2.5s, INP <200ms, CLS <0.1
Concurrency	—	10k+ concurrent users, <2s response (Gijima principle #13)
Availability	99.95% typical B2B	99.5% app / 99.99% infra (principles #18–19)
Accessibility	EAA deadline passed Jun 2025 (EN 301 549/WCAG 2.1 AA); almost no procurement vendor publishes ACR	WCAG 2.1 AA (aim 2.2 AA) + published ACR

6.12 Key Findings & Recommendations

1. **Don't sell "AI"; sell speed - to - value + zero supplier fees + Africa compliance.** All incumbents have AI stories; none have 4–8 - week deployments at mid - market pricing with B - BBEE/POPIA/King V built in.
2. **MVP = RFx spine + AI layer + Teams surface.** Reuse ≈70% of RFx's proven requirements; concentrate new engineering on intake AI, RFx drafting AI, evaluation AI (completeness + weighted scoring + exec summaries), and Teams orchestration.
3. **Build the SA compliance rule pack early (Stage 2 at latest)** — it is the hardest - to - copy moat and unlocks SOE/public - sector lighthouse deals (Eskom, Transnet targets in business case).
4. **Align to Microsoft Agent Framework + MCP now** — distribution through Agent Store and interop with customer Copilot estates converts Microsoft from potential competitor to channel.
5. **Two - envelope + committee governance in the MVP data model** even if UI ships Stage 2 — retrofitting sealed - bid separation is expensive.

6. **Instrument the ROI story** — cycle - time and savings ledgers from sprint 1; every marketing claim ($\geq 30\%$ cycle reduction) must be provable in - product (Customer Zero: Gijima Procurement BU).
7. **Watch consolidation:** Coupa bought 4 AI - natives in 18 months. Remaining independents are feature blueprints and comps; a differentiated regional anchor play beats a me - too generalist.

6.13 Risks & Constraints

Risk	Likelihood	Impact	Mitigation
Microsoft first - party agents (Copilot Cowork) commoditize light bid comparison	High	Medium	Domain depth (committees, envelopes, B - BBEE); MCP interop rather than competition
Azure OpenAI model availability gaps in SA North	Medium	High (POPIA s72)	Verify per model; s72 contractual cover for cross - region; abstraction layer for model routing
Incumbent ERP lock - in (Ariba on SAP clients)	High	Medium	Overlay positioning: integrate, don't replace (business - case hybrid strategy)
AI evaluation outputs challenged in tender disputes	Medium	High	Human - in - the - loop, rationale + citations, immutable audit of overrides
PPA commencement timing uncertainty	Medium	Low	Configurable rule packs; PPPFA default, PPA switch
9–18 month enterprise sales cycles strain runway	High	Medium	Customer Zero (Gijima internal) + lighthouse SOE pilots (business case strategy)
Coupa/suite acquisition of remaining challengers narrows partner options (e.g., supplier - data feeds)	Medium	Low	Multi - source data strategy (EcoVadis, D&B, CSD)

6.14 Sources

Key sources (full URLs in agent research log): SAP News & ERP Today (Joule agents, May 2026), Coupa newsroom (Navi, Agent Studio, Tonkean/Scoutbee/Rossum acquisitions), GEP QUANTUM/QI, Ivalua IVA Studio PR (Jun 2026), Jaggaer JAI (Businesswire May 2026), Zip Series D & Superagents (Jun 2026), Keelvar Gartner Market Guide 2026, Arkestro, Globality, mysupply SAP Store, Everest Group consolidation analysis (Mar 2026), HICX supplier experience survey, Microsoft Agent Framework/Learn, Azure regions/TechCentral (SA investment), OpenAPI Initiative (3.2), POPIA s21/s72 (Michalsons, Securiti), IoDSA King V Code (Oct 2025), eTender Portal & ProTenders (PPPFA 80/20–90/10), CDH & SmartProcurement (Public Procurement Act 2024, draft regs Apr 2026), Level Access (EAA), BlazingCDN/Parachute (performance benchmarks).

Quality Gate — Research:

- ✓ All 5 research questions answered with evidence
- ✓ Competitor table: 5 legacy + 8 challengers (13 entries)

- ✓ Technology recommendations justified against Gijima's 20 principles
- ✓ Architecture - focused; quantitative where available

Next: Product Owner agent → product - brief.md

7. IMPLEMENTATION READINESS REPORT — GO/NO - GO

Source artifact: implementation - readiness - report - 2026 - 07 - 24.md

The formal quality gate: artifact inventory, programmatic cross - reference validation, findings and the GO decision at 95.7/100.

Project: IntelliSource — AI - Driven Sourcing & Contracting Platform

Author: BMad Readiness Check Agent

Date: 2026 - 07 - 24

Pipeline: Research → Product Owner → Business Analyst → Architect → UX Designer → Scrum Master → Readiness

7.1 Executive Summary — Decision: ✓ GO

All six planning artifacts are present, internally consistent, and pass their quality gates at or above minimum thresholds. Requirement traceability is 100% (124/124 FRs mapped to 79 stories, verified programmatically — zero orphans). Count integrity between summary tables and enumerated content was explicitly verified (the RFx lesson), including one discrepancy caught and corrected during planning (FR count 106→124). No blockers. Two minor findings and three notes are logged for the implementation team.

Aggregate Quality Score: 95.7/100

Artifact	Gate	Min	Target	Actual	Status
research - brief.md	Research questions answered	All	All	5/5 + 13 - vendor landscape	✓ Pass
product - brief.md	artifact_completeness	90	95	96	✓ Pass
prd.md	testability	90	95	100 (124/124 FRs testable) · doc score 95	✓ Pass
architecture.md	Testing strategy defined + completeness	90	95	95	✓ Pass
ux - design - specification.md	Accessibility checklist (a11y enabled)	5/5	5/5	5/5 · doc score 96	✓ Pass
epics.md + sprint - status.yaml	FRs mapped / ACs testable	100% / 90	100% / 95	100% / 97	✓ Pass

7.2 Artifact Inventory

Artifact	Size	Modified	Status
research - brief.md	26.3 KB	2026 - 07 - 24	✓ Complete — 5 legacy + 8 challenger vendors, SA compliance research, tech landscape, benchmarks
product - brief.md	15.2 KB	2026 - 07 - 24	✓ Complete — 4 personas, 10 must/5 should MVP

Artifact	Size	Modified	Status
			scope, 7 KPIs with numbers+timeframes
prd.md	60.3 KB	2026 - 07 - 24	✓ Complete — 9 journeys (all with failure+edge cases), 124 FRs, 53 NFRs, 23 entities, API overview
architecture.md	27.4 KB	2026 - 07 - 24	✓ Complete — 16 ADRs, all 20 Gijima principles mapped, full testing/error/API/perf/a11y architecture
ux - design - specification.md	25.5 KB	2026 - 07 - 24	✓ Complete — light+dark tokens (exact values), 20 components all states, 12 pages, frontend - design ready
epics.md	84.3 KB	2026 - 07 - 24	✓ Complete — 12 epics, 79 stories, 348 ACs, traceability matrix, dependency graph, sprint plan
sprint - status.yaml	—	2026 - 07 - 24	✓ Complete — 79 stories in backlog (matches epics.md exactly)
dev - config.yaml	—	2026 - 07 - 24	✓ Complete — all flags, targets, tools propagated

7.3 Cross - Reference Validation (programmatically verified where possible)

Check	Result
Every PRD FR maps to ≥1 story	✓ 124/124 — comm diff between PRD FR set and epics matrix returned empty
Story counts consistent (epics.md headings vs sprint - status.yaml vs overview table)	✓ 79 = 79 = 79
Effort totals consistent (overview vs per - story sum)	✓ 182d (corrected during planning)
UX components ↔ architecture structure	✓ C01–C20 map to React - island structure (ADR - 02); E01 - S012 wires them
Data model supports all features	✓ 23 entities cover all 16 FR modules incl. new AI entities (AIInvocation, CompletenessFlag, Score, PromptProfile ref)
Testing strategy covers all story types	✓ Unit/integration/contract/E2E/load/security/a11y/AI - regression all tooled with gates
Error - handling consistency chain	✓ Brief Section 7 → PRD NFR - ERR - 01..06 → Arch Section 8 (Polly, circuit breakers, degradation matrix) → UX Section 5a patterns → error AC in every front - end story

Check	Result
API documentation chain (api_first)	✓ Brief (MVP deliverable) → PRD NFR - API - 01..06 (OAS 3.2.0) → Arch ADR - 09 (code - first, oasdiff, Spectral, Schemathesis) → OpenAPI AC on every API story
Performance budget chain	✓ Brief targets → PRD NFR - PERF/BUDGET (numeric) → Arch Section 11 (CI enforcement) → perf ACs on E01 - S010/S012, E03 - S011, E05 - S007, E10 - S003, E12 - S001
Accessibility chain (a11y enabled)	✓ Brief (WCAG 2.1 AA target) → PRD NFR - A11Y - 01..06 → Arch Section 12 (axe/Lighthouse/Pa11y/NVDA protocol) → UX Section 6 5/5 checklist → a11y ACs on all UI stories → E12 - S003 ACR
AI governance chain	✓ PRD FR - AI - 01..08 → Arch ADR - 10 (user - permission inheritance, prompt hashing, s72 router) → E01 - S011/E11 - S005 stories → audit integration (FR - AUD - 01/FR - AI - 05)
Frontend - design readiness (flag enabled)	✓ UX Section 7 checklist all boxes: exact hex/px/ms tokens, exhaustive states, complete dark palette, explicit breakpoints
Gijima Architecture Principles compliance	✓ All 20 mapped in Arch Section 2 with design responses
RFx reuse traceability	✓ Research Section 5 inventory → PRD FRs tagged "RFx reuse" → stories cite lessons (rate - limit tiers, DR targets, count integrity, a11y from day one, multi - approver chains)

7.4 Findings

7.4.1 [BLOCKER] Blockers — none

7.4.2 [MAJOR] Major — none

7.4.3 [MINOR] Minor (fix during implementation)

- 1. Azure OpenAI model availability in SA North unverified (PRD OQ3).** Owner: tech lead, Sprint 0 spike inside EPIC - 01 - STORY - 011. The model - router abstraction plus s72 policy already contains the risk; verification only tunes routing defaults.
- 2. Evidence - pack legal format not yet reviewed by Gijima Legal (PRD OQ4).** Owner: product owner; due before EPIC - 09 - STORY - 002 starts (Sprint 13 window). Structure (PDF chronology + CSV/JSON + manifest + verification statement) is standard; legal may add layout requirements.

7.4.4 [NOTE] Notes

- 1. Supplier MFA** deferred to fast - follow (PRD OQ2) — lockout, uniform - response, and password policy provide MVP baseline; Entra External ID option documented for Stage 2.
- 2. FR - S2 - QA (anonymous supplier Q&A)** is flagged as a pull - forward candidate if MVP capacity allows — J4 journey and notification engine already accommodate it.
- 3. Consolidation watch:** Coupa's acquisition pace may absorb supplier - data partners; multi - source strategy (EcoVadis, D&B, CSD) noted in research brief for Stage 2 planning.

7.5 Quality Score Detail

Dimension	Weight	Score	Weighted
Research completeness	10%	96	9.6

Dimension	Weight	Score	Weighted
Product brief completeness	15%	96	14.4
PRD testability & completeness	20%	97	19.4
Architecture completeness (incl. testing strategy)	20%	95	19.0
UX spec completeness (incl. a11y 5/5, frontend - design ready)	15%	96	14.4
Epics traceability & AC testability	20%	97	19.4
Aggregate	100%		95.7 / 100

7.6 Recommendations

Before Sprint 1:

1. Run **bmad - frontend - design** (frontend_design_enabled: true) to generate the Gijima design system (Tailwind config, tokens, component scaffolds, Storybook) from UX spec Section 2–3 — EPIC - 01 - STORY - 012 then wires the output.
2. Execute the Sprint - 0 spike: Azure OpenAI/Foundry model availability in SA North (Minor #1) and confirm Adobe Fonts kit ID for Proxima Nova (PRD D5).
3. Confirm Customer Zero (Gijima Procurement BU) pilot dataset and SME availability (PRD A3).

During implementation:

1. Enforce the count - integrity discipline that caught the 106→124 discrepancy: CI check comparing traceability - matrix FR set against PRD FR set on every artifact change.
2. Schedule Gijima Legal review of evidence - pack format by Sprint 12 (Minor #2).
3. Re - run **bmad - plan** (not manual edits) for any mid - project requirement changes, then re - run this readiness check.

Handoff sequence: bmad - frontend - design → bmad - develop (stories in sprint - status.yaml order, EPIC - 01 first) → bmad - quality per story/epic → bmad - user - manual after final retrospective (user_manual_mode: true).

✓ **Readiness: IntelliSource**

Decision: GO

Aggregate Quality Score: **95.7/100**

Blockers: 0 | Major: 0 | Minor: 2 | Notes: 3

All artifacts ready for implementation sprint.